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BS-ZOOLOGY

ZOO102- Principles of Animal Life-I



HANDOUTS TOPIC NO 1 TO 270

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1 Concept of Zoology

Defining Zoology

Zoology is the branch of biology which deals with the study of animal life. It includes all the aspects of animals' life. The term zoology is combination of two Greek words "zoon" means animals and "logos" means to study.

History of Zoology

History of zoology is as old as history of man. Greeks and Romans were the first to document animal life. Muslim scientists gave many discoveries in the field of zoology. Modern Scientists revolutionized the knowledge about animal life.

What do we study in zoology?

Animal life is very fascinating to human being. It includes a great variety of animals ranging from microscopic invertebrates to very large size vertebrates. Beautiful domestic pets, gorgeous wildlife and stunning aquatic animals are all studied under the heading of zoology. When we see living organisms different questions rise in our mind like;

How and where they live?
What they eat?
What they do?

The answer to all questions is given by observing the animals living in natural environment, studying their activities (feeding, growth and reproduction) structure and function of different body parts up to molecular level.

2 Significance of Zoology

The knowledge of zoology is important for survival of man and other animals on this earth. It helps us to know about different types of animals living on earth. It tells us how life is sustaining on this earth. The knowledge of zoology has helped mankind to improve environment, health and food quality.

Improving Health

The knowledge of zoology is helping to improve the health of human and other animals. We are able to understand the mechanism of diseases and their modes of transmission. Many diseases like cancer, hepatitis and AIDS which were previously thought to be uncontrolled can now be prevented and cured. Many others like diabetes and cardiovascular diseases can be avoided by adopting health life style. Many medically important hormones and proteins are now days are obtained from transgenic animals.

More and Better Quality Food

Rapidly increasing population of world demands more and better food. More food is required to feed more than 7 billion population of the world. Many breeds of animals have been developed which provide more meat, milk, eggs. Dairy industry, poultry and aquaculture are well established and running at commercial scale. Quality of food has been improved with more nutrition value. Selective breeding and genetic engineering are two lines of research for production of more and better quality food.

Improving Environment

More than 1.5 million species of animals have been documented. Animals have critical role in maintaining ecosystem. Beautiful birds and animals also have esthetic value. The knowledge of zoology is important in maintaining their natural habitat and protecting them from extinction. Many animals play their role in cleaning the environment. For example vulture and other scavengers eat dead bodies of other animals. Similarly small aquatic animals remove toxic metals and other pollutants from water bodies. Many animals are sensitive to pollutants and act as bioindicator of different pollutants.

3 Applications of Zoology

Zoology covers the areas ranging from basic structure of animals to their industrial application. There are four major areas of with direct applications of zoology:

1. **Medical Research**

In medical research the first step is the diagnosis of a disease. Many diseases are due to animals which cause and transmit them. Advancement in techniques and protocols have enabled the scientists to understand that how a person is affected by a disease and what are the underlying mechanism for specific disease. For example *Aedes* mosquitos are responsible for spread of dengue. Once the mode of action is known, the disease may be controlled by taking preventive measures and proper medication. New drugs are designed, their mode of action and effects are first studied on animal models then they are used by human.

2. Agriculture

Study of zoology helps us to understand the life of different insects, pests, rodents, birds and mammals. Many crop pests of are the food of insects and birds. They may be used to control the pest. The phenomenon is called **biocontrol**. In comparison to chemical control, there is no risk of developing resistance in crop pests. **Animal husbandry** is the farming of animals to produce meat, milk and eggs while **aquaculture** is the culturing of aquatic animals and plants. Beekeeping for the production of honey and wax is called **apiculture** while rearing of silkworm for production of silk is called **sericulture**.

3. Industry

Billion dollar medical industry is based on many animal and plant products. Vaccines which typically contain weak or killed disease causing agents are used to immunize human, farm animals and pets against different disease causing agents. Antibodies are also used for passive immunity where immune system is too weak or needed not to be activated. Growth hormones are used in children with suppressed growth. Leather industry products like bags, shoes, jackets sofas and hundreds of other goods are made from hides of animals.

4. Environmental Role

Biodiversity of an ecosystem is the total number of different species of plants and animals. Biodiversity of earth is decrease rapidly, mainly of habitat destruction by human (anthropogenic) activities. It is the knowledge of zoology which is helping to reduce the biodiversity loss by habitat protection through national parks, game reserves and other protective places. Zoo and safari parks have great aesthetic value.

Animals have also an important role in controlling environmental pollution. Scavengers are the animals which keep the environment clean by eating up dead bodies of animals. Many small soil and aquatic animals remove toxic pollutants. This process is called bioremediation.

4 Fundamental Properties of Life

"The ability to reproduce is Life." What would be if there is a single male or female rabbit?? Life has no perfect definition but certain characteristics.

1. Living organisms have unique and complex composition of biomolecules which differ from other molecules in their structures.
2. Living organism have ability to reproduce (sexual and asexual reproduction). They inherit different properties from their parents.
3. Living organisms have genetic material in the form of DNA or RNA which is used to encode different enzymes and other proteins.
4. Metabolism is all types of chemical changes in the body during which animals derive energy, synthesize molecules of body need and detoxify the harmful molecules.
5. Living organisms interact with their environment. The branch of biology in which we study the interaction of living organisms with their environment is called ecology.

5 One World View

Although life appears very diverse but there is a structural and functional unity in all forms of life. Life shares same properties.

a. Genetic Unity

Almost all forms of life have same genetic blue prints (DNA). DNA of bacteria plants and animals has same chemical composition. It carries genetic information. Genetic code is almost same.

b. Common Structural Organization

All the prokaryotes and eukaryotes are made up of cells. Cell is both the structural and functional unit of life. Cells of all living organisms have striking similarities.

c. Common Evolutionary Forces

All the animals are uniformly acted upon by same type of evolutionary forces like genetic drift, inbreeding, selection and mutation. This is one reason of close similarity in the living organisms.

d. Common Environment

All the organisms living on this earth have same type of physicochemical environmental factors. They face same type of environmental conditions. They get food from same environment.

Due to basic similarities in living organisms scientists are of the view that life has a common origin. Evolutionary theory is also based on the idea that all organisms evolved from a common ancestor.

6 Specializations in Zoology-I

Specializations on the basis of field of study

1. **Anatomy:** It is the study of the structure of complete organisms and their body parts.
2. **Physiology:** The study of the function of organisms and their body parts.
3. **Cell biology (Cytology):** Study of the structure and function of cell and its components.
4. **Molecular Biology:** Molecular biology is the study of structure and function of the macromolecules (e.g. proteins and nucleic acids, Lipids) essential to life.
5. **Histology:** Study of organisms at tissue level is called histology.
6. **Embryology:** Study of development of embryo from fertilized egg.
7. **Genetics:** Study of the mechanisms of transmission of traits from parents to offspring.

7 Specializations in Zoology-II

Specializations on the basis of animal group

1. **Protozoology:** Protozoology is the study of protozoa. It deals with the characterization, taxonomy and medical importance of protozoa.
2. **Nematology:** Nematology is the scientific discipline concerned with the study of nematodes, or roundworms.
3. **Entomology:** Entomology is a branch of zoology which deals with the study of insects.
4. **Malacology:** Malacology is the branch of invertebrate zoology that deals with the study of the mollusca.
5. **Ichthyology:** Ichthyology is the branch of zoology concerned with the study of fishes.
6. **Herpetology:** Herpetology is the branch of zoology that deals with the study of amphibians and reptiles.
7. **Ornithology:** It is the branch of zoology concerned with the study of birds.
8. **Mammalogy:** It is the branch of zoology that deals with the study of mammals.

8 Animal Diversity

Animal diversity is the number of different types of animals living on this earth. The total diversity of life documented is about 1.74 million.

Group	Described	Estimated
Chordates	**64,788**	**~80,500**
Mammals	5,487	~5,500
Birds	9,990	>10,000
Reptiles	8,734	~10,000
Amphibians	6,515	~15,000
Fishes	31,153	~40,000
Invertebrates	**~1,359,365**	**~6,755,830**
Echinoderms	7,003	~14,000
Arthropods	~1,149,248	~5,750,000
Mollusca	~85,000	~200,000
Annelida	16,763	~30,000
Nematods	25,000	~500,000
Platyhelminthes	20,000	~80,000
Cnideria	9,795	……
Porifera	6,000	~18,000

Scientists estimate species range from 5 to 30 million. However, the more accurate estimate about current diversity of earth is between 8.7±1.3 million. It means about 80% species have to be described yet. Every species survives on earth for approximately 1 million year. Scientists estimate that more than 99% species have got extinct from this earth.

9-10 Hierarchy of relatedness

Animal hierarchy is the arrangement of animal into different groups and subgroups on the basis of their relatedness/similarities. The branch of science that deals with the naming and classification of organisms into different groups and subgroups on the basis of relatedness is called **taxonomy**. Each group of classification is called a taxon. Overall sequence of hierarchy of classification of animals is as follow:

- Kingdom
- Phylum
- Class
- Order
- Family
- Genus
- Species

The smallest group of classification system is species.

a. Species

Species is a group of animals which can interbreed and produce fertile offspring. It is the group of animals with maximum similarities. e.g., dogs, cats, horses and elephants.

b. Genus

Closely related species form a genus. Genus name starts with a capital letter and italicized e.g., *Panthera. Parnthera* is genus of five living wild cats *i.e.,* lion, tiger, jaguar, leopard and snow leopard

c. Family

Closely related genera form a family. e.g., Felidae and Hominidae.

d. Order

Closely related families form an order. e.g., Carnivora and Primates

e. Class

Similar orders are grouped together form a class. e.g., Amphibia, Reptilia, Aves and mammalia.

f. Phylum

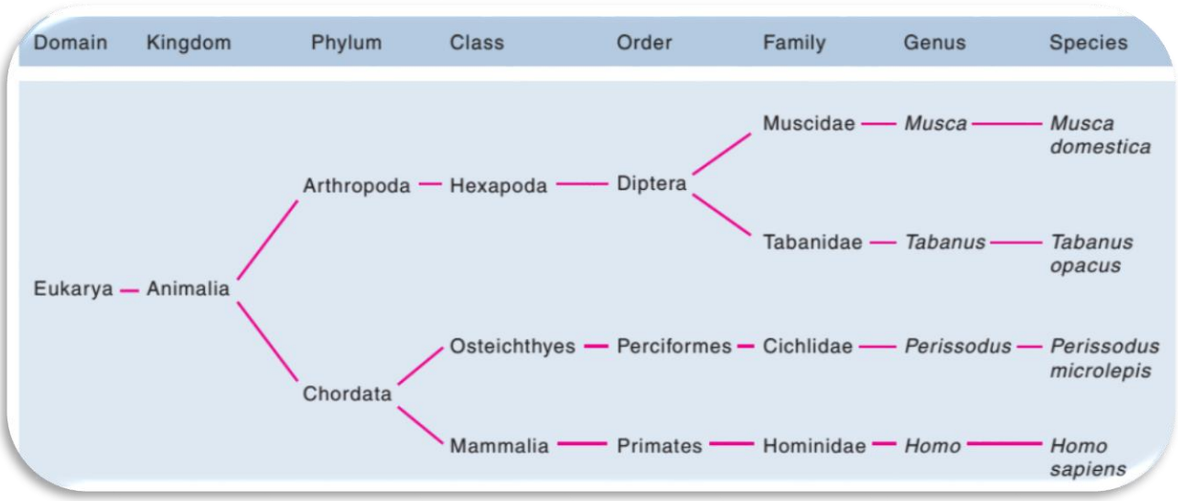
It is the principal category of classification. Classes with some common features are grouped to form a phylum. e.g., echinodermata and chordata.

g. Kingdom

Kingdom is the highest rank of classification. All the living organisms are distributed into five kingdoms *i.e.,* monera, protista, fungi, plantae and animalia. All the animals are included in kingdom aminalia

h. Domain

In 1977 Carl Woese divided the cellular organisms in three domains. i.e., archaea, bacteria and eukarya.



11-12 World Resources

Resource is any kind or stock or material that can be used effectively. **Natural Resources** are the resources which exist in nature without the involvement of mankind. They include all the valuable characteristics of nature. Magnetic gravitational force, sunlight, atmosphere, water, plants, animals etc. are important natural resources.

Natural resource can be divided into two main types: Renewable resources and Non-renewable resources.

Renewable resources are those resources which are constantly available. After using they are reasonably replaced or recovered e.g., water, sunlight, air and biomass. **Non-Renewable Resources** are those which once exhausted are no more available. They are not replaced or recovered e.g., oil, gas and minerals

Threats to Resource

Human overpopulation is a major threat to natural resources. Although modern technology is helping to produce more and more food but extensive use of minerals and oil reserves will exhaust these resources. Although biodiversity is a renewable resource but different animals and plants living on this earth are at the risk of extinction because of overexploitation and destruction of their natural habitat.

Different strategies can be used to protect the resources. Conservation is one of the important methods to save natural resources. Resources may be conserved by increasing efficiencies of resource utilization, using alternative resources and recycling of materials. A control on population will have direct impact on resource conservation.

13 Population of the World

In 70,000BC about 1000 to 10,000 people were living on earth while the current population of the World is more than 7.7 billion. Most of the population is living in less developed countries. The number of people in less developed countries will increase in future. This will further increase the disparity between rich and poor nations.

Growth rate is the number of person increasing annually. The current population growth rate is 1.07% (means 82 million more people every year).The population growth rate was maximum in 1960's. Since

then it is decreasing. An overview of growth rate and population of world is evident from the following table.

Table: Human population growth rate of the world. The growth rate shows that world population will be stabilized in 2100 while at that time the total population will be 10.9 billion.

Time span	Population growth rate
1960's	2%
2018-2019	1.07%
2023	<1%
2052	<0.5%
2076	<0.25%
2100	0.09%

14 The Scientific Method: Observations

Scientific method is a series of steps followed by scientists to answer a specific natural problem. It is based on facts and measurable evidences. The main steps of scientific method are given in the flowchart.

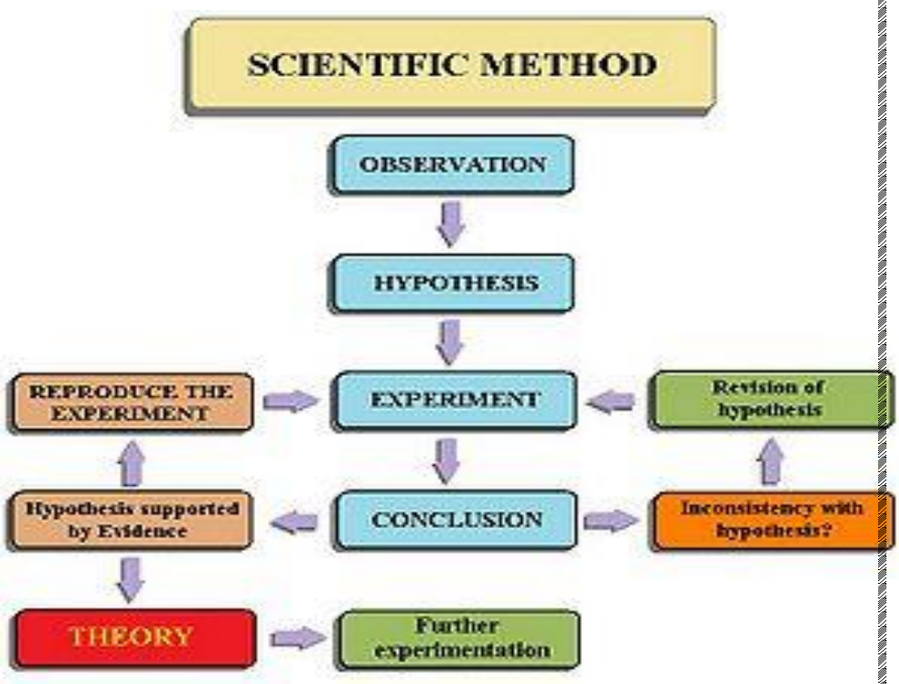


Fig: Flowchart of scientific method. Investigation of scientific problems usually involves different steps in a sequence.

a. Observations

Observation is the collection of information about a particular problem. Observations are the authority in science. Man uses his five senses to collect information and take the help of scientific instrument when required. For example, thermometer is used for accurate measurement of temperature while microscope is used to observe very small objects. The information obtained from observations is called **data**. Data is of two types: quantitative data and qualitative data. Quantitative data has numerical value like height, weight and temperature while qualitative data do not have numerical value like odor, shape and colour.

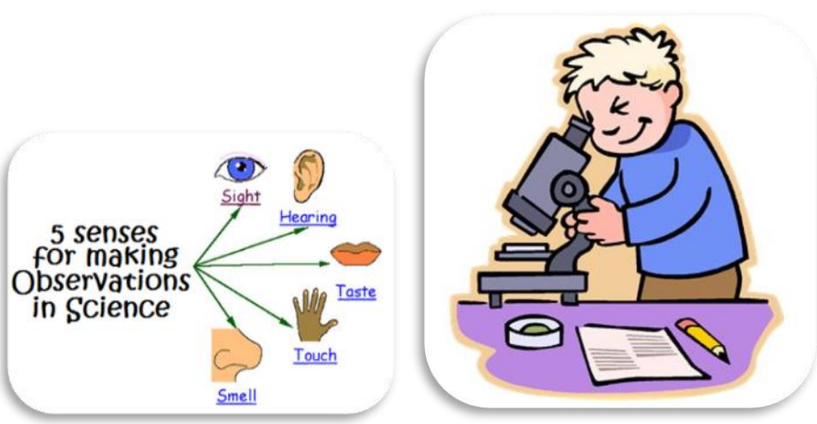


Fig: Observation is the collecting information using senses and specific instruments.

15 The Scientific Method: hypothesis

b. Hypothesis

On the basis data collected through observation and their previous knowledge scientists give a reasonable explanation to the natural question that is called a hypothesis. A hypothesis may base on deductive reasoning or inductive reasoning. A **deductive reasoning** hypothesis is that conclude something from general information while the **Inductive reasoning** hypothesis given some general conclusion from very specific information.

c. Experiment

A hypothesis is tested by experiments. An experiment has three set of variables. An **independent variable** is parameter being tested in the experiment. A **dependent variable** is the parameter used to determine the outcome of an experiment while a **control variable** is that which can influence the results so it is kept constant. For example, an experiment was set to measure the effect of day light length on reproductive readiness of mice. Here, daylight length is independent variable, frequency of mating is dependent variable while temperature is a control variable. All experiment contain at least two groups *i.e.,* experimental group and control group. A control group is kept at same conditions with independent variable omitted. Control group is used for comparison.

Any difference between control and experimental group is the result of experiment. Regardless of the outcomes of the experiment further experiments are needed and the results must be shared with other scientists.

16 The Scientific Method: Theory

d. Theory

When a hypothesis confirmed by repeated experiments and other related hypothesis support then it is converted to a theory. “A theory is a tested and widely accepted model that helps to explain and predict natural facts and which is based on observations, experiments and reason”. It is generalized term very different from commonly known things. The major criteria of theory are: Observable, repeatable and predictive/ Explanatory. A theory is always provisional and may be modified in future. Example: Theory of evolution

17 The Scientific Method: Scientific Law

e. Scientific Law

A scientific law is a statement based on repeated experimental observations that describes some aspect of the nature. It is applied under certain conditions. Scientific law is represented by certain statements and formulas. Experimental results can be predicted by formula. A theory becomes law when the experiments agree with the theory and no contradictory opinion is arises. The law has universal acceptance and cannot be challenged unless solid experimental evidences given. Law of segregation, Law of independent assortment and Hardy-Weinberg law are the examples of scientific law.

18 Experimental Vs Evolutionary Sciences

When we study about living organisms many question arise in our mind. e.g., How cells divide? Why specific number of chromosomes in each species? How genetic information encode proteins? Why some animals live together while others solitary?

All these questions may be divided into two categories: Proximate causes and Ultimate causes. **Proximate cause** is an event immediately responsible for causing some observed result. Normally related to functioning of biological system for a particular time. E.g., metabolic and physiological functions. **Ultimate causes** are those reasons which have produced system and its characteristic through evolutionary time. It also called the real reason. E.g., Evolutionary forces that caused birds to acquire migratory behavior.

Biological science that addresses the questions of proximate causes is called **experimental science**. It proceeds through experiments. E.g., process of maintaining body temperature, process of cell division.

Biological science that addresses the questions of ultimate causes is called **evolutionary science**. It proceeds through comparative methods E.g., comparison of hemoglobin structure of human and fish.

19-20 Origin of life: Synthesis of organic molecules and Their Polymers

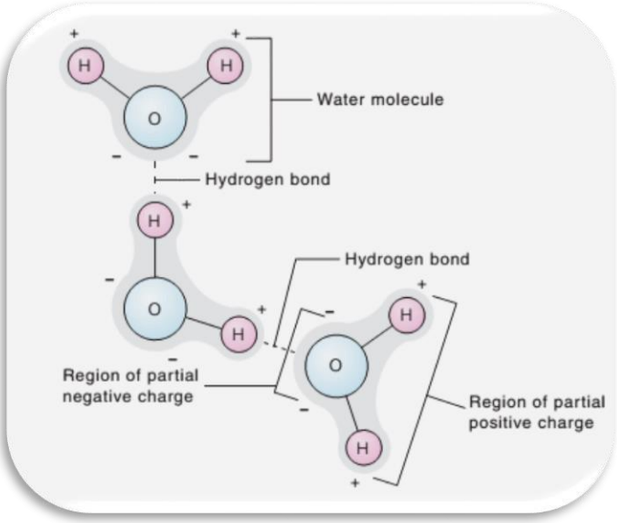
Life evolved about 4.0 billion year ago. According to theory of chemical evolution simple molecules were first converted into monomers of carbohydrates, proteins, lipids and nucleic acids. These monomers later joined to form polymers. All these polymers were suspended in oceans giving up a soup like consistency. Polymers later arranged to form a primitive type of cell called Protocell. This was a simple self replicating system. It was a simple bacteria like primary heterotroph organism. As food availability became limited, natural selection acted upon these simple cells and self replicating systems were more successful. Cells began to evolve and genetic code was established and protein synthesis mechanism was developed. Fossil record shows that life was existing about 3.8 billion years ago. Eukaryotes (membrane bounded nucleus and organelles) evolved about 1.5 billion years ago.

The debate arise that what was the controlling center of these protocells? There are two hypotheses. The first called RNA world hypothesis says that initially RNA was having main role in the cell. RNA molecules were working as enzyme as well as genetic material. The other hypothesis says that it was protein which was having dominant role in controlling cell.

21 CHEMISTRY OF LIFE

Contents

1. Elements
2. Compounds
3. Types of bonding
4. Acids, bases and buffers

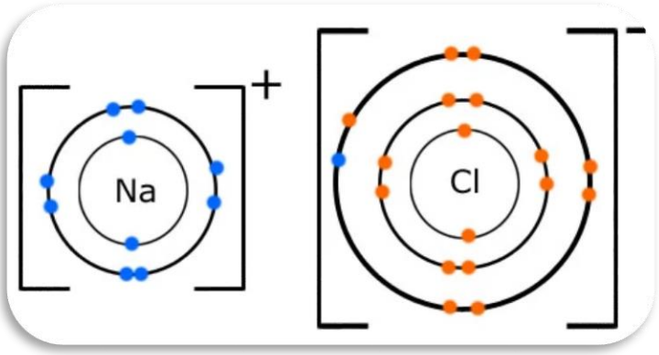


1. Elements

- A species of atom having the same number of protons in their atomic nuclei.
- About 25 elements make living organisms

2. Compounds

- Formed by atoms of two or more elements.
- Organic molecules
- Inorganic molecules
- E.g., Water, glucose, amino acids, fatty acids, nucleotides



3. Types of bonding

- Atoms and molecules are held together by chemical boding.
- Mainly three types of bonds:
  - Covalent bond
  - Hydrogen bond
  - Ionic bond

4. Acids, bases and buffers

- Acids release H ions
  - Amino acids, fatty acids, hydrochloric acid
- Bases release OH ions
  - Nitrogen bases
- Buffer stabilize pH

4. Acids, bases and buffers

- Buffers stabilize the internal environment of cell. (pH 7.35- 7.45)
- Too much fluctuation in pH can destroy cell.
- E.g., Carbonic acid- bicarbonate system

22 THE MOLECULES OF LIFE: BIOMOLECULES

Contents

1. Biomolecules
2. Characteristics of Biomolecules
3. Chemical Bonding in Biomolecules
4. Functions of Biomolecules
5. Types of Biomolecules

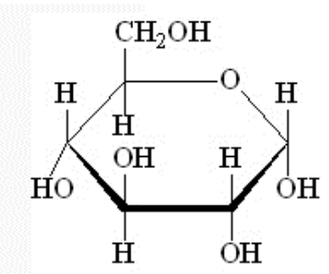
1. **Biomolecules**

"Biomolecules are the molecules produced by cell or living organisms"

- Large biomolecules include proteins, glycogen and starch
- Small biomolecules include monosaccharides, amino acids and intermediates of different metabolic pathways.

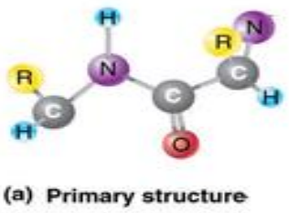
2. Characteristics of Biomolecules

- They make all forms of life
- Chemically they are consist of carbon, hydrogen, oxygen, nitrogen, sulfur phosphorus
- Biomolecules have specific structure

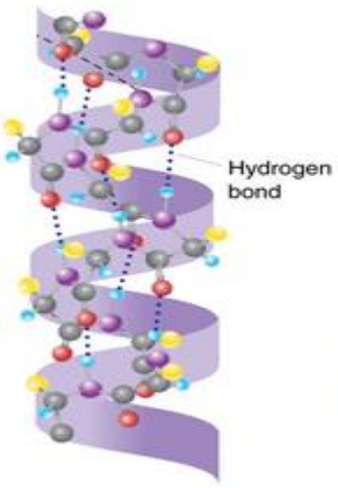


3. Chemical Bonding in Biomolecules

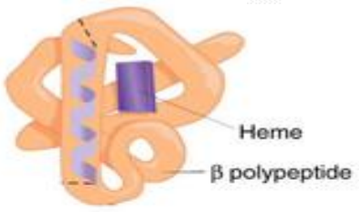
- Atoms in biomolecules are held together mainly by covalent interaction.
- Others include hydrogen bonding, ionic bonding, van der Waal forces



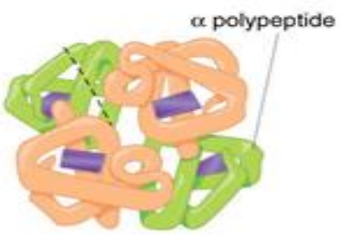
(a) Primary structure



(b) Secondary structure



c) Tertiary structure



(d) Quaternary structure–

4. Functions of Biomolecules

- Basic building blocks of living organisms
- Source of energy
- Performing biochemical reactions
- Storing genetic information
- Maintaining homeostasis

5. Types of Biomolecules

- Mainly four classes
    - Carbohydrates
    - Proteins
    - Lipids
    - Nucleic acids

23 CARBOHYDRATES: Chemical Composition and Occurrence

Contents



1. Defining Carbohydrates
2. Chemical Composition
3. Occurrence
4. Classification of Carbohydrates

1. **Defining Carbohydrates**

"Carbohydrates are polyhydroxy aldehydes, ketones or complex molecules giving aldehyde or ketone on hydrolysis"

- Most abundant biomolecules on earth
- Part of diet in most parts of the world

2. Chemical Composition

- Principally they are made up of C, H and O.
- N, S, P may also present in derivatives.
- General formula is: $(CH_2O)_n$
- Also called hydrated carbons
    - hydrogen and oxygen (2:1)

3. Occurrence

- Bakery items: bread, cakes, pastries etc.
- Dairy products: milk, yogurt, ice cream
- Vegetables and fruits: all vegetables, fruits, cereals (wheat, rice, corn), legumes
- Candies, soft drinks, desserts
- Meat also contain some amount



4. Classification

- Three main classes
    - Monosaccharides
    - Disaccharides/Oligosaccharides
    - Polysaccharides

24 Different Classes of Carbohydrates: Monosaccharides

Contents

1. Monosaccharides
2. Physicochemical Properties
3. Types on the Basis of Size

4. Types on the Basis of Functional Group
5. Chain Ring structures
6. Importance

1. **Monosaccharides**

“The simplest carbohydrates which can not be further hydrolyzed into simpler carbohydrates”

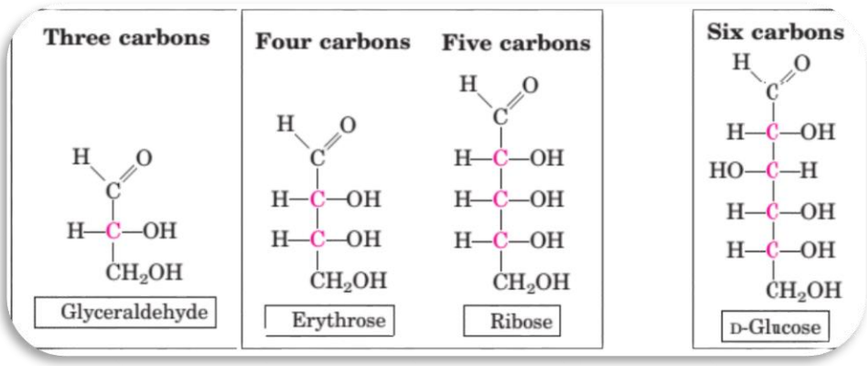
- Glucose is the most common type of monosaccharides

2. Physicochemical Properties

- Sweet in taste
- Readily soluble in water
- Crystalline at room temperature
- They are reducing sugar

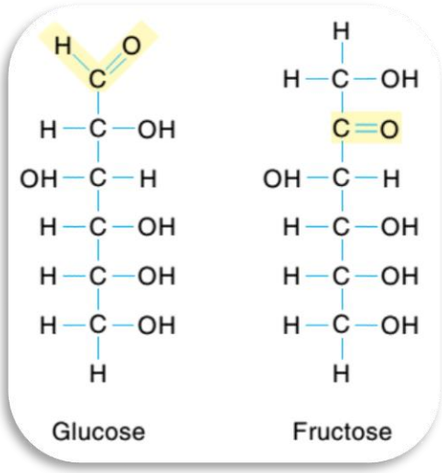
3. Types on the Basis of Size

- Normally 3-7 carbons
- Triose (3C)
- Tetrose (4C)
- Pentose (5C)
- Hexose (6C)
- Heptose (7C)



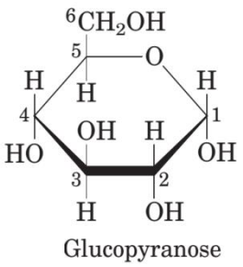
4. Types on Basis of Functional Group

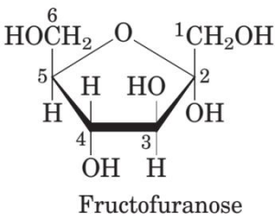
- Aldoses
- Ketoses



5. Chain and Ring Structures

- Carbon atoms form linear chain
- Four or more carbon form a ring structure
- Pyranose
- Furanose


Glucopyranose


Fructofuranose

5. Importance

- As a <u>source of energy</u>
  - Glucose as energy source (C-H bond)
- As a <u>building material</u> for large carbohydrates
  - Cellulose
  - Starch
  - Glycogen

25 Different Classes of Carbohydrates: Disaccharides

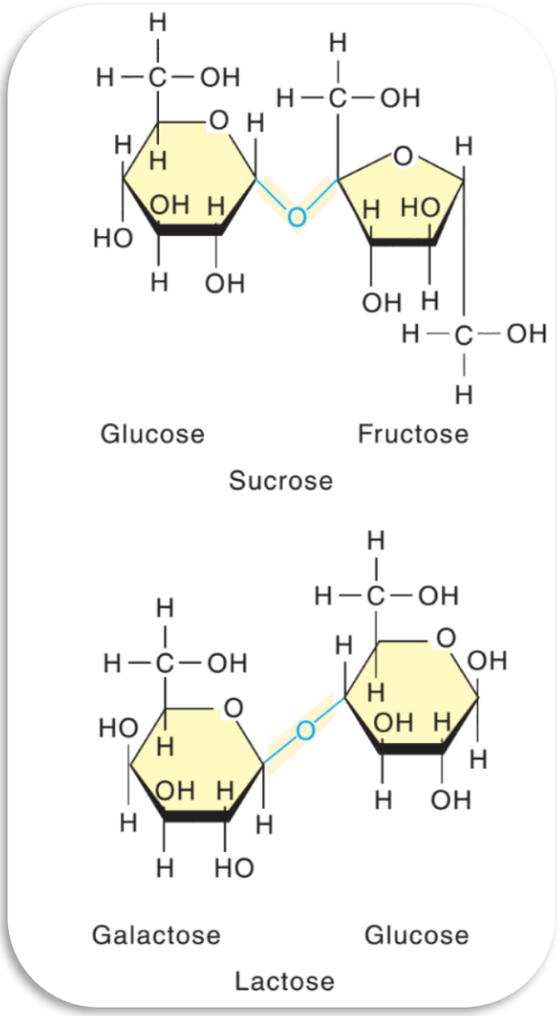
Contents

1. Disaccharides
2. Some Common Disaccharides
3. Isomers
4. Glycosidic Bond

1. **Disaccharides**

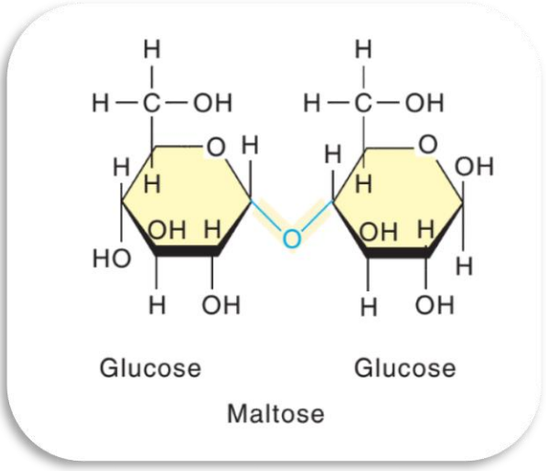
“Two monosaccharides combine together to form a disaccharide”

- Soluble in water
- Less sweet in taste
- Crystalline solids
- Some reducing, some non-reducing
- $C_x(H_2O)_y$



2. Some Common Disaccharides

- Sucrose
- Lactose
- Maltose

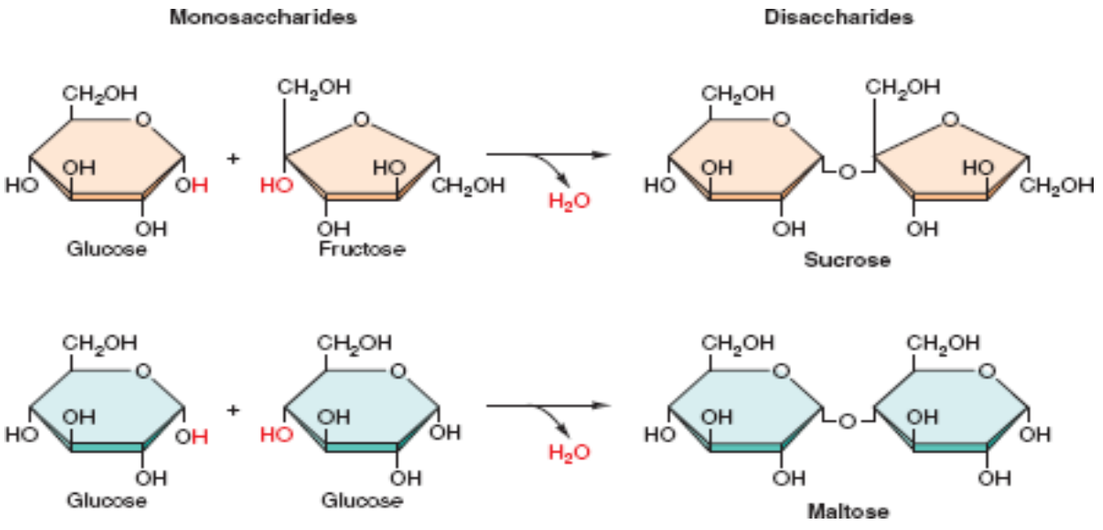


3. Isomerism

- Molecules with same molecular formula but different structural formula
- All disaccharides are isomers of each other.
- Molecular formula $C_{12}H_{22}O_{11}$

4. Glycosidic Bond

- The bond joining monosaccharide subunits.



26 Different Classes of Carbohydrates: Polysaccharides

Contents

1. Polysaccharides
2. Physicochemical Properties
3. Some Common Polysaccharides
4. Role in Living Organisms

1. Polysaccharides

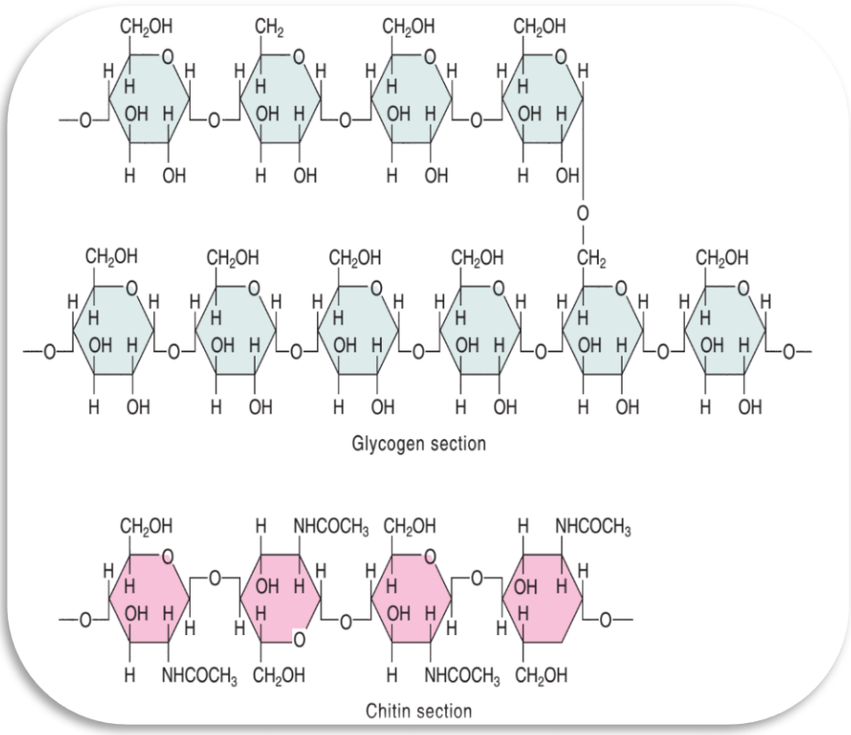
"When many monosaccharides join together through glycosidic bod, they form a polysaccharide"

2. Physicochemical Properties

- Not sweet in taste
- Very little solubility in water
- Non reducing
- General formula $C_x(H_2O)_y$ or $(C_6H_{10}O_5)_n$

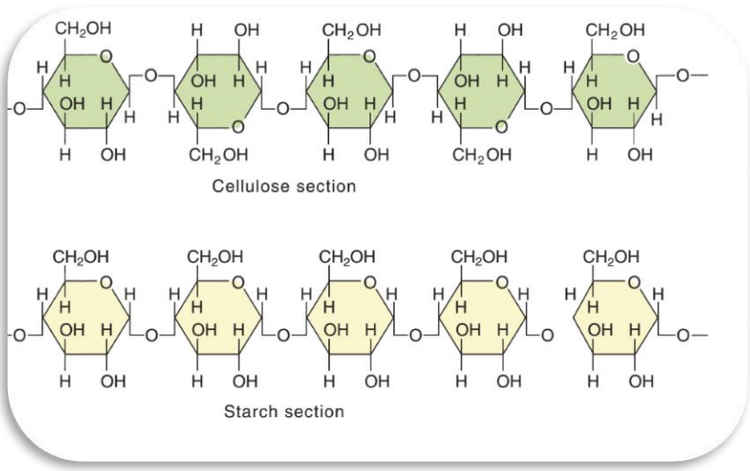
3. Common Polysaccharides

- Glycogen
- Starch
- Cellulose
- Chitin



4. Role in Living Organisms

- Fuel storage:
  - Glycogen
  - Starch
- Structural role:
  - Chitin
  - Cellulose



27 Carbohydrates as energy source and structural support

Contents

1. As Energy Source:
   a) Monosaccharides
   b) Disaccharides
   c) Polysaccharides
2. As Structural Support
   a) Chitin
   b) Cellulose
   c) Glycocalyx
   d) Extracellular material

1. **As Energy Source**

- Main source of daily energy needs
- Provide 60-65% of energy
- 225-325 gram per day

1. **As Energy Source**

a) **Monosaccharides**

- Glucose is primarily synthesized by plants during photosynthesis.
- It is most important source of energy for all organisms.
- Fructose and Galactose also provide energy

1. **As Energy Source**

a) **Disaccharides**

- Important sugar which provide energy
- Sucrose from sugar cane and sugar beet.
- Lactose present in milk sugar
- Maltose is present in sweet potatoes, candies
- Broken down by intestinal enzymes into monosaccharides

1. **As Energy Source**

a) **Polysaccharides**

- Starch
  - Digested into glucose.
- Glycogen
  - Quickly breaks to provide glucose in cells and blood
- Cellulose
  - Broken down into glucose

2. As structural support

- **Chitin**
- Exoskeleton of different arthropods. e.g. insects, crustaceans
- Provide support
- Protection from predators

2. As structural support

- **Cellulose**
- Providing strength to cell wall of plants

2. As structural support

- **Glycocalyx**
- Making a whorl outer to cell membrane.
- Make recognition sites
- Work as protective factor

2. As structural support

- **Extracellular material**
- Hold the cells together
- Present in:

- Fluid of joints
- Vitreous humor
- Tendons
- Ligaments
- Skin

28 Lipids: Chemical composition and occurance

Contents

1. Defining Lipids
2. Chemical Composition
3. Occurrence
4. Types of Lipids
5. Classification of Lipids

1. **Defining Lipids**

- A diverse group of biomolecules which are water insoluble
- Non-polar
- Readily soluble in organic solvent

2. Chemical Composition

- Like carbohydrates lipids are mainly made up of C, H and O.
- N, S, P also present.
- Much smaller amount of oxygen than carbohydrates.
- They have highest number of C-H bonds

3. Occurrence

- Dairy: Milk, cheese, butter, cream, yogurt, margarine
- Meat, fish eggs,
- Dry fruits: peanut, almond, walnut,
- Cooking oils: sunflower, soybean, corn, olive oil
- Fried foods



4. Importance

- Richest source of energy
- All cellular membranes
- Hormones
- Protection
- Insulation
- Cell signaling

4. Types of Lipids

- Important types include
    - Triglycerides
    - Phospholipids

- Steroids

29 Fatty acids

Contents

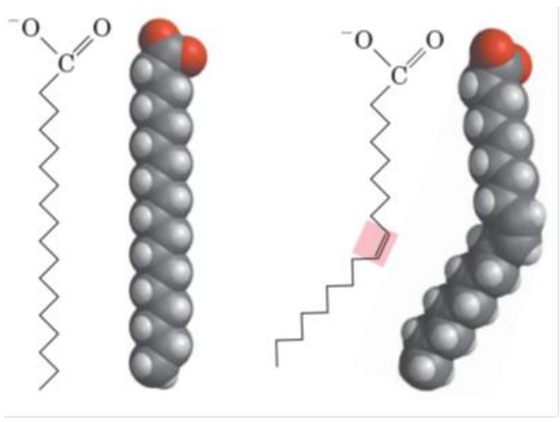
1. Defining Fatty Acids
2. Chemical Composition
3. Melting Point and Solubility
4. Carbon Chain in Fatty Acids
5. Saturated and Unsaturated Fatty Acids

1. **Defining Fatty Acids**

- They are long chain hydrocrabons bonded to carboxylic (-COOH) group.
- They have maximum number of C-H bonds.
- Oxygen is present only in the form of COOH group

2. Carbon Chain in Fatty Acids

- Mostly even number fatty acids
- Size varies from 4-36 carbons
- Most common are 14-20 carbons



3. Saturated and Unsaturated Fatty Acids

- Having single covalent bond between all carbons are saturated.
- Those having one or few double bonds are unsaturated fatty acids.
- Saturated solidify at room temperature
- Unsaturated liquid at room temperature

4. Melting point and Solubility

- Melting point increase with increase in number of carbon atoms
- Solubility decreases with increase in carbon chain size
- Melting point also decrease with addition of double bonds

30 Fatty acids examples

Contents

1. Palmitic acid
2. Stearic acid
3. Arachidic acid
4. Palmitoliec acid
5. Oleic acid
6. Linoleic acid
7. Linolenic acid

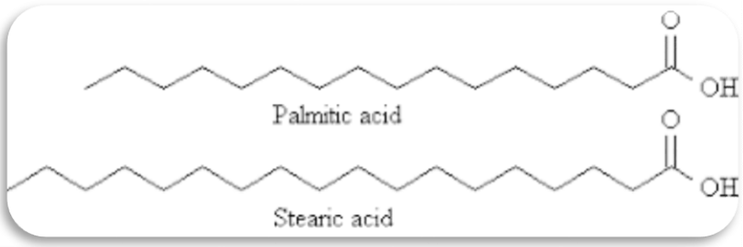
1. **Palmitic acid**

- Most common fatty acid found in animals, plants and microorganisms
- 16 carbon fatty acid with molecular formula $C_{16}H_{32}O_2$

- Saturated
- Melting point 62.9

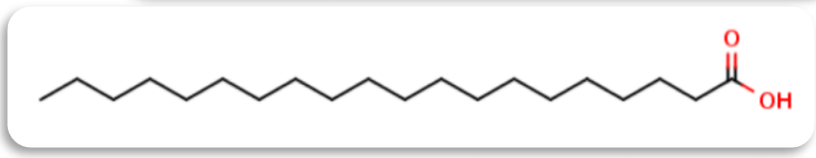
2. Stearic acid

- 18 carbon
- Saturated
- Found in animal fat
- Waxy solid at room temperature
- Melting point 69.6°C



3. Arachidic Acid

- 20 carbon
- Saturated
- Present in peanut and corn oil.
- Melting point 76.5

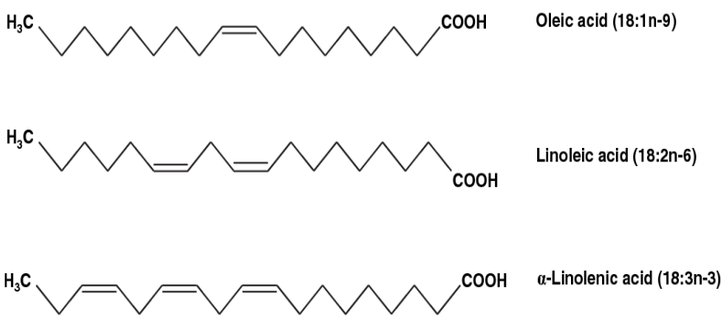


4. Palmitoleic acid

- 16 carbon
- Unsaturated, double bond between C9 and C10
- Olive oil
- Melting point 13.4°C

5. Oleic acid

- 18 carbon
- Unsaturated
- Double bond between C9 and C10
- Human adipose tissues
- Melting point 13.4°C



6. Linoleic acid

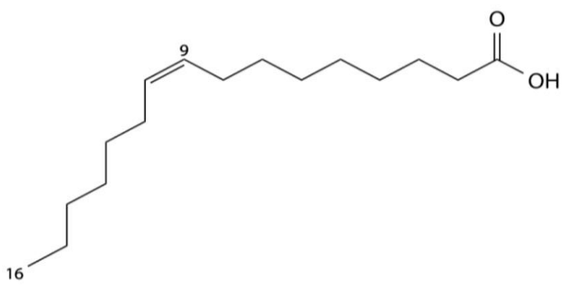
18 carbon

Polyunsaturated fatty acid

Omega 6 fatty acid

Essential

Melting point 1-5°C



7. Linolenic acid

18 carbon

Polyunsaturated fatty acid

Also called omega-3

Essential fatty acid

Melting point -11°C

31 Lipids Types: Fats and Oils

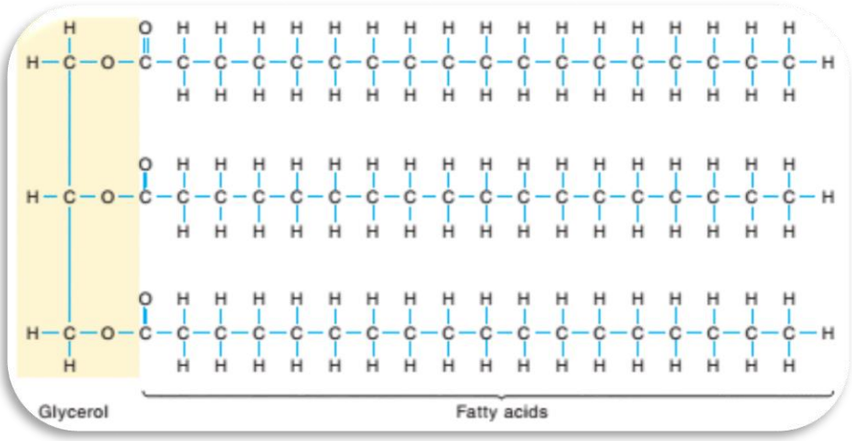
Contents

1. Triglycerides
2. Chemical composition

3. Functions
4. Fats
5. Oils

1. **Triglycerides**

- Simplest group of lipids
- Also called neutral fats and triacylglycerol



2. Chemical Composition

- Glycerol backbone
- One fatty acids attached with each of three carbon of glycerol.
- Ester linkage
- Fatty acids mostly 16-20 carbon
- Simple and mixed triacylglycerol

3. Functions

- Energy storage
  - In adipocytes or fat cells
  - Provide more energy than carbohydrates
  - Require less water for storage
- Insulation against heat loss

4. Fats

- Triacylglycerol solid at room temperature.
- They contains all three saturated fatty acids.
- Animal triacylglycerols are fats.

5. Oils

- Triacylglycerol liquid at room temperature.
- They contains unsaturated fatty acids.
- Plant triacylglycerols are oils.

32 Lipids Types: Phospholipids

Contents

1. Phospholipids
2. Molecular structure
3. Functions
4. Types

—	Phosphatidic acid
Ethanolamine	Phosphatidylethanolamine
Choline	Phosphatidylcholine
Serine	Phosphatidylserine
Inositol	Phosphatidylinositol

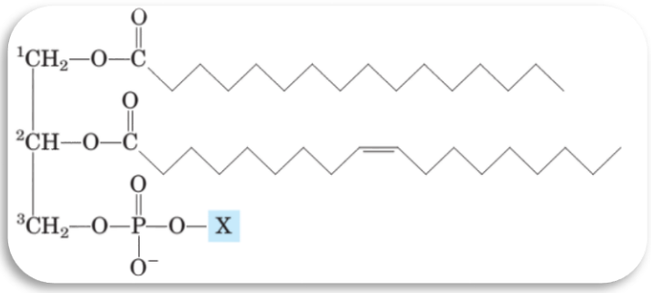
1. **Phospholipids**

- Phospholipids are the molecules in which glycerol molecules combines with two fatty acids and a phosphate group along with nitrogen base.

2. Molecular Structure

- Two Fatty acids are attached to glycerol through ester/acyl linkage.
- On 3rd carbon of glycerol there is a phosphate group

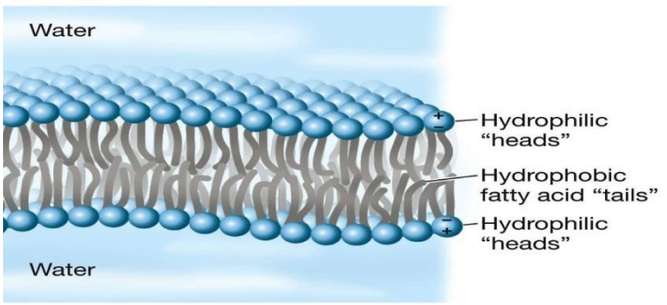
- Nitrogen base is attached to phosphate
- Polar head
- Nonpolar tail



3. Functions

- Form membranes of cells and organelles.
    - Due to hydrophilic head and hydrophobic tail form lipid bilayer.

Phospholipid bilayer



4. Types

- Phospholipids are of four types.
- Phosphatidic acid

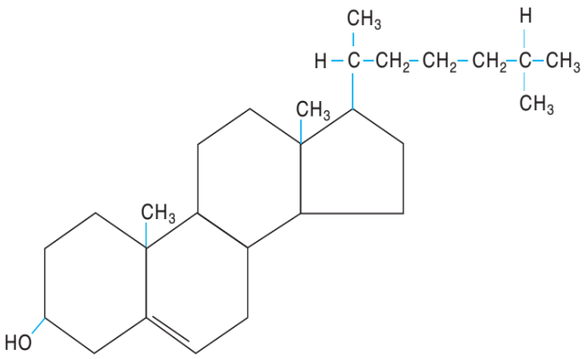
33 Lipids Types: Steroids

Contents

1. Steroids
2. Molecular Structure
3. Functions

1. **Steroids**

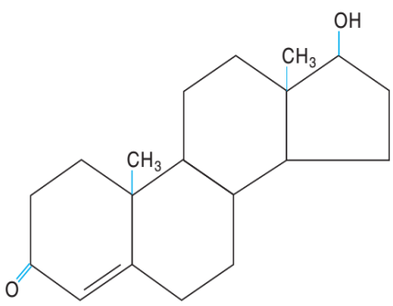
- Naturally occurring lipids which contain four fused carbon rings.
- e.g. vitamin D, cholesterol, aldosterone, estrogen and testosterone



Cholesterol

2. Molecular Structure

- Four fused rings.
- Three are 6-sided and one is 5-sided
- Total 17 carbon steroid nucleus



Testosterone

3. Functions

- Cholesterol in cell membrane
    - Testosterone for secondary sex characteristics in males
    - Estrogen for secondary sex characteristics in females
    - Vitamin D uptakes and regulates calcium level.

34 Proteins: Chemical Composition

Contents

1. Proteins
2. Occurrence
3. Chemical Basis
4. Amino acids

5. Occurrence



1. **Proteins**

- Biomolecules having one or more chains of amino acids.
- e.g. hemoglobin, myoglobin, different types of enzymes

2. Protein Occurrance

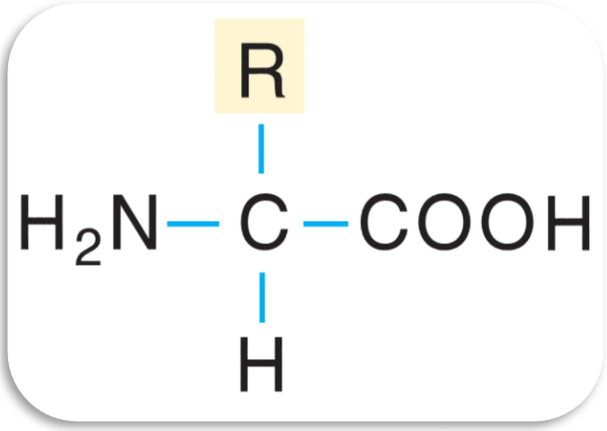
- Dairy: milk, yogurt and cheese
- Meat, poultry, fish, egg
    - Seeds, nuts, beans and logumes

3. Chemical Basis

- Chemically they are made up of carbon, hydrogen, oxygen and nitrogen
- Sulfur is also present in some proteins

4. Amino Acids

- Building blocks of proteins
    - Amino acids contain
        1. A central carbon
        2. A carboxylic group
        3. An amino group
        4. A hydrogen atom
        5. A functional group



5. Amino Acids

- About 20 types of amino acids are present in proteins
- Nature of R group dictate the chemical properties of proteins
    - May be polar, non polar, acidic, basic and aromatic amino acids.

35 Proteins: Importance

Contents

1. Structural part
2. Energy Source
3. Protection Against Diseases
4. Transportation
5. Enzymes
6. Movement
7. Chemical Messengers

1. **Structural role**

- All the animal body structures.
- Bones, muscles, visceral organs, hair, nails

2. Source of Energy

- Proteins are degraded to amino acids.
- Amino portion removed by deamination.

- Then oxidize to provide energy

3. Protection against diseases

- Antibiotics
- Antibodies

4. Transportation

- Hemoglobin carries oxygen from lungs to capillaries
- Myoglobin carries oxygen in muscles
- Movement across cell membrane

5. Enzymes

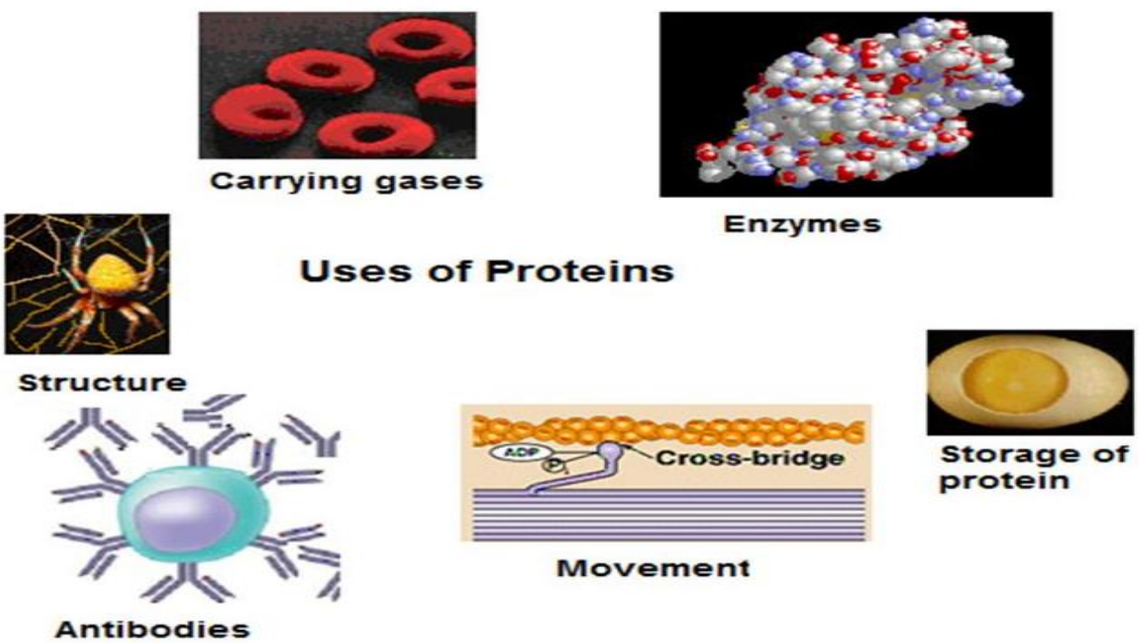
- Most of the enzymes are protein in nature.
- They catalyze biochemical reactions

6. Movement

- All types of movements in living organisms are due to proteins.
- Locomotion
- Heart beat
- Breathing process

7. Chemical messengers

- Hormones
- neurotransmitters
- Intracellular messaging



36 Structural Levels of proteins: Primary Structure

Contents

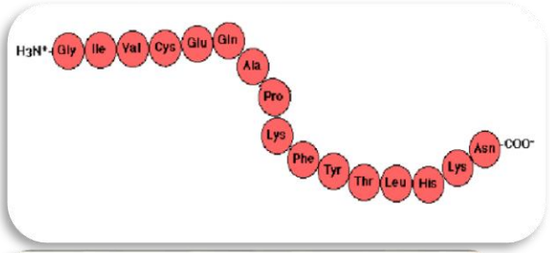
1. Primary Structure
2. Peptide Bond
3. Importance of Primary Structure

1. **Primary Structure**

- Linear chain of amino acids in the form of polypeptide chain
- Amino acids are linked through peptide bond

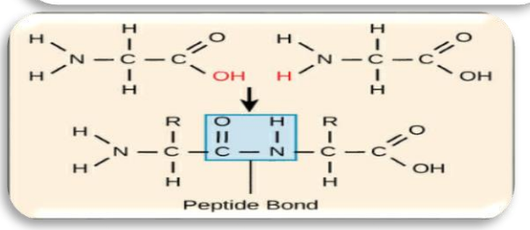
3. Peptide Bond

- Bond present between two amino acids
- COOH group of one amino acids reacts with NH2 group of another amino acid to form peptide bond



2. Importance of Primary Structure

- Size of protein/polypeptide molecules
- Sequence of amino acids in polypeptide chain.



37 Structural Levels of proteins: Secondary Structure

Contents

1. Secondary Structure
2. Alpha helix
3. Beta pleated sheets

1. Secondary Structure

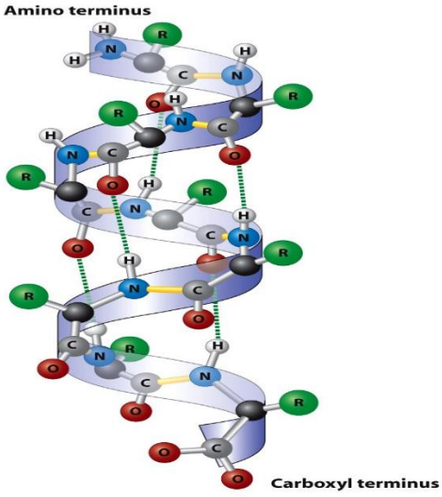
- It is local folding of polypeptide chain
- It is a repeating pattern of hydrogen bonds between amino acids.
- Two major types:
  - Alpha helix
  - Beta sheets

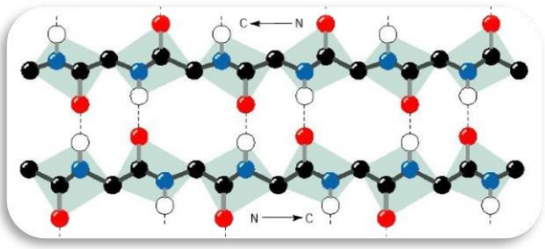
2. Alpha Helix

- α-helix is helical arrangement of polypeptide chain.
- This helical is right handed
- Amino hydrogen of one amino acid makes a bond with carboxylic oxygen of fourth amino acid.

3. β-pleated Sheets

- Polypeptide chain form sheets.
- Sheets of polypeptides are attached together through hydrogen bonds







4. β-turns and random coil

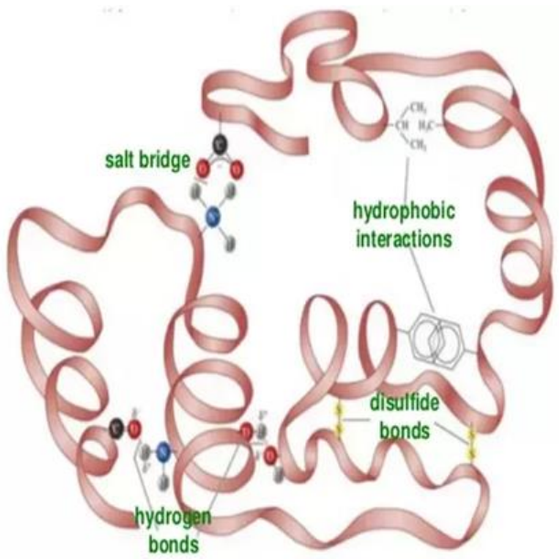
38 Structural Levels of proteins: Tertiary Structure

Contents

1. Tertiary Structure
2. Bonding Involve in Tertiary Structure
3. Examples of Protein Tertiary Structure

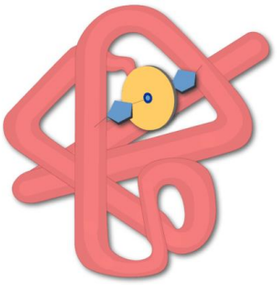
1. **Tertiary Structure**

- Overall folding of polypeptide chain into a 3D form is called tertiary structure.



2. Bonding Involve in Tertiary Structure

- Hydrogen bond
- Ionic bond
- Hydrophobic interaction
- Disulfide bond



3. Example of Proteins with Tertiary Structure

- Myoglobin
- Salivary amylase

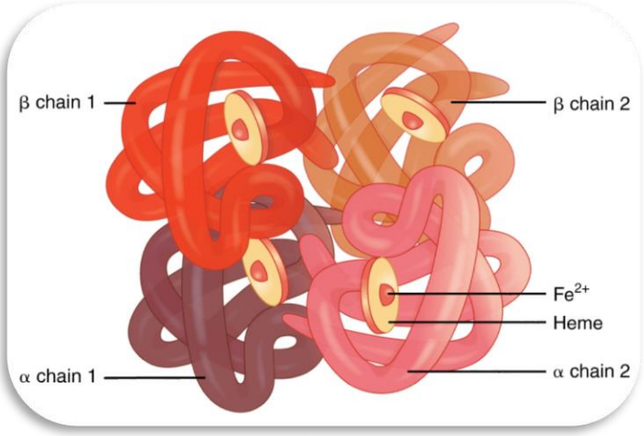
39 Structural Levels of proteins: Quaternary Structure

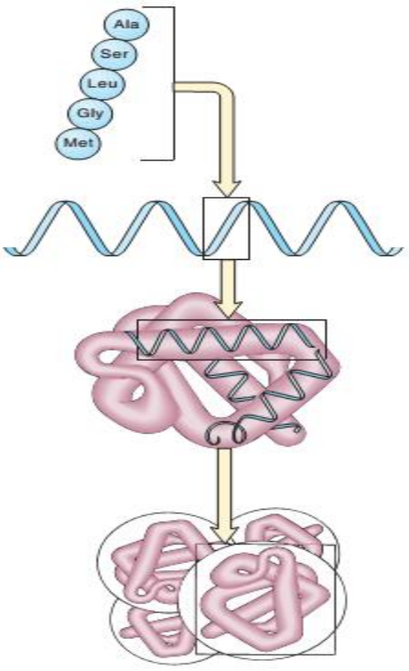
Contents

1. Quaternary Structure
2. Bonding Involve in Quaternary Structure
3. Examples of Protein with Quaternary Structure
4. Fibrous Proteins
5. Globular Protein

1. **Quaternary Structure**

- When two or more polypeptide chains fold together they form a quaternary structure.





2. Bonding Involve in Quaternary Structure

- Hydrogen bond
- Ionic bond
- Van der Waal forces
- Sometime covalent bond

3. Example of Proteins with Quaternary Structure

- Hemoglobin
- pepsin
- Keratin

4. Fibrous Proteins

- Polypeptide chain in long fibers

- Structural support
- Water insoluble
- Keratin in nails, hair, skin
- Collagen in bones, cartilage, tendons
- Silk fibroin in silk

5. Globular Protein

- Globular structure
- Water soluble
- Functional role
- All Enzymes
- Hemoglobin
- Myoglobin

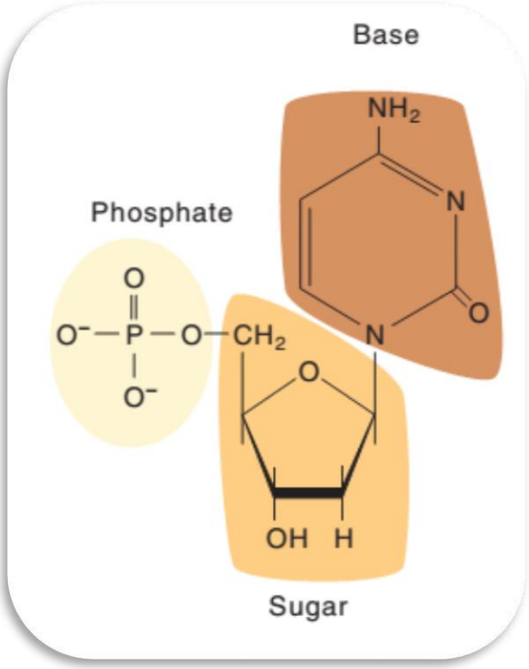
40 Nucleotides: chemical composition

Contents

1. Nucleotides
2. Parts of Nucleotides
3. Types of nucleotides

1. **Nucleotides**

- Important biomolecules essential to life
- Perform different cellular activities
- They are also building blocks of nucleic acids



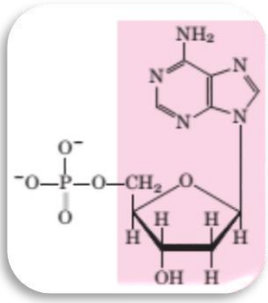
2. Parts of Nucleotides

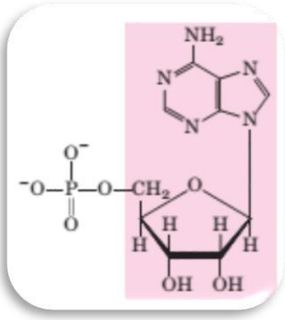
- **Three components**

1) Five carbon/pentose sugar/ (ribose, deoxyribose)
2) Phosphate group
3) Nitrogen containing base

3. Types of nucleotides

- **On the basis of sugar**
- Deoxyribonucleotide
- Ribonucleotides





3. Types of nucleotides

- **On the basis of nitrogen base**
- Adenine
- Guanine
- Cytosine
- Thymine/Uracil

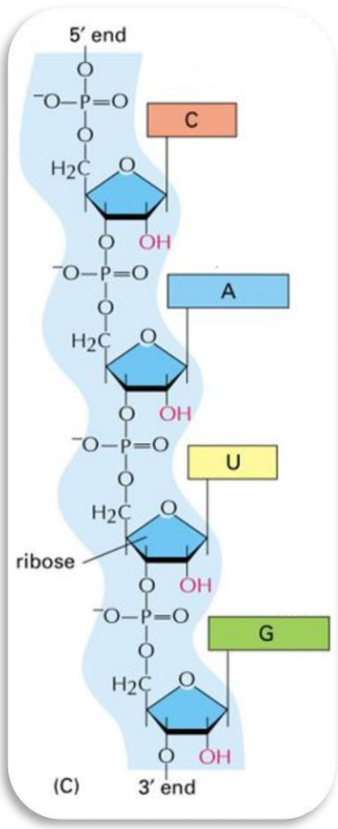
41 Nucleotides: Role in cells

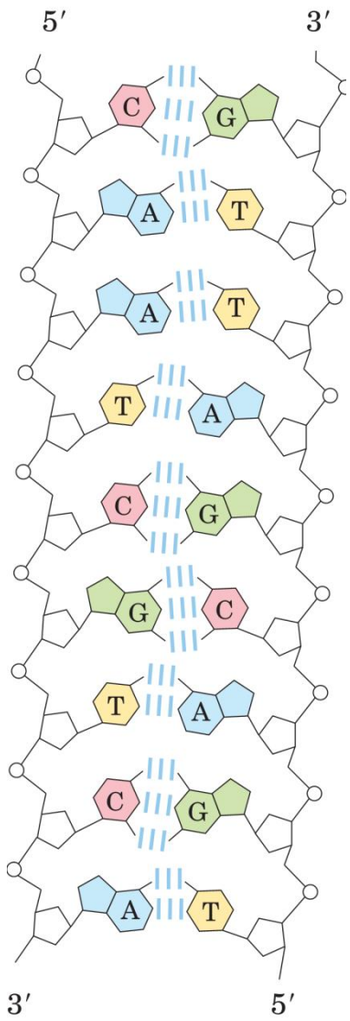
Contents

1. As DNA/RNA Building Blocks
2. As Coenzymes
3. Chemical Messenger
4. Energy Carrier
5. Energy Transfer

1. **DNA/RNA Building Blocks**

- **For DNA**
- dATP, dGTP, dCTP, dTTP
- **For RNA**
- ATP, GTP, CTP, UTP





2. As coenzymes

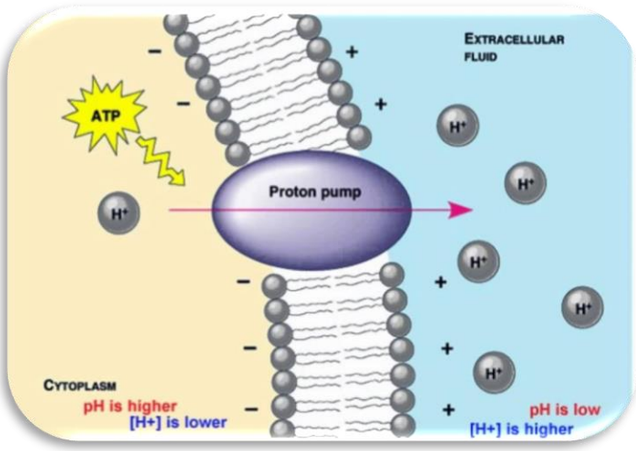
- Coenzymes of different enzymes
- Nucleotide coenzymes: NAD, NADP

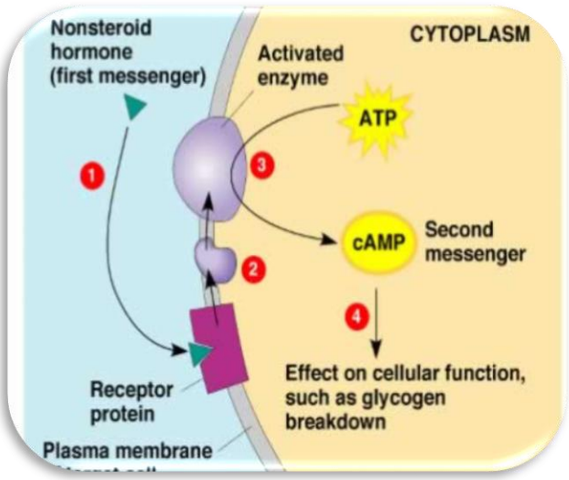
3. As Chemical Messenger

- cAMP work as chemical messenger within the cell

3. As Energy Carrier

- ATP
- GTP





42 Nucleic Acids: DNA

Contents

1. Chemical Composition
2. Structure
3. As Heredity material

1. **Chemical composition**

- **Four types of deoxyribonucleotides**
  - dATP
  - dGTP
  - dCTP

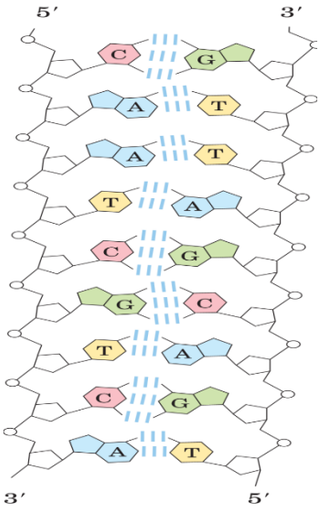
- dTTP

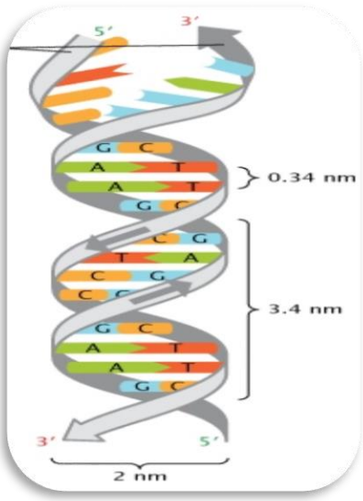
2. Structure

- Double helical structure
- Adenine pairs with thymine
- Guanine pairs with thymine

3. Hereditary material

- Store genetic information
- Transfer to next generation





43 Nucleic Acids: RNA

Contents

1. Chemical Composition
2. Structure
3. Types of RNA
4. Role in cell

1. **Chemical composition**

- **Four types of ribonucleotides**
  - ATP
  - GTP
  - CTP
  - UTP

2. Structure

- Single stranded polynucleotide chain

3. Types of RNA

- Three types of RNA
  - rRNA
  - tRNA
  - mRNA

4. Role in cell

- Part of ribosomes
- mRNA copies information from DNA to synthesize proteins
- Genetic material of some viruses

44 Structural Hierarchy in Multicellular Animals

“A hierarchy is an organizational structure in which items are ranked according to levels of importance”. The living organisms have following structural hierarchy.

i. Atoms

Atom is first level of structural hierarchy. It is the smallest unit of matter.

ii. Molecules

Atoms combine together to form molecules. Molecules exist independently and take part in a chemical reactions. Small size molecules are called micromolecules (H_2O, CO_2) while large size molecules are called macromolecules (DNA and proteins).

iii. Organelles

Organelle is the small cellular structure performing specific function in the cell. Some are membrane bounded (Vacuole, Golgi bodies) while some are without membrane (ribosome).

iv. Cells

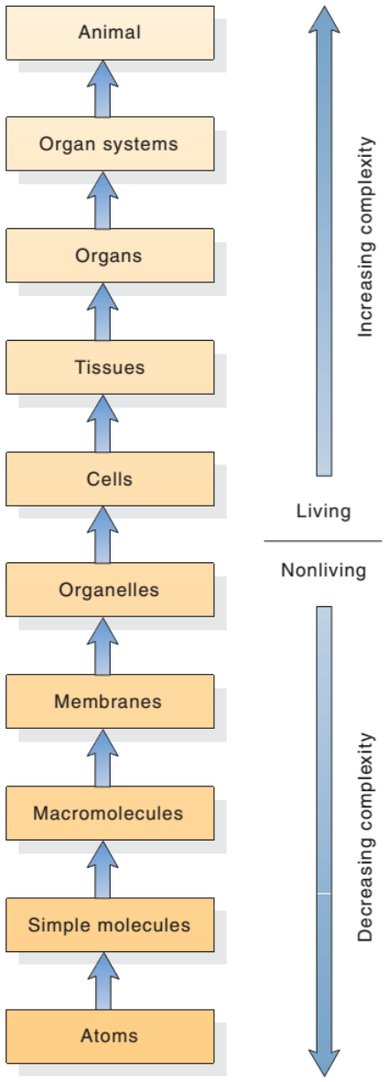
Cell is the structural and functional unit of life. It means all living organisms are made up of cells and all functions of living organisms are also performed by the cell. Cells are two basic types: Prokaryotic cell (without nucleus) and Eukaryotic cell (have true nucleus).

v. Tissues

Tissue is a group of cells performing similar functions. Animals have different types of tissues e.g., cardiac tissues, epithelial tissues and endocrine tissues.

vi. Organs

Organ a part of body which performs specific function. Stomach, intestine and heart are the examples of organs.



vii. System

Different organs performing related functions form a system. It system is designated to perform specific function. Major systems in our body include respiratory system, digestive system and circulatory system.

viii. Organism

Systems make an organism or individual. An organism has individual existence. It performs all the activities of life.

45 What is Cell?

Cell is the smallest independent unit of life. Cell is the structural and functional unit of life. Cells can grow, divide and synthesize biomolecules. All cellular activities like secretion, absorption and transportation are performed by the cell. Unicellular organisms are made up of only one cell while multicellular organisms have few to trillions of cells.

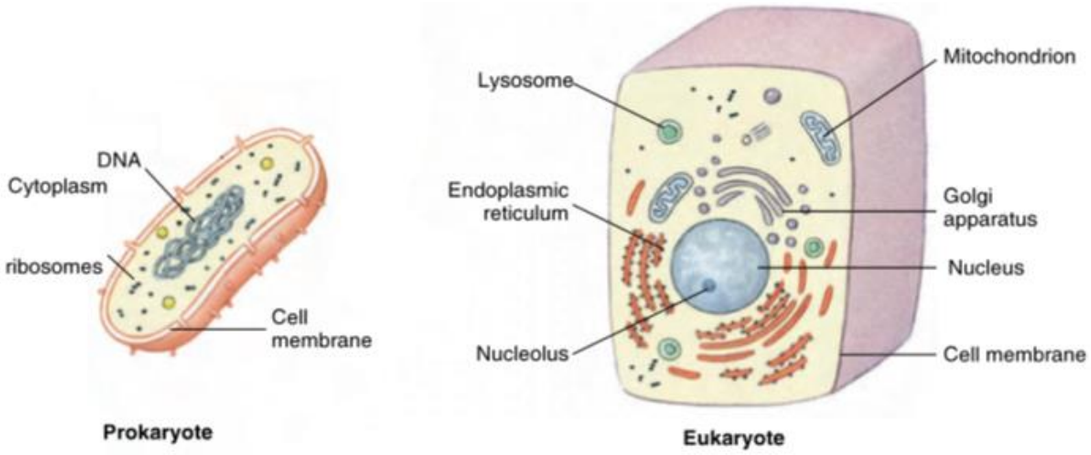
Cells are of two basic types: Prokaryotic cells and Eukaryotic cell.

46 Prokaryotic and eukaryotic cells

Prokaryote is a combination of two Greek word “pro” means before and “Karyon” means nucleus. Prokaryotic cells are those which do not have a membrane bounded nucleus. They also have no other membrane bounded organelles as well. Organisms with prokaryotic cells are called prokaryotes. E.g., bacteria and cyanobacteria.

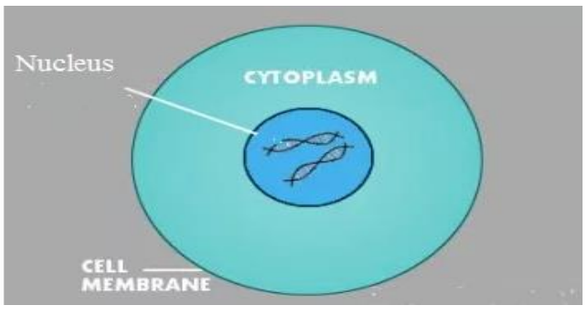
Eukaryotic Cell

Eukaryote is a combination of two Greek word “eu” means true and “Karyon” means nucleus. Eukaryotic cells are those which have a membrane bounded nucleus. They also have other membrane bounded organelles. Organisms with eukaryotic cells are called eukaryotes. E.g., animal and plant cells.



47 General Features of Eukaryotic Cells

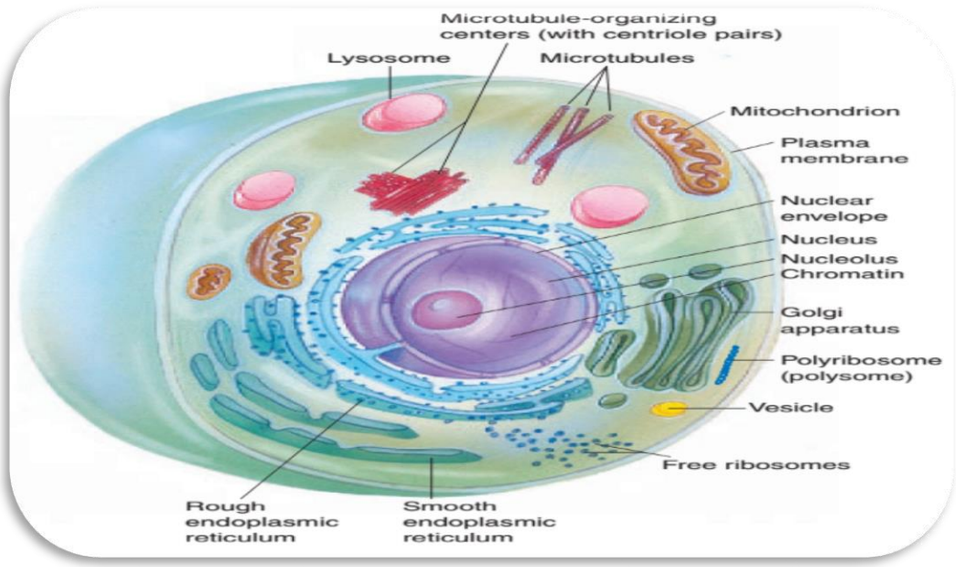
General Features of Eukaryotic Cell



There are three main components of eukaryotic cells:

1. Plasma membrane
2. Cytoplasm
3. Nucleus

Membrane bounded organelles, cytoskeleton and linear chromosomes are important features of eukaryotic cells. Presence of different organelles is also called compartmentalization. Each organelle performing specific job. For example Lysosome performs digestion, mitochondria produces energy and Golgi apparatus is involved in cell secretion.

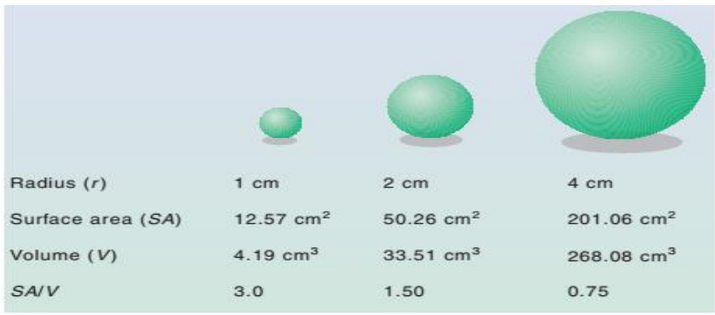


48 Why most of the cells are small?

Why Most Cell are Small?

Average size of Ostrich egg is 15 x 13 cm while neurons of human and elephant are about 1m and 4 m, respectively. However, most of the cells are very small and range 5-150µm. Why cells are so small? This is because an increase in cell size increases volume enormously. The increase in volume causes two major problems. Firstly, by increasing size the ratio between volume of nucleus to volume of cytoplasm decreases. As a result nucleus will lose its control on very large cytoplasm. Secondly, cell

volume increases much more than its surface area. This will result in reduced gases exchange, uptake of nutrients and removal of wastes.



Radius (r)	1 cm	2 cm	4 cm
Surface area (SA)	12.57 cm^2	50.26 cm^2	201.06 cm^2
Volume (V)	4.19 cm^3	33.51 cm^3	268.08 cm^3
SA/V	3.0	1.50	0.75

Fig. The increase in volume causes a decrease in surface area to volume ratio (SA/V).

49 Chemical Composition of Cell Membranes

CELL MEMBRANE

Composition

Cell/plasma membrane is the outer most boundary of a cell. It also makes the boundaries of membranous organelles. Chemically cell membrane is made up of phospholipids, cholesterol, proteins and carbohydrates.

Phospholipids form lipid bilayer. They have polar heads and nonpolar tails. Polar heads are oriented outward and to cytoplasm while tails are present between two head layers. **Cholesterol** is present in plasma membrane of eukaryotes only. Cholesterol is embedded interior of membrane. It makes cell membrane stable and less permeable to water. Membrane **proteins** are individual molecules present inner or outer surface (peripheral proteins) or embedded in the membrane (intrinsic proteins). Outer proteins are attachment site for carbohydrates while proteins on cytoplasmic side are attachment site for cytoskeleton. Some membrane proteins transport materials into and out of cell. **Carbohydrates** make less than 1% of the cell. They combine with proteins to form glycoproteins and with lipids to form glycolipids. Carbohydrates, proteins and lipids at the surface of cell membrane form cell coat called **glycocalyx**. Glycocalyx is involved in cell recognition and determination of cell behavior.

50-51Structure of Cell Membranes: Fluid Mosaic Model-I

Structure of Cell Membranes

Cell membrane separates the cell from external environment. All material move across the cell membranes. There were two models about structure of cell membrane: a). Unit membrane model b). Fluid mosaic model

a) Unit membrane model

According to unit membrane model cell membrane has four layers. Two lipid layers are present in the center and two outer protein layers. It means lipids are sandwiched between two protein layers. This model was however, unable to explain the movement of molecules across cell membrane.

b) Fluid mosaic Model

In 1972, S. Jonathen Sanger and Garth Nicolson presented fluid mosaic model of membranes. According to this model plasma membrane is fluid in nature where lipids form bilayer and proteins are embedded in a mosaic pattern. Lipids form bilayer due to hydrophobic tails. Phospholipids molecules form a sea in which protein molecules are floating like icebergs or ships. Being fluid the membrane is in a constant state of shifting and changing. However membrane maintain a uniform structure. Due to fluid nature membrane is not locked for transportation of materials.

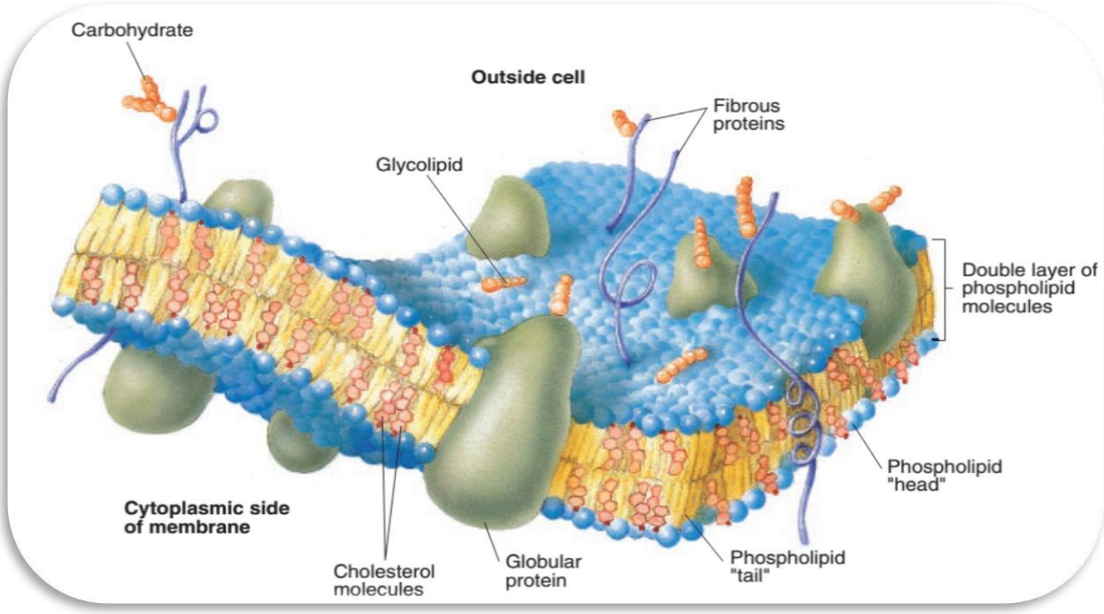


Fig: Fluid mosaic model of cell membrane

52 Functions of Cell Membranes

Functions of Cell Membranes

Some of the important functions performed by cell membrane are:

1. It separates the cells from one another and from external environment.
2. It separates various cellular organelles from general cellular environment. This property is called compartmentalization.
3. It regulates the transport of materials into and out of the cell. It also controls the movement of materials between different organelles.
4. Different Chemical reactions take place on the surface of cell membranes.
5. Cell membranes contain different types of molecules which serve as identification markers of different cells. i.e., antigens
6. It maintains the internal environment of cell (cellular homeostasis).

53 Movements Across Cell Membranes: Simple Diffusion

MOVEMENTS ACROSS CELL MEMBRANES

1. **Simple Diffusion**

"Movement of molecules from an area of high concentration to the area of low concentration is called simple diffusion". Diffusion is a passive transport of molecules. *i.e.,* no energy is required for this process. It is a random movement of molecules. Diffusion accounts for movement of molecules for short distance.

a) Movement of O_2 from lungs into blood stream
b) Movement of CO_2 from tissues into tissue fluid

54 Movements Across Cell Membranes: Facilitated Diffusion

1. **Facilitated Diffusion**

"Movement of molecules from an area of high concentration to the area of low concentration with the help of membrane proteins is called facilitated diffusion." There are two types of these proteins:
a) Channels b) Carrier

Like simple diffusion, facilitated diffusion is also a passive transport method. Large size molecules and polar or charge bearing molecules are transported across cell membrane by this method. Glucose transporter, Na^+ channels and K^+ channels transport glucose, Na^+ and K^+ through facilitated diffusion mechanism.

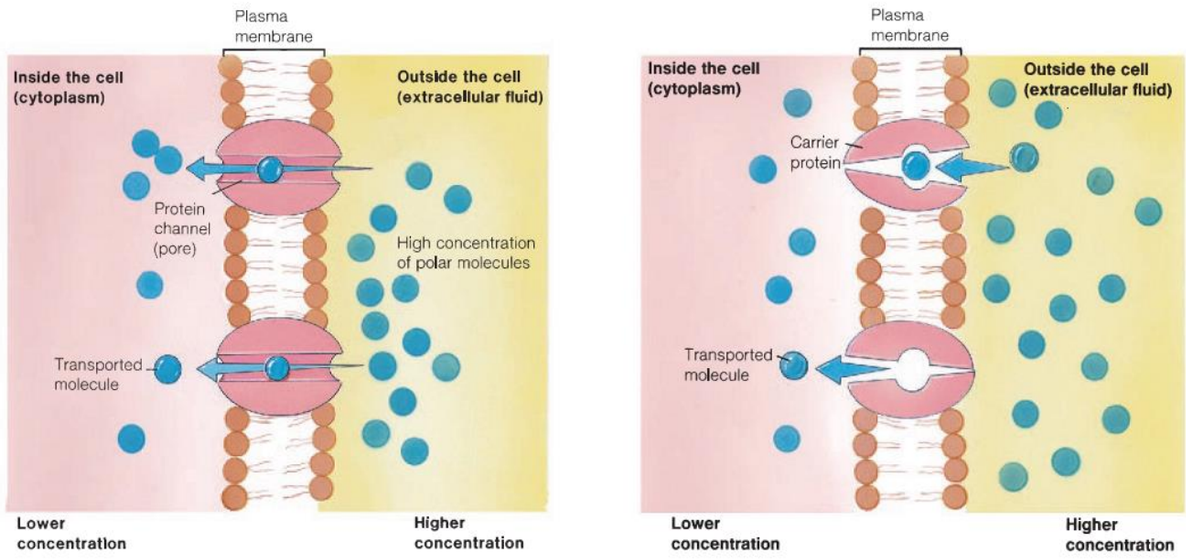


Fig: Facilitated diffusion

55 Movements Across Cell Membranes: Osmosis

1. **Osmosis**

"It is the movement of water molecules across a selectively permeable membrane from an area of higher concentration to the area of lower concentration". The minimum pressure required to prevent the inward flow of its pure solvent across a semipermeable membrane is called **osmotic pressure**. The concentrated solution surrounded by dilute environment is called **hyperosmotic**. e.g., a cell place in pure water, cell is hyperosmotic with reference to surroundings. If the solution have same concentration of solute with reference to surroundings, it is called **isosmotic**. e.g. if a cell is place in normal salt concentration medium, the cell is isosmotic with reference to surroundings. The solution with lower concentration of solute than surroundings is called **hypoosmotic**. e.g., when a cell surrounded by concentrated environment, the cell is hypoosmotic with reference to external environment.

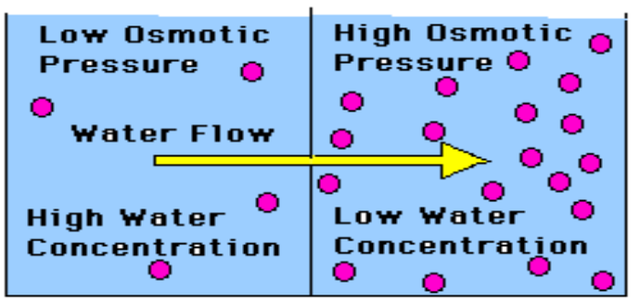


Fig. osmotic pressure is high where water molecules are in low concentration

Osmosis maintains ionic balance and osmotic equilibrium. Minerals and nutrients are uptaken by the process of osmosis.

56 Movements Across Cell Membranes: Filtration

1. **Filtration**

"Filtration is a process that forces small molecules across selectively permeable cell membrane using hydrostatic pressure". It is the pressure exerted by fluid at any point within the fluid or on the walls. Hydrostatic pressure increases with increasing depth and it acts in all directions. E.g., Filtration also takes place in glomerulus of nephron in kidney which is first step of urine formation.

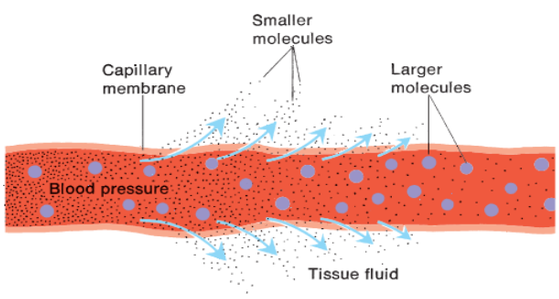


Fig. Filtration across capillary walls

57 Movements Across Cell Membranes: Active transport

1. **Active Transport**

"Movement of molecules from a selectively permeable membrane against concentration gradient is called active transport". It is the movement of molecules from low concentration to high concentration with the expenditure of energy (ATP). Like passive transport carrier protein are involved. These proteins are also called pumps because they work from low to high concentration. e.g., sodium/potassium pump which helps to maintain K^+ and Na^+ gradient in nerve cells. Ca^{2+} is kept hundreds of time low concentration in cytosol than endoplasmic reticulum.

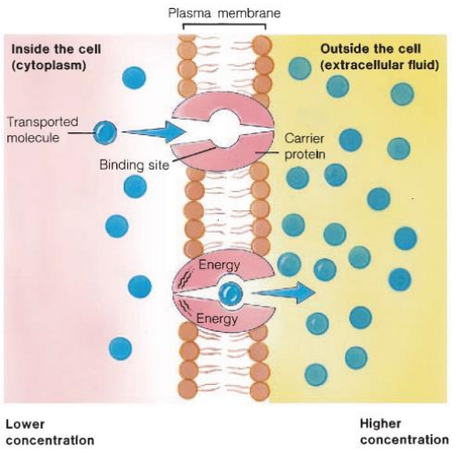


Fig. Active transport mechanism

58 Movements Across Cell Membranes: Endocytosis

2. **Endocytosis**

"The bulk movement of molecules into the cell across plasma membrane". There are three forms of endocytosis:

a. Pinocytosis b. Phagocytosis c. Receptor mediated endocytosis

a. Pinocytosis

It is the uptake of liquid from extracellular environment. Liquid is up taken in the form of vesicle. A small solid dissolved into the liquid is also taken into the cell.

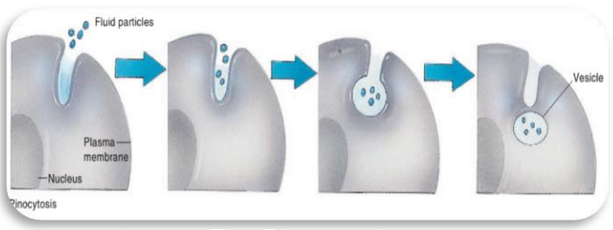


Fig. Pinocytosis

b. Phagocytosis

It is the uptake of solid materials by the cell. Cell membrane invaginates to take solid material in the form of a vesicle. Within cell this vesicle combines with lysosome to form **phagolysosome** for digestion.

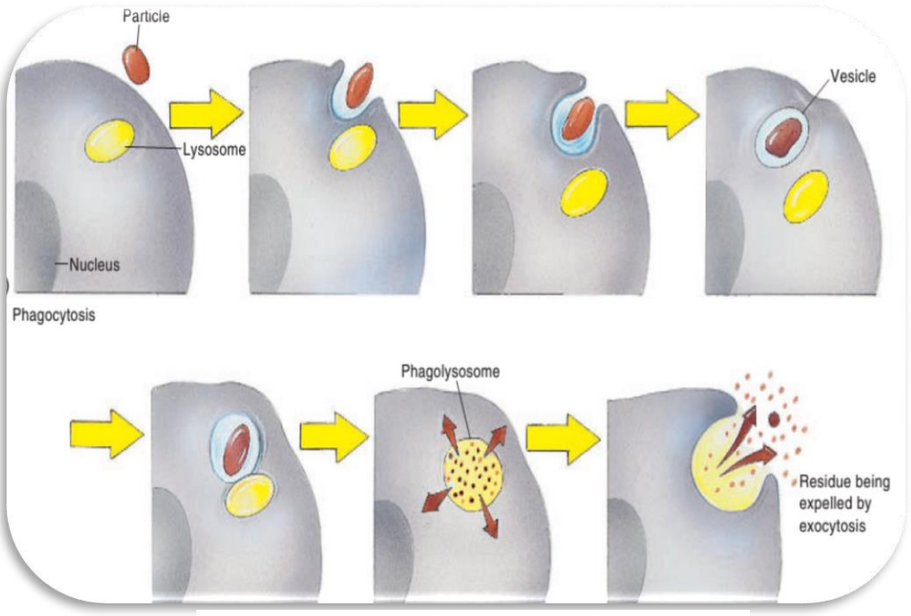


Fig. Phagocytosis by an animal cells

c. Receptor Mediated Endocytosis

It is the uptake of molecules when they bind with specific receptor on cell membrane. After this membrane invaginates and vesicle formation occurs. e.g., transport of cholesterol across cell membrane.

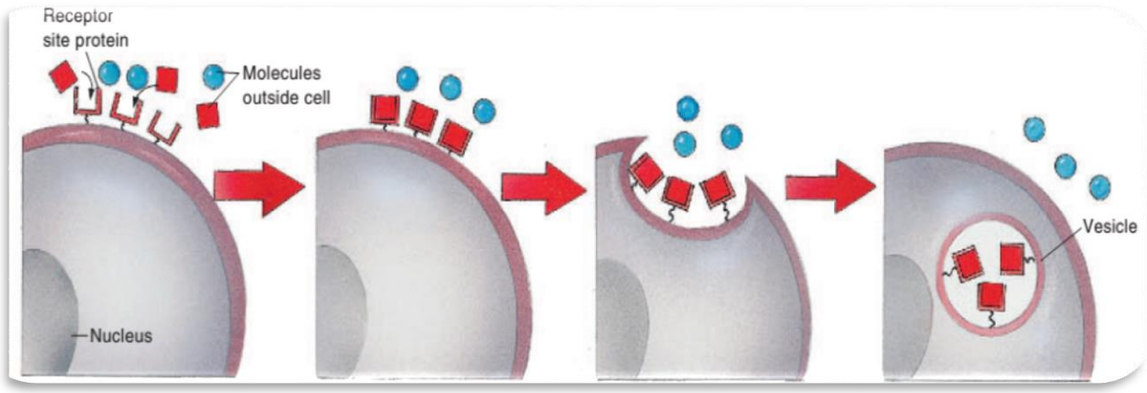


Fig. Receptor mediated endocytosis

59 Movements Across Cell Membranes: Exocytosis

1. **Exocytosis**

"It is the process by which cellular material are packed up into vesicles and transported out of cell". This is opposite to endocytosis. Different proteins which have to be transported out of cell are brought to cell surface via exocytosis. E.g., release of extracellular enzymes and removal of undigested food by amoeba and paramecium.

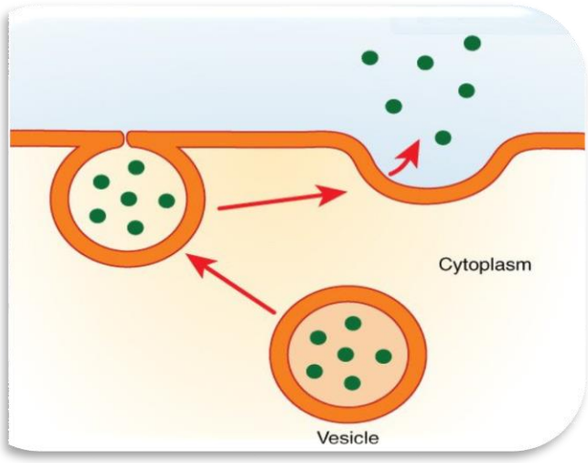


Fig. Exocytosis

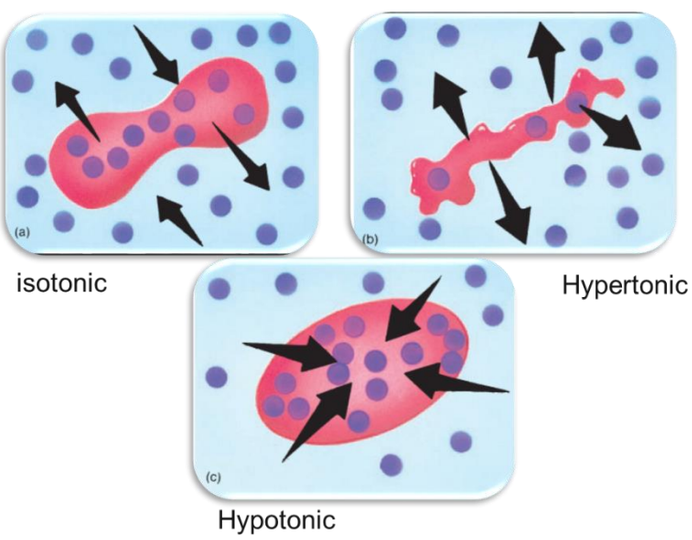
Exocytosis adds new fragments to cell membrane. This is the compensation of membrane lost during endocytosis.

60 Tonicity

TONICITY

"It is the relative concentration of solutes dissolved in solution inside and outside the cell". Water molecules move into and out of cell due to difference in tonicity. A may be present in three different types of solutions.

a. **Isotonic Solution:** When solute concentration is same outside and inside of the cells. Water molecules move inside and out of the cell at same rate.
b. **Hypertonic Solution**: When solute concentration is higher outside the cells than inside. Water molecules move out and the cell shrinks. This condition is called crenation.
c. **Hypotonic Solution:** When solute concentration is lower outside the cell than inside. Water molecules move into the cell which causes the cell to swell and burst.



61 Cytoplasm and Its Functions

CYTOPLASM AND ITS FUNCTIONS

Cytoplasm is the portion of cell present between nucleus and plasma membrane. **Ectoplasm** is the clear, outer layer of cytoplasm which is immediately below the plasma membrane. While the inner, dense portion of cytoplasm is called **endoplasm**. Cytoplasm has semi-fluid portion called **cytosol** while membrane bounded and non-membranous structures are called **organelles**. Organelles of cytoplasm perform different functions such as protein synthesis, digestion, ATP synthesis etc. Cytoplasm has the storage of molecules required for different cellular activities.

The important cellular structures are: ribosomes, endoplasmic reticulum, lysosomes, Golgi apparatus, mitochondria, vacuoles, cytoskeleton, cilia and flagella.

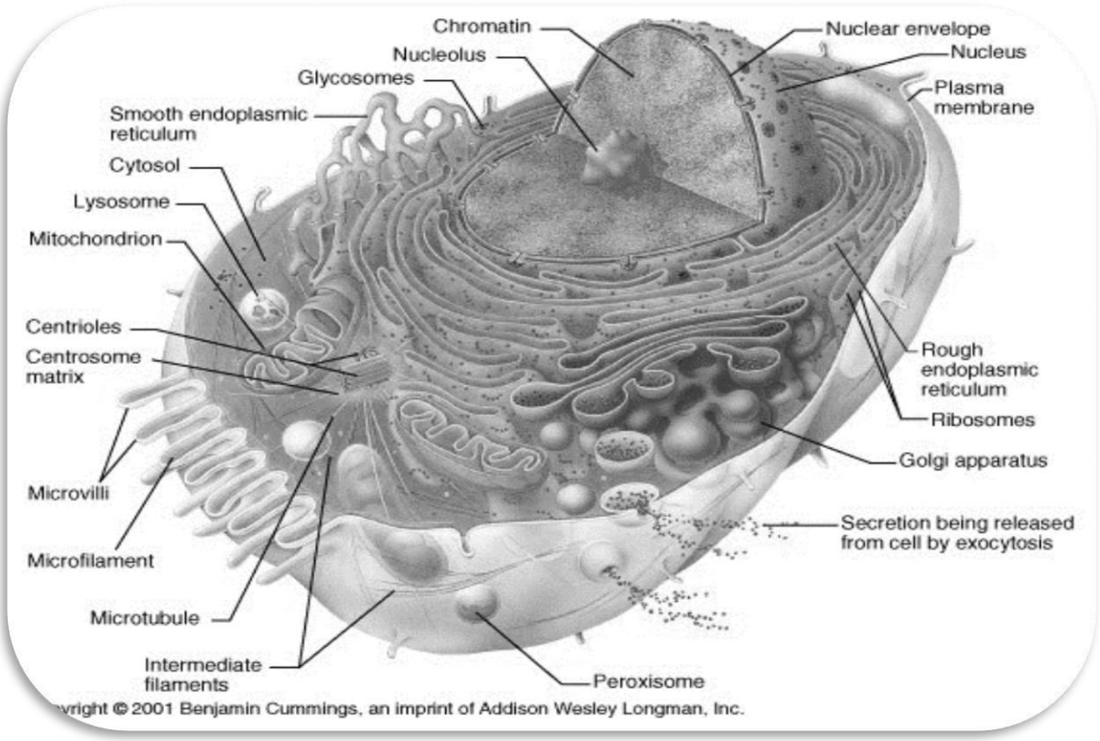


Fig. General structure of an animal cell

62 Cytoplasmic Organelles: Ribosomes

1. Ribosomes

Ribosome is the non-membranous structure of cytoplasm involve in protein synthesis. Prokaryotes like bacteria have small size ribosome (70S) while eukaryotes have large size ribosome (80S). Ribosome consists of two subunits: small ribosomal subunit and large ribosomal subunit. Chemically ribosomes are made up of equal amount of rRNA and proteins. They may be free or attached with endoplasmic reticulum. Ribosome get attached with mRNA and synthesize protein molecule. Sometimes many ribosomes are attached to single mRNA and the resulting structure is called **polysome**.

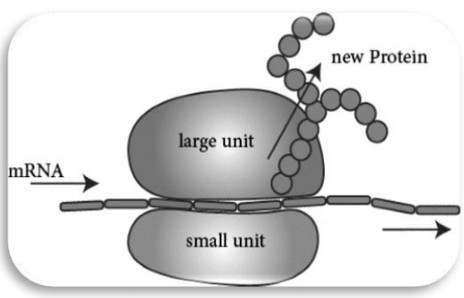


Fig. A single ribosome during protein synthesis

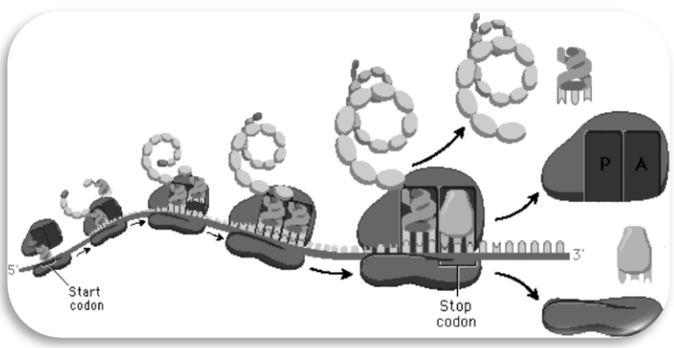
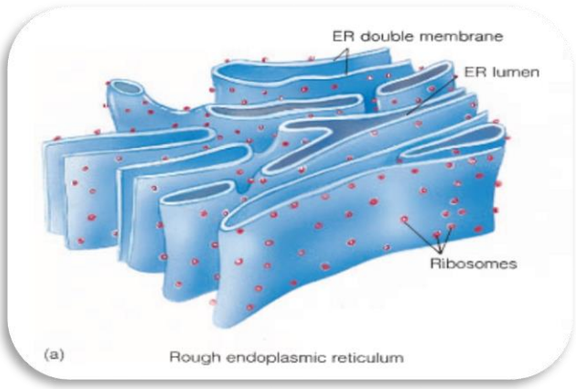


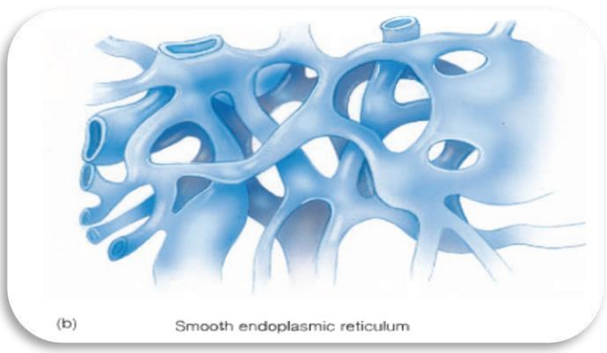
Fig. A polysome

63-64Cytoplasmic Organelles: Edoplasmic reticulum I

1. Endoplasmic Reticulum

Endoplasmic reticulum (ER) is a complex network of membrane bounded sheets, sacs and tubules present throughout the cytoplasm. At one side attached with outer membrane of nucleus while on other side attached with plasma membrane. Tubules of endoplasmic reticulum are called cisternae. There are two types of endoplasmic reticulum: Rough endoplasmic reticulum (RER) and Smooth endoplasmic reticulum (SER). Endoplasmic Reticulum with attached ribosomes on their surface are called **rough endoplasmic reticulum** (RER).Presence of ribosomes give them rough appearance. While endoplasmic reticulum without ribosomes on their surface are called **smooth endoplasmic reticulum**. Due to absence of ribosomes on their surface they appear smooth. Important functions of ER are: mechanical support, material circulation, storage of enzymes, proteins and calcium, lipid production, detoxification of organic molecules and calcium storage.



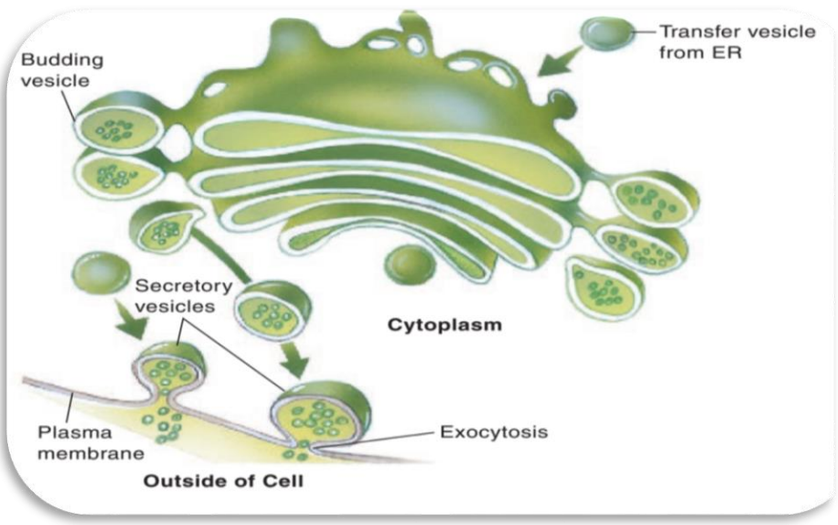


65 Cytoplasmic Organelles: Golgi apparatus

1. Golgi Apparatus

It is a collection of membranes within cytoplasm involve in cell secretion and intracellular transport. Golgi apparatus is also called Golgi complex. It was first reported by Camilo Golgi in 1898. Golgi Apparatus is composed of 3-7 cisternae and some golgi vesicles. The side of golgi apparatus towards nucleus is called forming face while the side away from golgi apparatus is called maturing face. Golgi complex is involved in sorting, packing and secretion of proteins and lipids. Proteins are concentrated, chemically modified and sorted for different destinations. It is most abundant in cell involve in secretion.

Proteins are synthesized in the lumen of endoplasmic reticulum from where they are sealed off in small packets called **transfer vesicles** and send to Golgi apparatus. After modifications in Golgi apparatus proteins are packed into small sacs called **secretory vesicles** and move towards cell membranes for cells secretions



66 Cytoplasmic Organelles: Lysosomes

1. Lysosomes

Lysosomes are single membrane bounded organelles which contain hydrolytic enzymes for intracellular digestion. They digest carbohydrates, lipids, proteins and nucleic acids. They can selectively digest and recycle their own cytoplasm or organelles. Lysosomes contain acid hydrolases which digest food under acidic conditions. A phagocytic food vacuole and lysosome fuse to form phagosome. In

phagosome the ingested is digested and absorbed. Undigested food is excreted out by the process of exocytosis.

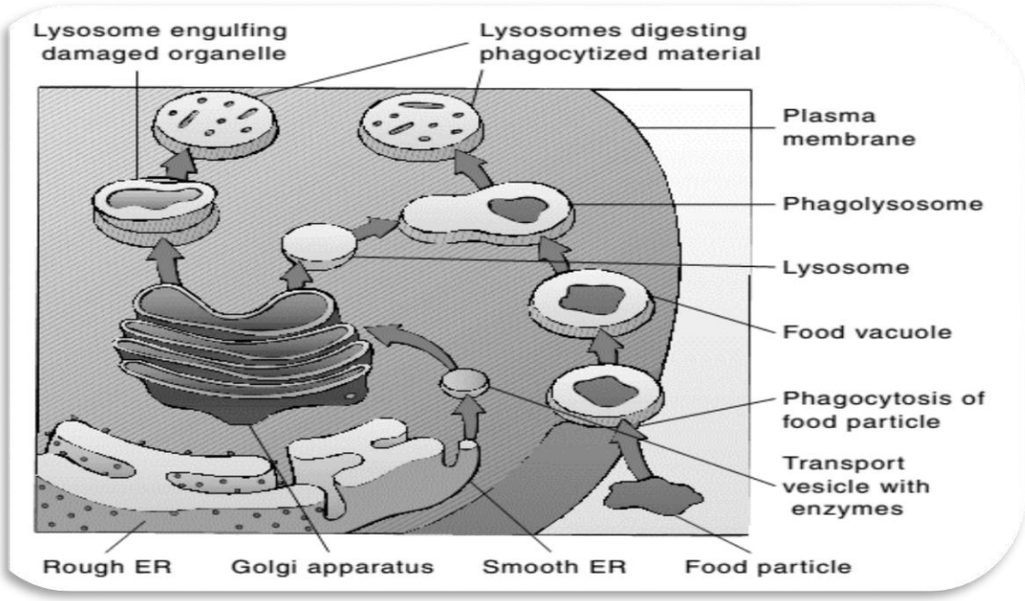


Fig. formation and function of lysosome

67 Cytoplasmic Organelles: Mitochondria

2. Mitochondria

- Mitochondria are double membrane bounded organelles of cell involve in energy production. They are spherical to elongate in shape. Their outer membrane is flat while inner membrane is folded and forms finger like projections called cristae. Matrix is present between cristae. Matrix contain ribosomes, circular DNA, and other materials.

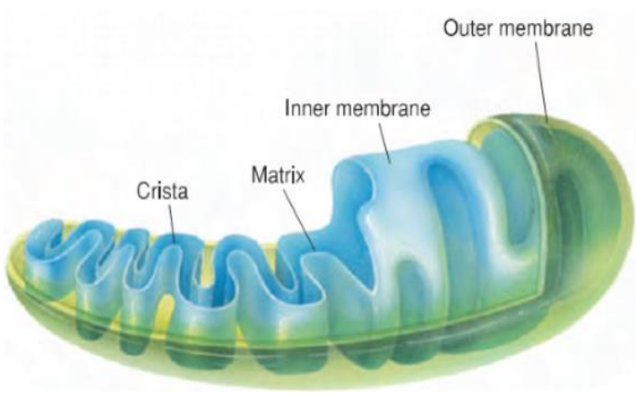


FIg. Inner structure of mitochondria

Mitochondria are “Power house” or “power generators” of the cell. Processes of ATP synthesis, Krebs cycle, electron transport chain and fatty acid oxidation take place in mitochondria. The number of mitochondria vary according to energy requirements of cells.

68 Cytoplasmic Organelles: Vacuoles

1. Vacuole

Vacuoles are single membrane bounded organelles mainly involve in storage of water and salts in the cell. They vary in shape and number. Animal cells contain more than one small vacuoles. Membrane of vacuole is specialized in transportation of water and salts. Contractile vacuole in protozoa and sponges remove waste water from the cell while food vacuole is involved in digestion of food.

69-70Cytoskeleton I

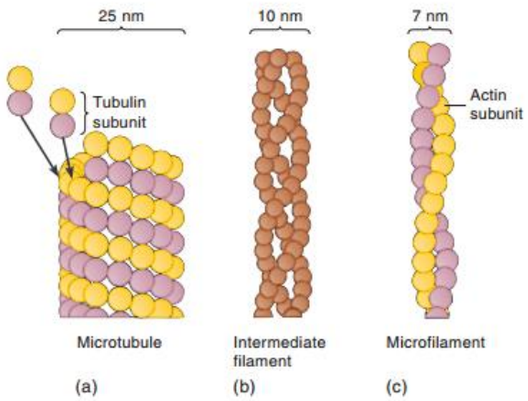
1. Cytoskeleton

It is the network of protein fibers extending throughout the cytoplasm present in eukaryotic cells only. It is provides mechanical support and connects various cellular organelles. Cytoskeletal is made up of three types of fibers: Microtubules, Intermediate filaments and Microfilament.

Microtubules are hollow, slender, cylindrical structures. They are largest cytoskeletal elements. Each microtubule is about 25nm in diameter and made up of tubulin protein. Microtubules help in movement

of organelles and chromosomes. They are part of transport system (e.g., secretory vesicle movement) in the cell and are involved in overall shape changes of cell during specialization. Microtubules make different structures like mitotic apparatus, cilia, flagella and centrioles.

Intermediate filaments are heterogeneous group of cytoskeletal fibers. They are made up of more than 70 different types of proteins. They are of intermediate size (10nm diameter). Intermediate filaments help to maintain cell shape, maintain spatial arrangement of organelles and promote mechanical activities in cytoplasm.

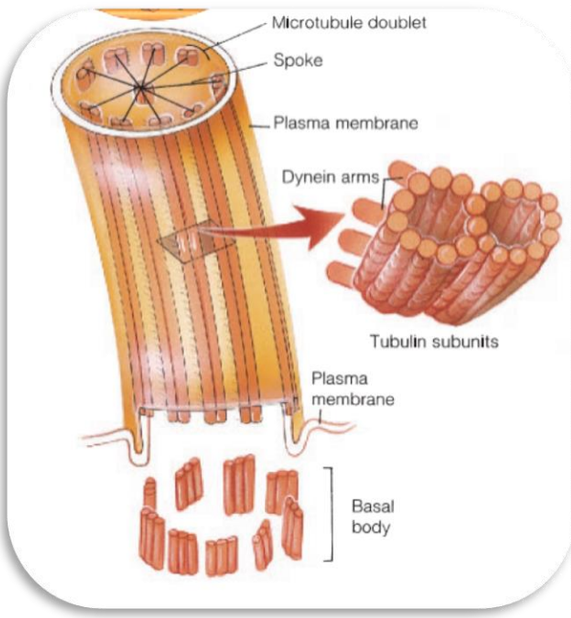


Microfilament are the smallest cytoskeletal fibers with 7nm diameter. They are made up of actin subunits. Microfilaments in muscle are highly developed and called as myofibrils. Main functions of microfilaments are muscle contraction, mechanical support in non-muscle cells and contractile movements of cell (E.g., amoeboid movement)

71-72 Cilia and Flagella: Occurance and Function

1. Cilia and Flagella

Cilia and flagella are motile appendages present on the surface of some cells. **Cilia** are 5 to 20 times smaller than cilia and present on cells of trachea, fallopian tubes, protozoa and invertebrate larva. In trachea they push mucus and dust particles toward mouth, in fallopian tubes they moves egg towards uterus and in protozoa and invertebrate larvae they are responsible for locomotion. **Flagella** are present on flagellated protozoans, bacteria and Gametes of algae, fungi, plants and animals. They are mainly involved in locomotion and have sensory role.



Both cilia and flagella are structurally same. Cell membrane form continuous outer membrane. Material inside the membrane is called matrix. The central microtubule structure of cilia and flagella is called **axoneme** or axial filament. Axoneme has nine pairs of microtubules arranged in a circle and two central tubules. This is 9+2 pattern. Dynein arms project towards neighboring doublet. Nine pairs are connected with central tubules through **radial spokes**.

A microtubules structure present at the base of cilia ad flagella is called basal body. It is made up of nine triplets of microtubules and no central pair of microtubules.

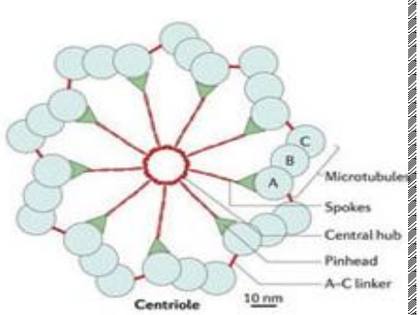
73 Centrioles

LESSON 73

1. Centrioles

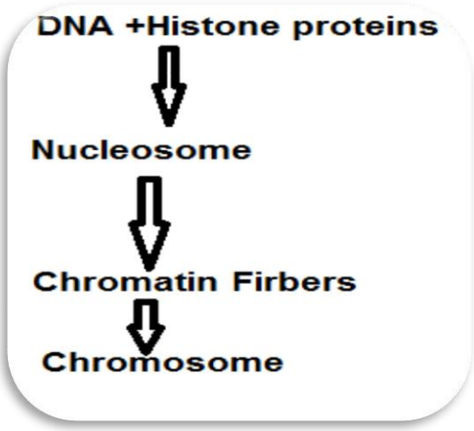
Centriole is a cylindrical organelle present near to nucleus and involved in cell division process. Nine triplets of microtubules. They radiate from the center like spokes of wheel. There is pair of centrioles in the cell. Both are present right angle to each other. Centriole duplicate before cell division. During cell division they form mitotic apparatus and help chromosomes to get separate. They are also involved in the formation of cilia and flagella.





74 Microtubules Organizing Centers

1. Microtubule Organizing Center



Microtubule organizing centers (MTOC) are the structures from where microtubules emerge. When observed under electron microscope, microtubules appear to originate from single point. Centrosome and basal body are two MTOC in the cell. **Centrosome** is a complex region present close to nucleus. It has two main components: centriole and pericentriolar material. Microtubule formation takes place in two steps. A small proportion of microtubules is initially formed slowly. This is called nucleation. In second step microtubules length increase. It is a faster step. Microtubule nucleation starts at centrosome and radiate out into all cytoplasm. **Basal Bodies** are the MTOC for cilia and flagella. Basal bodies are structurally identical to centrioles.

75 Nucleus and its Functions

NUCLEUS

Nucleus is double membrane bounded organelle of cell which contain DNA and is main controlling center of eukaryotic cell. Nucleus Controls all activities of cells, stores genetic information and transfer it to next generation. Cells may be uninucleate (one nucleus), binucleate (two nuclei) or multinucleate (many nuclei). Important components of nucleus are: Nuclear envelope, chromatin material, nucleolus and nucleoplasm.

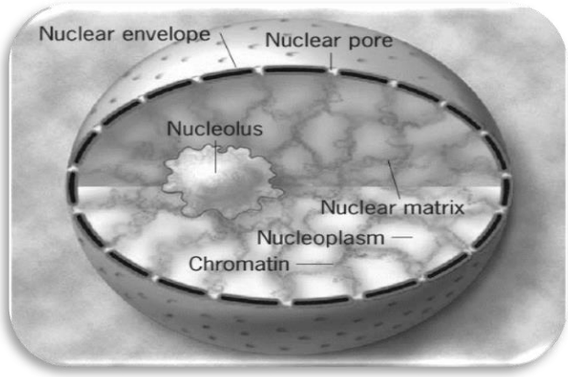


Fig. The nucleus

76 The Nucleus: Nuclear Envelope

Nuclear Envelope: Nuclear pore is a double membrane structure that separates nucleus from cytoplasm. Both outer and inner nuclear membranes are separate from each other. They are directly connected at the nuclear pores. Outer nuclear membrane is in continuation with endoplasmic reticulum. Nuclear envelope bears **nuclear pores**. The number of nuclear pores vary from few to many thousands (3000 in egg). Nuclear pores is composed of globular and filamentous proteins. It controls the transport of materials between nucleus and cytoplasm.

77 The Nucleus: Nucloeplasm

Nucleoplasm: Nucleoplasm is the soluble portion of nucleus enclosed in nuclear envelope. It is composed of ground substance of nucleus *i.e.,* nucleotides, RNA molecules, proteins, and enzymes for DNA replication and Transcription. Nucleoplasm is the site for DNA replication, Transcription and RNA processing.

78 The Nucleus: Nucleolus

Nucleolus: Nucleolus is non-membrane bounded structure in nucleus of non dividing cells. There are two or three nucleoli in most cells. Some cell may have thousands (e.g. amphibian egg). Nucleolus contain two regions the outer granular region and inner fibrillar region. Nucleolus is the site for rRNA synthesis and assembly of ribosomes.

79 The Nucleus: Chromosomes and Chromatin Material

Chromosomes and Chromatin Material: All animals have DNA as genetic material. Different segments of DNA encode specific proteins and polypeptides. The part of DNA which encode specific protein or polypeptide chain is called a gene. Genes also encode tRNA and rRNA molecules. DNA is packed along with histone protein and form a network. This network of genetic material is called **chromatin material**. At the time of cell division chromatin material condenses and attain the form of **chromosome.** Every species of animals have fixed number of chromosomes in their body.

80 Tissues: General Concept

TISSUES

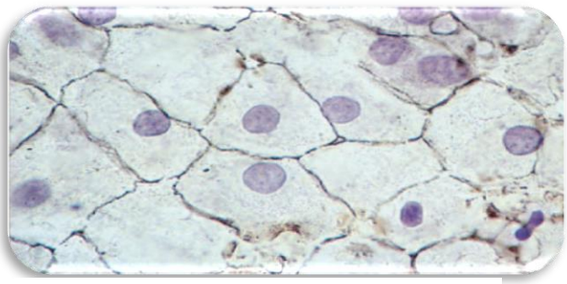
A tissue is an aggregate of cells performing common function. They are specialized for specific functions. Cells of tissue differentiate together during development. Tissues which are made up of made up of single type of cells are called **simple tissues** while those which are made up of more than one types of cells are called **compound tissues.** Animal Tissues are classified into four major classes: epithelial tissues, connective tissues, muscle tissues and nervous tissues.

81-82 Epithelial Tissues: Types and Functions-I

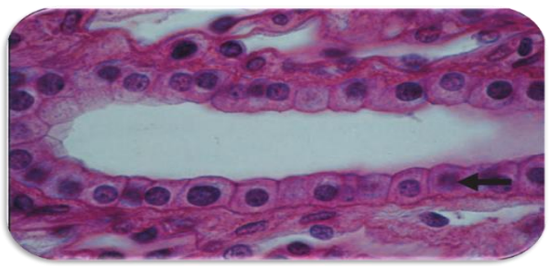
A. Epithelial Tissues

They are renewable sheets of cells covering different body parts or line the tubular portions. They form outer layer of our skin, visceral organ like heart, kidneys and lining of tubular portions such as digestive system. Epithelial tissue is separated from underlying tissues by basement membrane. The shape, size and arrangement of epithelial tissues is directly related to their function. Important functions of epithelial tissues are absorption (Intestine lining), Transport (capillaries, kidney tubules), Excretion (sweat glands), Protection (skin), sensory function (taste buds, hearing) and secretion (exocrine glands).

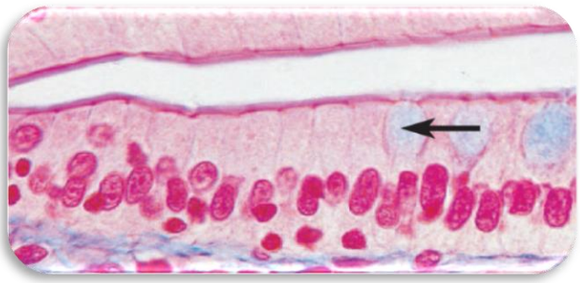
Epithelial tissues are classified on the basis of shape and number of layers. **Squamous epithelial tissues** are single layered, tightly packed, flatten disk shaped located in lungs air sacs, linings of heart, blood vessels and lymphatic vessels and kidney glomeruli. They are involved in smooth flow, diffusion and filtration process. **Cuboidal epithelial tissues** are tightly packed and cube shaped. They are located in kidney tubules, ducts, small glands and surface of ovaries. They are involved in secretion and absorption. **Columnar epithelial tissue** are single layer of elongated cells. They line digestive tract, gallbladder and excretory ducts of some glands. They are involved in absorption and enzyme secretion. **Pseudostratified ciliated columnar epithelial tissue** have tuft of cilia on columnar cell. Cells appear stratified due to multilayers of nuclei. They are located in lines of bronchi, uterine tubes and some region of uterus. They propels mucus and reproductive cells by the action of cilia. **Stratified squamous epithelium tissue c**ontains many layers of cells. They form keratinized surface of skin, lines of esophagus, mouth and vagina. Their main function is protection from abrasive damage.



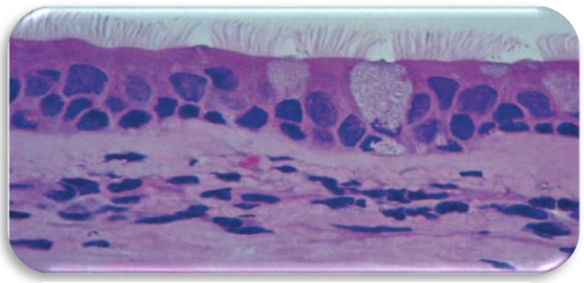
Squamous epithelium



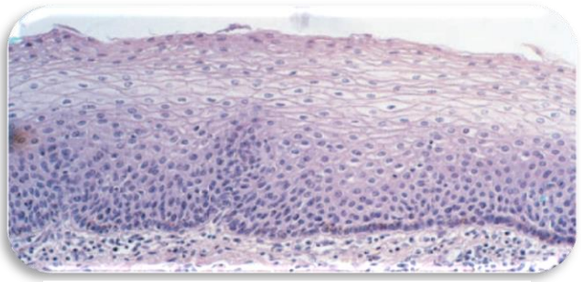
Cuboidal epithelium



Columnar epithelium



Pseudostratified columnar epithelium



Stratified squamous epithelium

83-84 Connective Tissues: Types and Functions-I

B. Connective Tissues

These are diverse types of tissues which serve both supportive and binding function. They are wide spread in animal body and have relatively few cells and much more extracellular fibers and matrix. Matrix may be in liquid or solid form. There are two general types of tissues: Loose connective tissues and hard connective tissues.

Loose connective tissues are strong flexible fibers of collagen which are interwoven with elastic fibers. They are located in epithelia of skin and throughout body. They wrap and cushion organs. In **fibrous connective tissues** collagen fibers densely packed, parallel. They are located in dermis of skin; submucosa of digestive tract; fibrous capsule of organ and joints. Their major function is to provide structural strength.

There are following types of connective tissues on the basis of structure and function:

i. Adipose tissue
ii. Cartilage
iii. Bone
iv. Blood

i. Adipose Tissues

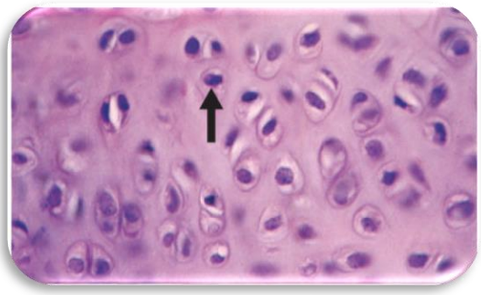
These are fat storing tissues. Large lipid droplets in adeposite cells push nucleus to one side. They are present around kidneys, under skin, in bones, breast and abdomen. Their main function is fuel storing, insulation, support and protection.

ii. Cartilage

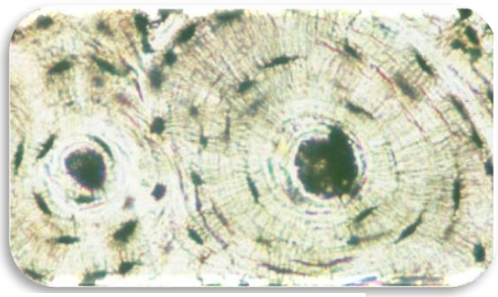
This is a hard and flexible tissue. In cartilage, chondrocytes cells are present in lacunae surrounded by matrix. Matrix and collagen fiber provide strength and elasticity. Cartilage is present in **e**xternal ear, epiglottis, bone ends, disks of joints and complete skeleton of sharks. Their main function is to **s**upport, Shock absorber and shape the organs/body with flexibility.

iii. Bone Tissues

Bone cells (osteocytes) are present in lacunae. They are heavily impregnated with calcium phosphate making bones hard and strong. They make over all structure of body, provide protection, act as lever system for muscle movement, store calcium and provide a site for blood cells formation.



Cartilage



Bone

iv. Blood

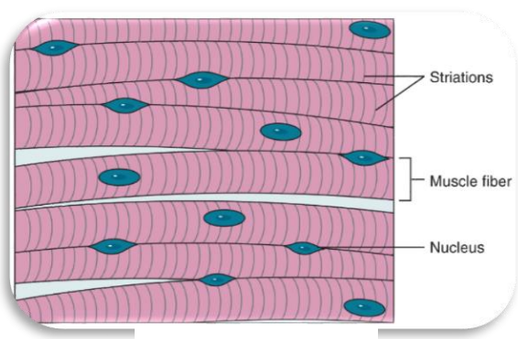
It is a fluid form of connective tissues. Blood is composed of plasma with suspended RBCs, WBCs and platelets. Blood plays an important role in transportation of gases, nutrients, hormones, minerals, vitamins and wastes within the body.

85 Muscles Tissues: Skeletal muscles

C. Muscle Tissues

These are the tissues having ability to contract and relax thus responsible for movement and locomotion in animals. There are three different types of connective tissues:

i. Skeletal muscles ii. Smooth muscles iii. Cardiac muscles



Skeletal muscles

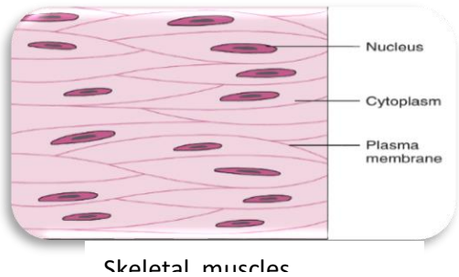
i. Skeletal Muscles

These are the muscle tissues attached to the skeleton bones. Skeletal muscles cells (muscle fibers) are long, cylindrical, striated and contain many peripheral nuclei. They involve in voluntary movements (movement of finger, hands, arms, head, legs and back) and locomotion (*i.e.,* walking, running and flying). They are found in both vertebrates and invertebrates.

86 Muscle Tissues: Smooth Muscles

i. Smooth Muscles

These are the muscle tissues making most of the internal organs especially hollow organs. Smooth muscles cells are spindle shaped, non-striated and contain single nucleus. They are involuntary muscles and have main role in movements of food through digestive system, urination and uterine contraction during birth. They are arranged together to form sheets. Smooth muscles are found in both vertebrates and invertebrates.

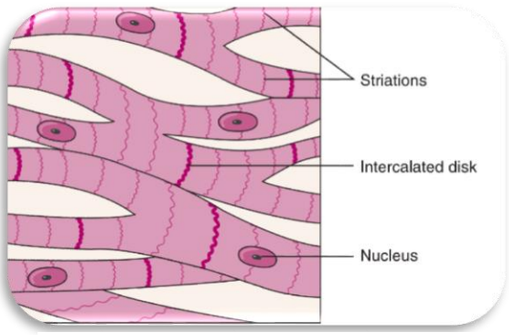


Skeletal muscles

87 Muscle Tissues: Cardiac Muscles

i. Cardiac Muscles

These are the specialized tissues of heart found in vertebrates only. Cardiac cells are branched, striated, contain single nucleus and connected through intercalated discs. Cardiac muscles are involuntary muscles. They circulate the blood in body and transport nutrients, gases and wastes. Average heart beat for human is 72 times/min or 100,000 times/day. In the life of 70 years heart beats for about 2.5 billion times.

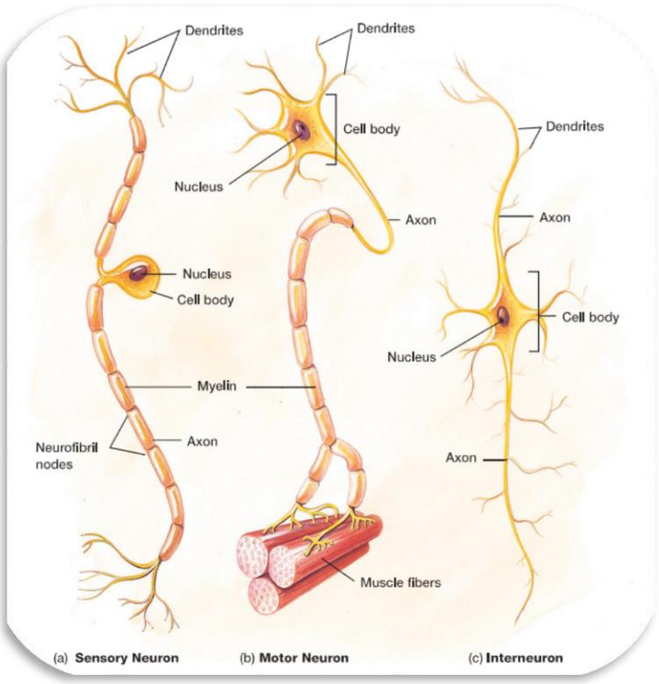


Cardiac muscles

88 Nervous Tissues

D. Nervous Tissues

These are the specialized tissues responsible for coordinating and controlling the body activities. They are found in brain, spinal cord and nerves. Nervous tissues are made up of neurons, neuroglia and peripheral glial cells. Neuroglia protects and nourish CNS neurons while peripheral glial cell protect and nourish peripheral neurons. The Neurons transmit electrical signals to other neurons, muscles and glands. A neuron has three main parts dendrite, cell body and axon. On the basis of functions there are three types of neurons:



Sensory neuron transfer information from receptors to central nervous system. **Motor neurons** transfer information from CNS to effectors and **Inter neurons** are present in brain and spinal cord.

89 Organ

Organ

Organ is a part of animal body adapted to perform specific function. Organ is made up of different types of tissues. These tissues work in a coordinated way to perform specific function. Heart, stomach, liver, kidney and Lungs are some examples of organs in animals.

90 Organ System

Organ System

Organ system is a association between different organs to complete a process. An organs system is composed of different organs having related functions. For example digestive system is made up of

different organs: esophagus, stomach, small intestine, large intestine, pancreas and liver. Eleven organ systems make a human

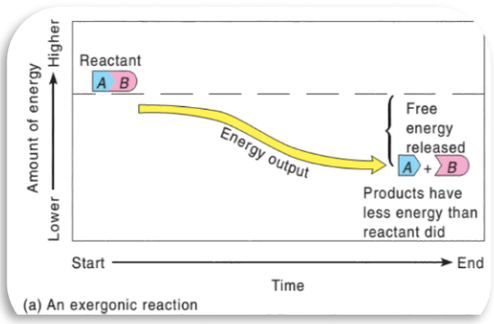
1. Integumentary system
2. Muscular system
3. Skeletal system
4. Digestive system
5. Circulatory system
6. Respiratory system
7. Endocrine system
8. Reproductive system
9. Nervous system
10. Lymphatic system
11. Urinary system

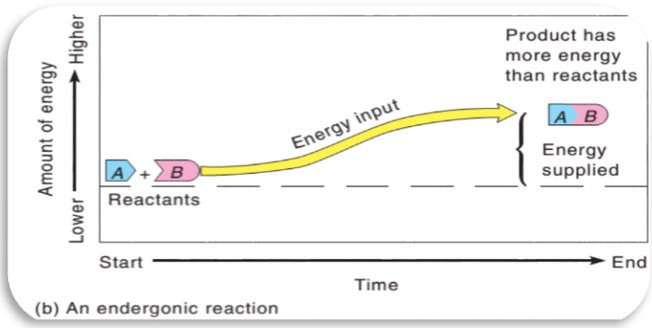
91 Energy, Free Energy

ENERGY

The ability to do work is called energy while work is transfer of energy. e.g., photosynthesis in plants transform solar energy into chemical energy. Energy exists in two basic states: **Kinetic energy** and **Potential energy**. Energy stored in chemical bonds of biomolecules is called potential energy. Energy has many forms e.g., heat energy, sound energy and electromagnetic energy.

Reactions in which energy is released are called **exergonic reactions** while reactions in which energy is absorbed are called **endergonic reactions**. The usable energy is called **free energy**. It is the available energy to do work. The SI unit of energy is joule while mostly it is measured in calories. Energy required to raise the temperature of 1g of water through 1ºC is called a calorie.





92 Laws of Energy Transformation

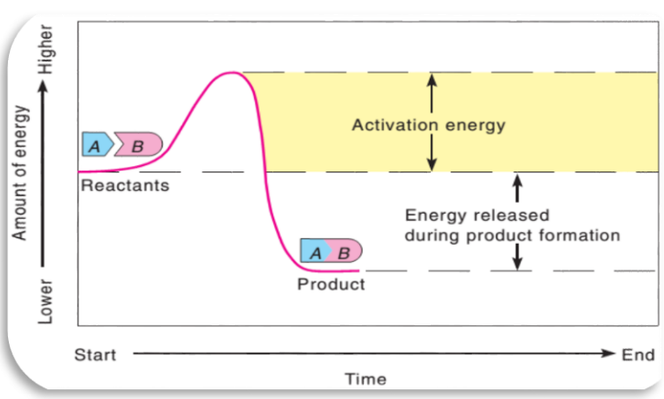
Laws of Energy Transformation

Brach of science which deals with deals with relationship between heat and other forms of energy is called thermodynamics. In this we measure the energy change during a biochemical reaction. There are two laws of thermodynamics. According to **first law of thermodynamics**, "Energy can neither created nor destroyed, however it can be transformed from one form to another form". e.g., electrical energy used for heat production; we eat food and do work (chemical to kinetic energy). According to **second law of thermodynamics**, "All objects in the universe tend to become disordered and the total amount of disorder in universe is continuously increasing. Things tend become more stable by acquiring low energy state (disordered). The measure of degree of disorganization is called **entropy**. For example, when natural gas burns to give heat energy, some of it cooks food, other released into environment. Unusable energy is increasing entropy.

93 Activation Energy

Activation Energy

The amount of energy required to start a chemical reaction is called activation energy. This is the energy barrier between reactants and products. The top hill of the energy barrier is called transition state. In exergonic reactions, the reactants are at high energy state while in case of endergonic reactions, reactants are at low energy state. None of the reactions occur spontaneously. The amount of reactants converted into products in a given time is called is called **reaction rate**. Reaction rate do not depend upon how much energy is released. It depends upon activation energy. Catalysts such as enzymes increase reaction rate by lowering activation energy.



94 Enzymes: Biological Catalysts

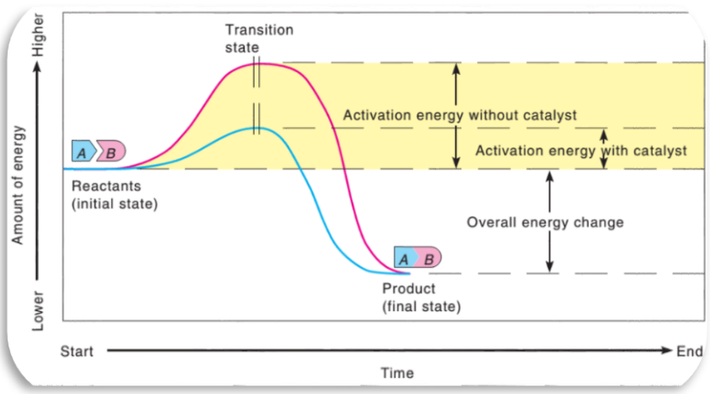
ENZYMES

Enzymes are molecules which catalyze biochemical reactions. They increase reaction rate enormously. Their catalytic power is much more than chemical catalysts. Life is a series of chemical reactions catalyzed by enzymes. Metabolism proceeds because enzymes channel molecules through specific chemical pathways. Without enzymes the biochemical reaction will be too slow to continue life. Enzymes are very selective for their substrate. They catalyze specific chemical reactions.

95 Enzymes Structure

Enzymes Structure

Most of the enzymes are protein in nature. They are globular proteins may present as soluble molecules or a part of membrane. An enzyme may be composed of single polypeptide or more than one polypeptide chains (multisubunit). Few RNA molecules in nature work as enzymes are called **ribozymes.**



Enzymes contain special pocket or cleft called active site where substrate bind and catalytic activity is performed. Active site has two parts: Binding domain and Catalytic domain. When a substrate binds to an enzyme it induces a change in shape of enzyme called induce fit. When product is released, enzymes resume its initial shape. Active site have optimized environment for converting a substrate into product.

96 Enzymes Function

Enzymes Function

Catalytic activity starts with binding of substrate to enzyme which forms enzyme substrate complex (ES complex). Different types of bonding like covalent bonding, hydrogen bonding etc. are involved in binding. Mostly it is a weak non-covalent interaction between enzyme and substrate. The weak interactions during binding release energy called **binding energy**. Binding energy is one of the major source of energy which lower activation energy. It also make reaction more specific. Enzymes do not change the nature of product. They simply increase reaction rate.

97 Factor affecting enzyme activity

Factors Affecting Enzyme Activity

The conditions that alter the three dimensional structure of enzyme will effect the enzyme activity. Mainly there are two factors: Temperature and pH

a. Temperature

The temperature at which enzyme activity is maximum is called optimum temperature. Proteins and nucleic acids maintain their 3D structure by hydrogen bonding. At low temperature enzyme lose their flexibility and their activity is reduced. High temperature easily disrupt this bonding and enzymes will denature. This will stop their activity. Birds and mammals enzymes work best between 35 to 40°C.

b. pH

Enzymes are more sensitive to pH than temperature. Change in pH causes ionization of enzymes. The pH at which enzyme activity is maximum is called optimum pH. Normally optimum pH of enzymes is according to the environment where they are present.

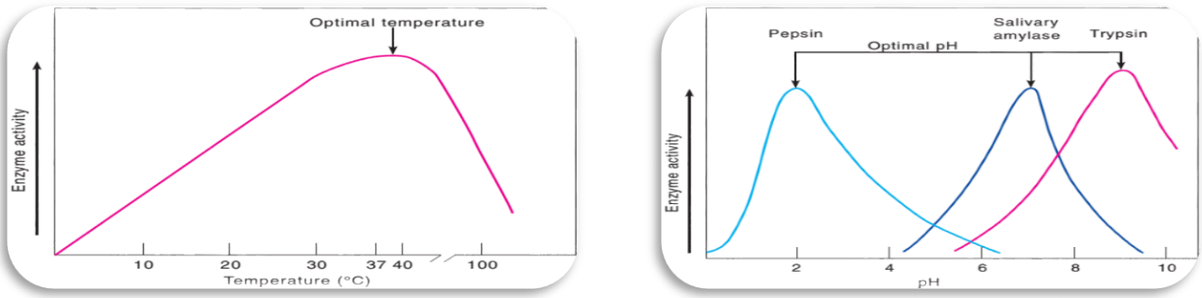


Fig. Effect of temperature and pH on enzyme activity

98-99 Cofactors and coenzymes

Cofactors and Coenzymes

Some enzymes require one or more metal ions for their activity called **cofactors**. e.g., Ca^{2+}, Mg^{2+}, Mn^{2+}, Cu^{2+} and Zn^{2+}. Zn is the cofactor of more than 70 enzymes. Cofactors have following three roles:

a) Activate active site b) Changing in shape of enzyme to bind with substrate

c) Temporarily bonds between enzyme and substrate in ES complex

The non-protein organic molecules require for enzyme activity is called **coenzyme**. e.g., vitamins (niacin, riboflavin); nucleotides (NAD^+, FAD). They normally transfer electrons or temporary carriers of functional group. An enzyme without its cofactor or coenzyme is called **apoenzyme**. Apoenzyme is catalytically inactive. An enzyme with its attached cofactor or coenzyme is called **holoenzyme**. Holoenzyme is catalytically active. When cofactor of coenzyme is tightly bound, it is called a **prosthetic group**.

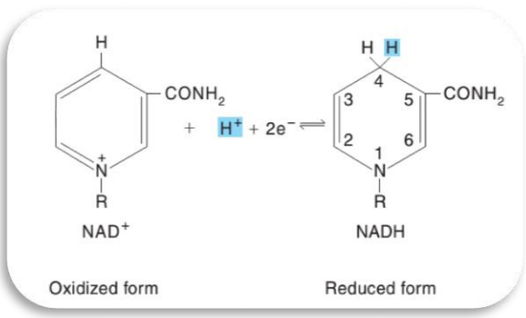


Fig. NAD^+ is one of the most important coenzyme in the cell

100 ATP: the cells energy currency

ATP: The Energy Currency of Cell

ATP is the major currency of the cell. Energy needed for different processes in living organisms is provided in the form of ATP. As ATP is used by all type of cells, it must have appeared in early life history. Ability of ATP to store energy lie in its structure. ATP has three subunits: adenine, ribose and three phosphate groups. Energy is property of entire molecule. The covalent bonds connecting phosphate group are high energy bonds. Hydrolysis of one phosphate group of ATP releases 7.3kcal. It is very unstable molecule and survives for a short time only. So, it can neither stored for long time in the cell nor transported from one cell to another cell.

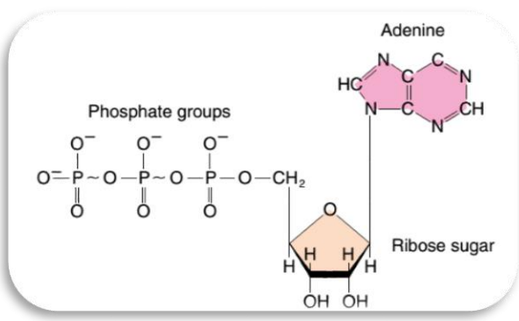


Fig. structure of ATP molecule

101 How cells harvest energy: substrate level phosphorylation

HOW CELLS HARVEST ENERGY?

Animals synthesize ATP from energy stored in C-H bonds of nutrient or stored materials. Animals make ATP in their cells by two method:

a. Substrate level phosphorylation

b. Chemiosmosis

a. Substrate Level Phosphorylation

In substrate level phosphorylation, ATP is synthesized by transferring P_i from a substrate molecule to ADP with the help of enzyme. The phosphate group (P_i) of substrate breaks more easily than ATP. These reactions are strongly exergonic. e.g., ATP synthesis during glycolysis. It is one of the most fundamental in all ATP synthesis reactions.

102 How cells harvest energy: Chemiosmosis

b. Chemiosmosis

Chemiosmosis is the synthesis of ATP by flow of H^+ across the membrane. In animals H^+ flow from outside of inner mitochondrial membrane to inside and ATP is synthesized. A proton gradient is required for chemiosmosis. Concentration of H^+ should be higher outside the membrane. When protons move from higher concentration to lower concentration, ATP is produced.

Proton gradient is generated when high energy electron move through electron transport chain. Hydrogen ions (protons) are pumped outside the inner mitochondrial membrane. When these hydrogen ions comeback, ATP synthetase use their potential energy to synthesize ATP.

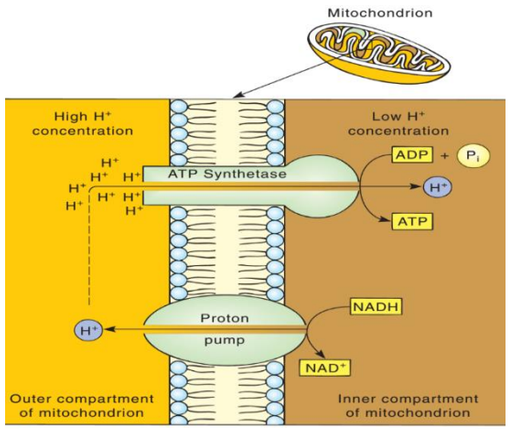


Fig. ATP synthesis by chemiosmosis

103 Metabolism

METABOLISM

All chemical reactions in living organisms are collectively known as metabolism. In a cell metabolism is used to harvest energy from biomolecules, synthesis of biomolecules and remove the toxic wastes. Metabolism is of two main types: catabolism and anabolism.

a. Catabolism

The type of metabolism in which complex molecules are broken down into simple molecules. e.g., digestion and respiration. Catabolism is important for harvesting energy and synthesis other molecules.

b. Anabolism

The type of metabolism in which simple molecules combine to form complex molecules. e.g., synthesis of glycogen, lipids and proteins.

104 Evolutionary Perspective on Glycolysis

GLYCOLYSIS

Evolutionary perspective

The breakdown of glucose into pyruvate through a series of chemical reactions is called glycolysis. Glycolysis is the combination of two Greek words: *glykys,* "sugar" and *lysis,* breakdown. It is the universal pathway of glucose metabolism present in all organisms from bacteria to human. Glycolysis is older than aerobic respiration. By this method harvest energy in the absence of oxygen. During early life history organisms/cells with glycolysis were at advantage than organisms having no glycolysis. Later when animals got more efficient methods to harvest energy, glycolysis was retained as stepping stone. Glycolysis is the sole source of metabolic energy in erythrocytes, renal medulla, brain and sperms. Many anaerobes are entirely dependent on glycolysis.

105-106 Glycolysis: Prepartory Phase

Glycolysis takes place in cytoplasm. There are total 10 steps of glycolysis. First five steps constitute **preparatory phase.** Two ATP are required in this phase. Reactions of step 6 to 10 constitute **payoff phase.** Total 4 ATPs are produced during this phase. Net output of glycolysis is 2ATPs. Two NADH are also yielded during this process.

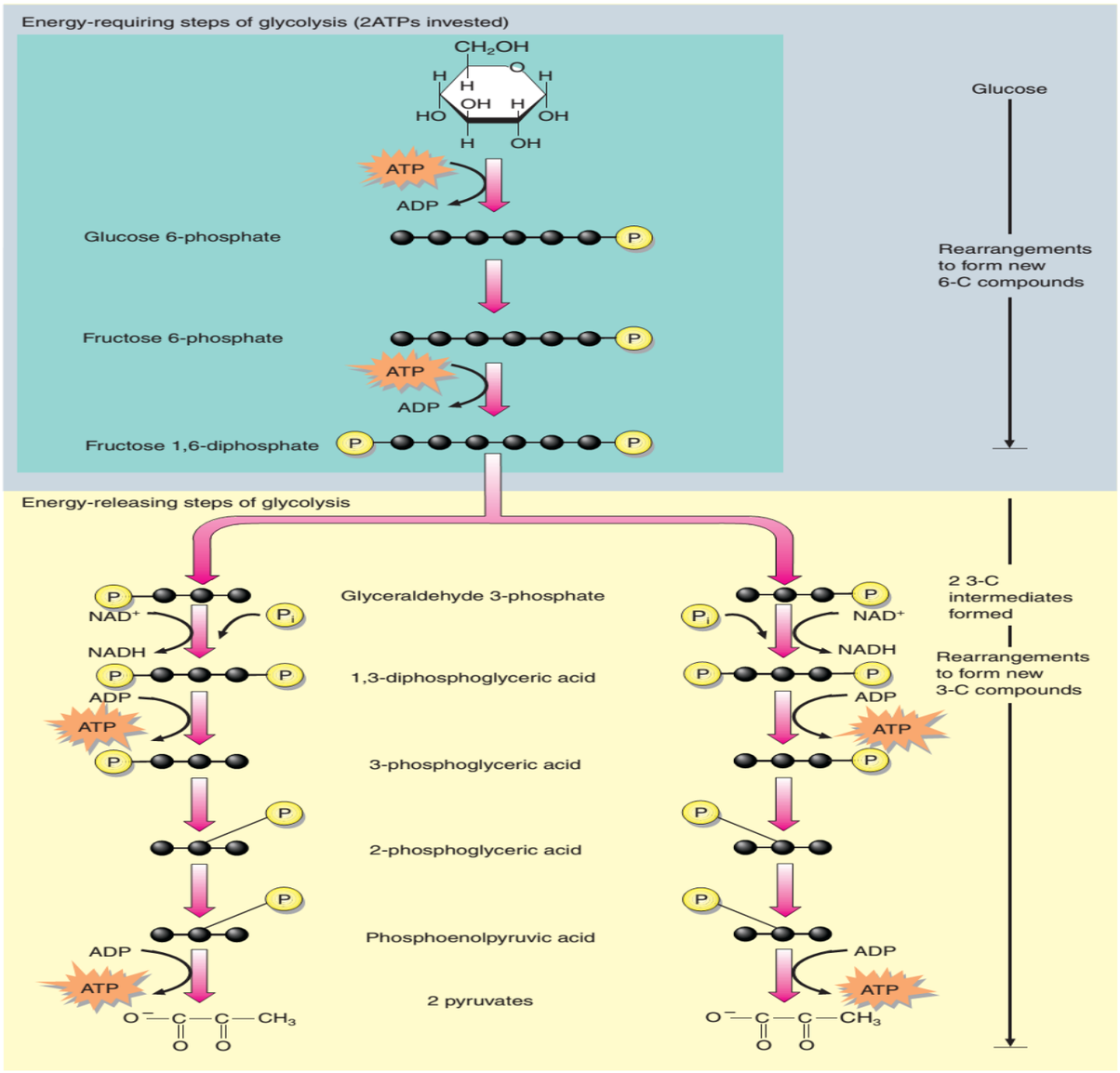


Fig. Reactions of glycolysis. (Ref. Miller and Harley, Zoology)

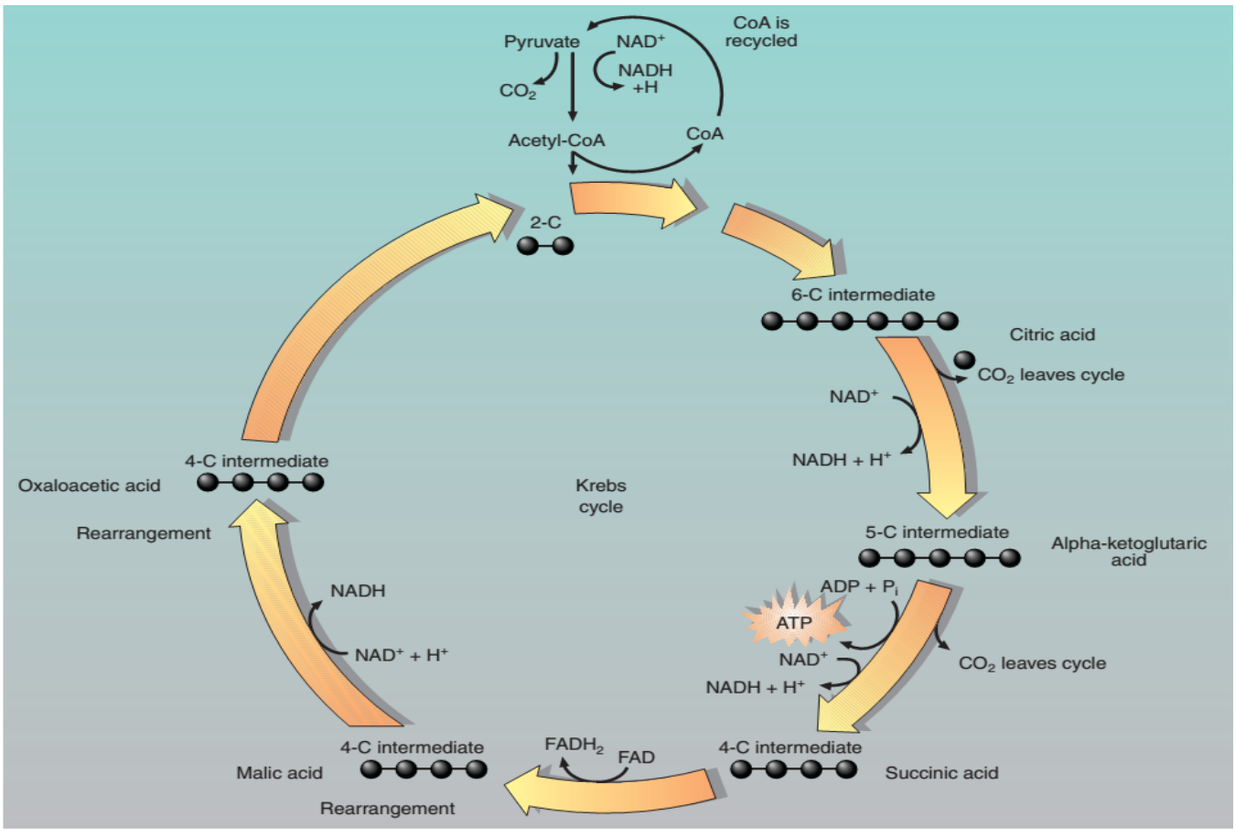
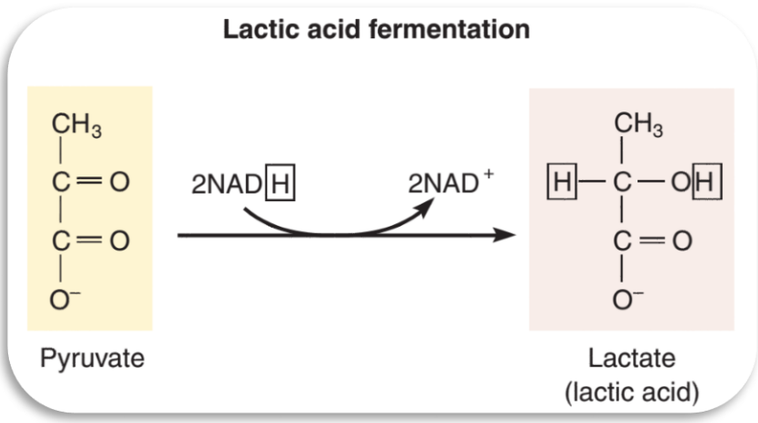


Fig. The Krebs cycle (ref. Miller and Harly, Zoology)

107 Fermentation

Contents

1. Defining Fermentation
2. Types of fermentation
3. Lactic Acid Fermentation
4. Alcoholic Fermentation
5. Why fermentation?



1. **Defining Fermentation**

"It is the process in which pyruvate (product of glycolysis) is metabolized in the absence of oxygen"

$$\text{Pyruvate} + \text{NADH} \rightarrow \text{reduced compound} + \text{NAD}^+$$

2. Types of Fermentation

- Two types
- Lactic acid Fermentation
- Alcoholic Fermentation

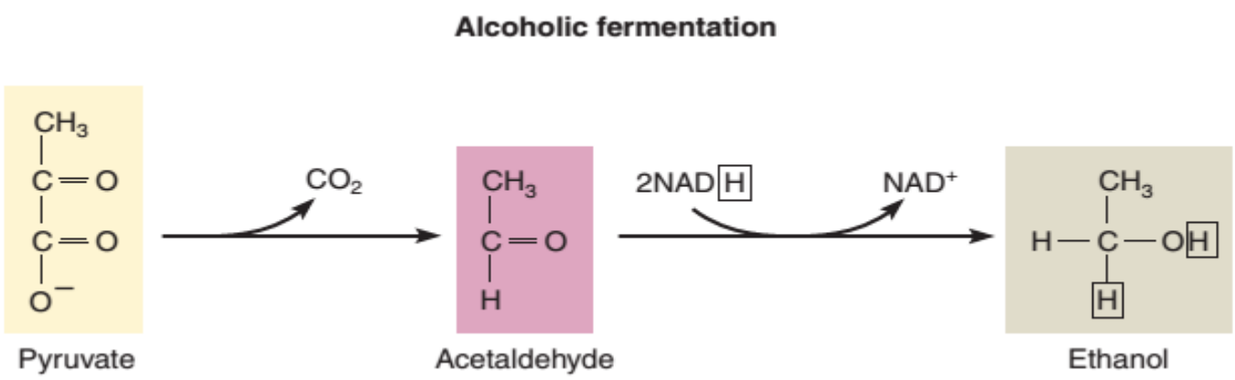
3. Lactic Acid Fermentation

- Pyruvate is converted into lactic acid
- Electrons and hydrogen atoms of NADH are used
- Examples: anaerobic bacteria and animal muscles
- **3. Lactic Acid Fermentation**

4. Alcoholic Fermentation

- Pyruvate is converted into alcohol
- CO_2 is released
- Electrons and hydrogen atoms of NADH are used
- Example: yeast, bacteria

4. Alcoholic Fermentation



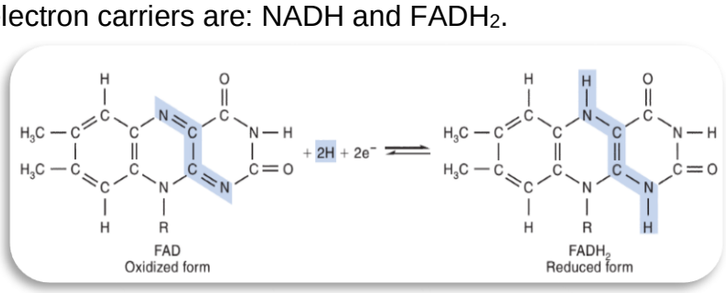
5. Why Fermentation?

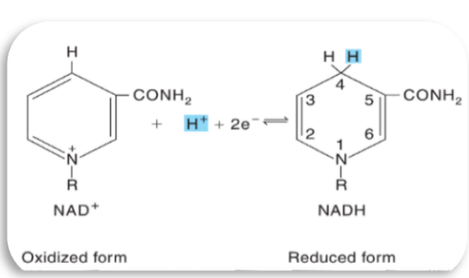
- There is no ATP production in fermentation.
- Fermentation serves only to regenerate NAD^+ from NADH

108 Nature of electron transfer: Structure of NADH and FADH2

Nature of Electron Transfer: Structure of NADH and $FADH_2$

Energy of organic molecules is trapped in chemical bonds. This energy transferred through electrons. In energy transfer the molecule which donates electrons is **oxidized** (loss of e^-) and the molecule which accepts electrons is **reduced** (gain of e^-). Overall it is oxidation-reduction reaction or **redox reaction.** Different types of electron carriers include metal ions, nucleotides and proteins. Two most important electron carriers are: NADH and $FADH_2$.





109-110 Krebs cycle-I-II

KREBS CYCLE

It is a series of chemical reactions in which acetyl-CoA is oxidized to CO_2 yielding ATP and high energy electrons. It was first discovered by Hans Krebs in 1930's. In eukaryotes Krebs cycle takes place in mitochondria. Carbons skeleton of organic molecules is completely oxidized during Krebs cycle. Before entering into Krebs cycle pyruvate is oxidized to acetyl-coA. In each Krebs cycle 3 molecules of NAD^+ are reduced to NADH, one FAD is reduced to $FADH_2$ and one molecule of GTP (ATP equivalent) is also formed.

In first step acetyl-CoA reacts with oxaloacetate to form Citric acid. Citric acid contain three carboxylic (COOH) groups. Due to this reason Krebs cycle is also called as citric acid cycle or Tricarboxylic acid cycle (TCA). Different molecules present at different steps of Krebs cycle (citric acid, alpha ketoglutaric acid, succinic acid, malic acid, and oxaloacetic acid) are called intermediates while molecules which are released during Krebs cycle are byproducts (CO_2, NADH and $FADH_2$).

111-112 Electron transport chain-I-II

ELECTRON TRANSPORT CHAIN

It is an organized set of electron carriers and enzymes with enhanced oxidation reduction reactions which extract the energy of electrons. Different molecules of electron transport chain make redox centers. Electron accepted from NADH and $FADH_2$ are passed through redox center. Energy harvested form these electrons is used to move H^+ across inner mitochondrial membrane to intermembrane space. Electron transport chain is located in inner mitochondrial membrane. Accept electrons from. Major components of electron transport chain are NADH dehydrogenase, flavoproteins (FAD, FMN), ubiquinone (Coenzyme Q), cytochromes and Fe-S proteins. At any given time the electrons will be accepted from one donor: either from $FADH_2$ or NADH. Later these electrons are given to O_2 which combines with H^+ to form water molecules.

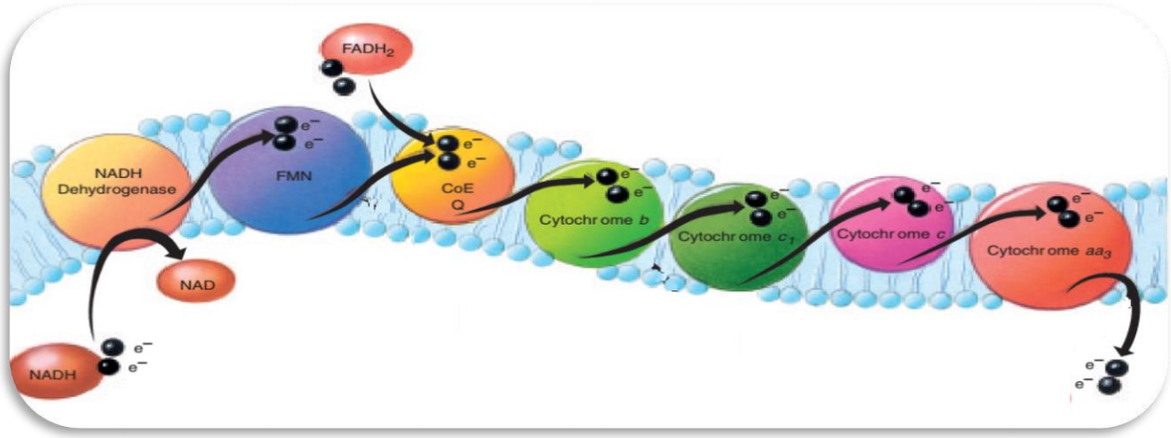


Fig. Arrangement of components of electron transport chain

113 ATP Generation/Synthesis

Contents

1. H^+ Ion Gradient
2. ATP Synthetase
3. Mechanism of ATP Synthesis
4. Fate of e^- and H^+

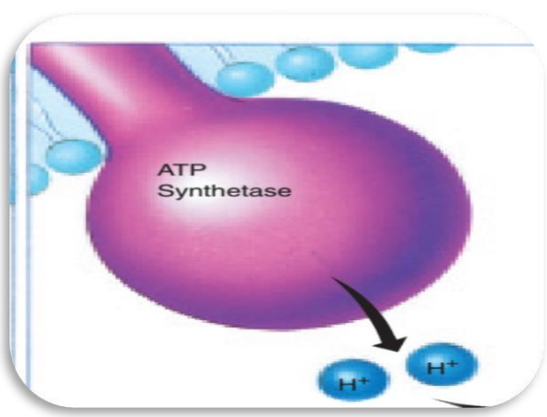
1. **H^+ Gradient**

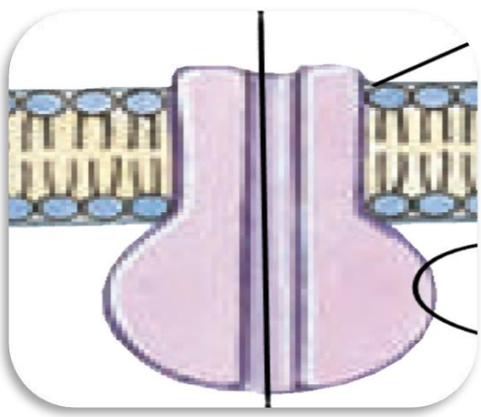
- Electron transport chain pumps H^+ from mitochondria to inter membrane space
- H^+ concentration is higher in inter membrane space than mitochondrial matrix

2. ATP Synthetase

- Inner mitochondrial membrane is selectively permeable
- H^+ can come back to matrix through a membrane protein "ATP Synthetase"

2. ATP Synthetase

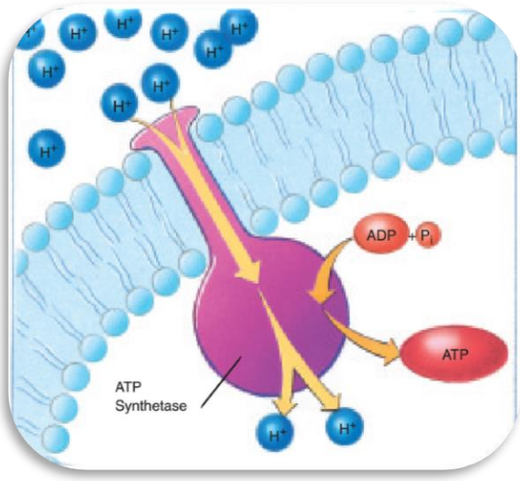




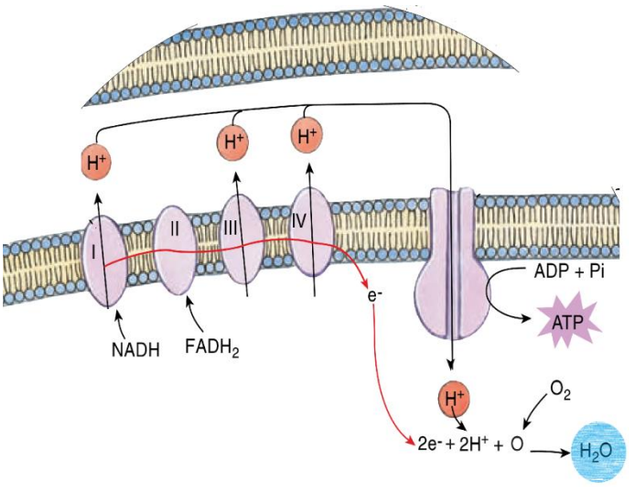
3. Mechanism of ATP Synthesis

- H^+ come back to matrix through ATP synthetase
- Potential energy of H^+ is used for synthesis of ATP

4. Mechanism of ATP Synthesis



3. Fate of e^- and H^+



Final product

114 Energy Yield in the Oxidation of Glucose

Contents

1. ATP Yield During Anaerobic Respiration
2. ATP Yield During Aerobic Respiration

1. **ATP Yield During Anaerobic Respiration**

- In the absence of oxygen
- Glucose is not completely oxidized
- Anaerobic respiration= Glycolysis+ fermentation
- Energy comes from glycolysis only

1. **ATP Yield During Anaerobic Respiration**

TABLE 32.1 APPROXIMATE ATP YIELD FROM THE COMPLETE ANAEROBIC OXIDATION OF ONE MOLECULE OF GLUCOSE

PROCESS	ATP PRODUCED	NADH, $FADH_2$ PRODUCED	ATP USED
Glycolysis	4 ATP	2 NADH	−2 ATP

2. ATP Yield During Aerobic Respiration

- In the presence of oxygen
- Glucose is completely oxidized into CO_2 and H_2O
- Aerobic respiration= Glycolysis+ Krebs cycle+ electron transport chain

2. ATP Yield During Aerobic Respiration

PROCESS	ATP PRODUCED	NADH, $FADH_2$ PRODUCED	ATP USED
Glycolysis	4 ATP	2 NADH	−2 ATP
Entrance of NADH to Krebs cycle		2 NADH	−2 ATP
Krebs cycle	2 ATP	6 NADH, 2 $FADH_2$	
Electron transport chain:			
3 ATP for each of 2 NADH generated via glycolysis	6 ATP		
3 ATP for each of 8 NADH generated via Krebs cycle and entrance to the cycle	24 ATP		
2 ATP for each of 2 $FADH_2$ generated via Krebs cycle	4 ATP		
Total	40 ATP		−4 ATP
Net ATP	36 ATP		

115-116 Metabolism of fats-I

METABOLISM OF FATS

Fats are important part of our diet. They are the basic part of cellular membranes and important source of energy to our body. Fats provide more than double amount of energy as compared to carbohydrates and proteins. The recommended amount of energy for our daily life is 30%. Some organs such as liver, heart and skeletal muscles meet half of energy requirement from fats while fats are sole source of energy in hibernating animals. Fats are actually triacylglycerides in which three fatty acids are attached with one glycerol molecule.

Fat digestion occurs in small intestine. Liver bile Emulsify the fat droplets while lipase completely digested the fats. Fats are absorbed in small intestine and converted into triglycerides again. Fats are transported in the body as chylomicron. Chylomicron are formed by the combination of fats, cholesterol and proteins. Fats are stored in adipose tissues and skeletal muscles.

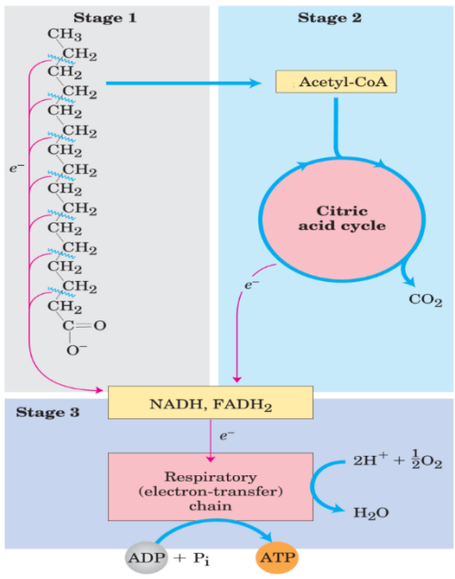


Fig. Fatty acids are oxidized to acetyl CoA, NADH and $FADH_2$. Acetyl CoA enters Krebs cycle while NADH and $FADH_2$ enter into electron transport

When required fats are hydrolyzed by enzymes into glycerol and fatty acids. Glycerol is converted to glyceraldehyde-3-phosphate which enters into glycolysis. Fatty acids oxidation mainly occurs in mitochondria while small amount is also oxidized in endoplasmic reticulum and peroxisome. Breakdown of fatty acids yields acetyl CoA, NADH and $FADH_2$. Acetyl CoA in Krebs cycle forms NADH and $FADH_2$ which enter into electron transport chain for ATP production.

One gram of fats give 2.5 times more energy than 1g of carbohydrates and protein. This is because they contain much more C-H bonds as compared to carbohydrates and lipids.

117-118 Metabolism of proteins-I

METABOLISM OF PROTEINS

Proteins are one the most important biomolecules for living organisms. They have both structural and functional role in our body. Important protein molecules include muscle fibers, bones, enzymes, hormones, antibodies, membrane transporters, lens proteins of eyes, hair, nails, feathers and spider web.

Proteins are digested in stomach and small intestine. Amino acids are final product of protein digestion which are absorbed in small intestine. They reach to cells through blood circulation and are used for protein synthesis. Excessive amino acids are degraded mainly in liver and skeletal muscles. Amino acid breakdown also occurs during starvation.

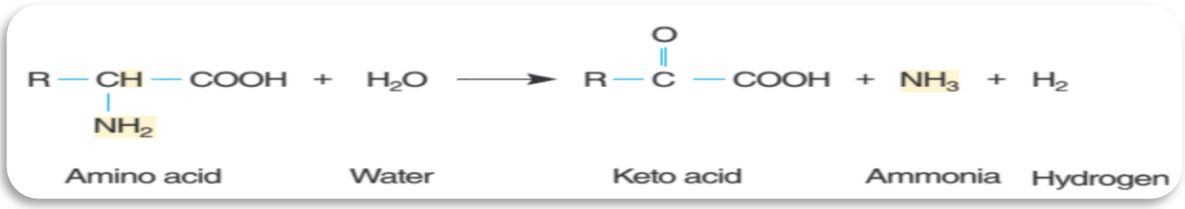


Fig. Amino acids catabolism

Amino acids break down to keto acid after deamination. Keto acids are funneled to Krebs cycle to yield ATP. One gram of proteins give approximately 4kcal energy which is approximately equal to glucose. Ammonia is formed during deamination of amino acid. In aquatic organisms ammonia is released from moist surfaces. For release of ammonia large amount of water is required. In land animals ammonia is converted to urea that is excreted through urinary system while animals with acute shortage of water, release uric acid.

119-120 Control of metabolism-I

CONTROL OF METABOLISM

Cells are very efficient and do not waste energy. If some molecule is in excess, its synthesis is switched off. The most common method for control of metabolism. In this method the product inhibits enzyme that catalyzes the key step. In anabolism cells expends energy for biomolecules synthesis. Anabolism is mainly controlled by regulating the enzyme activity at strategic points. For example, phosphofructokinase has receptor sites for ATP, ADP, AMP, citric acids. By controlling the activity of this enzyme cell regulates its entire metabolic process. ATP and citric acid inhibit phosphofructokinase in glycolysis. When a cell has sufficient ATP, its glycolysis is stopped by inhibition of phosphofructokinase. When ATP level is decreased in a cell and the level of ADP and AMP increase. This activates phosphofructokinase, as a result glycolysis is stimulated. When cell has enough citric acids produced in Krebs cycle. Citric acid in cytoplasm inhibits phosphofructokinase. At low concentration of citric acid glycolysis accelerates.

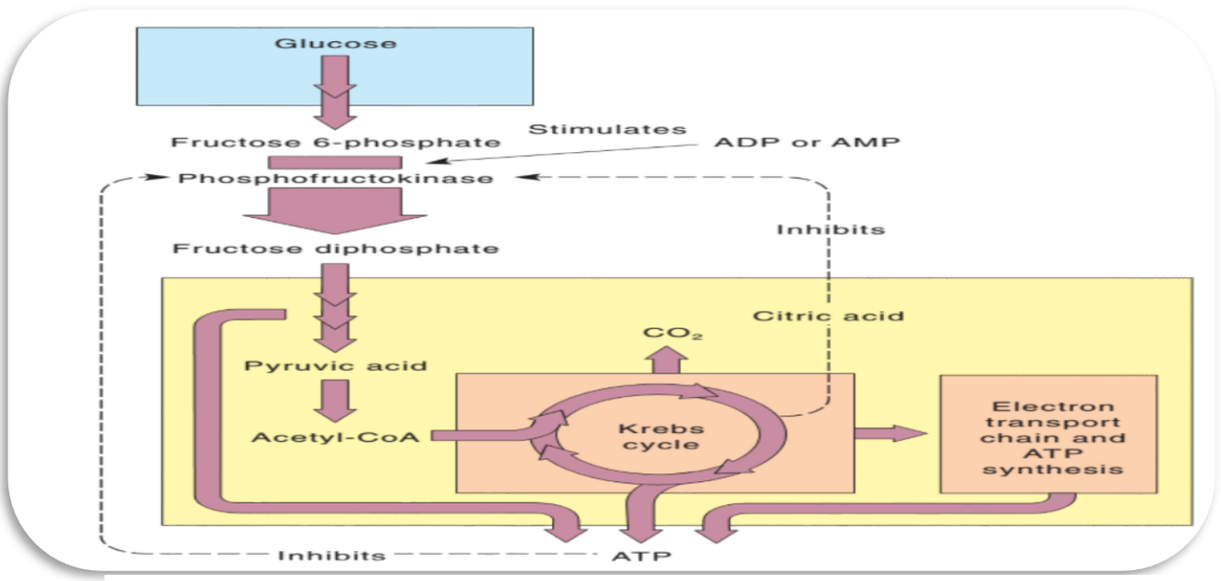


Fig. Flowchart for control of cellular respiration

121 Metabolic pool

METABOLIC POOL

All the substances of metabolism constitute metabolic pool. It is actually reservoir of molecules upon which enzymes act. These include substrates, products and reaction intermediates. All catabolic and anabolic reaction in cell use metabolic pool.

An open system is in which compound may enter or leave the biochemical pathway at various steps. Almost all the biochemical pathways in living organisms are open system. Metabolic intermediates of glycolysis and Krebs cycle may be withdrawn or put into by other metabolic pathways.

122 Reproduction and Its Importance

REPRODUCTION AND ITS IMPORTANCE

REPRODUCTION

With the passage of time animals get aged, new animals should take their place. Reproduction is the process of producing new individuals. It is the basic feature of all forms of life. It is essential for existence of species and also for continuity of life on earth. Capability to reproduce is called reproductive potential. All the animals have great reproductive potential. If resource are unlimited a single species of fishes may fill the all oceans.

Animals reproduce by two basic means: Asexual reproduction and sexual reproduction

123 Asexual Reproduction

A. Asexual Reproduction:

It is the type of reproduction in which offspring are produced without fertilization of gametes. In asexual reproduction the basic method of cell division is mitosis which involves no genetic recombination. So, the animals produced by asexual reproduction are identical to their parents and siblings. Asexual reproduction is common the common method of reproduction in protozoa (Amoeba, paramecium and euglena) and lower invertebrates (hydra, jellyfish, and flatworms). Asexual reproduction is correlated with ability to regenerate.

Early organisms used to reproduce by asexual method. This was the only method of reproduction in first 2 billion years of life. During this time life evolved slowly because of less genetic diversity.

Followings are the common method of asexual reproduction in animals:

a. Fission
b. Budding
c. Fragmentation
d. Parthenogenesis

124 Methods of asexual reproduction: 1/4: Fission

a. Fission

Fission (Lenten word *fissio, the act of splitting*) is a type of asexual reproduction in which one organisms splits into two or more organisms. There are two basic types of fission: Binary fission and multiple fission. The process in which one organism divides into two organisms is called **binary fission**. Both are approximately equal size. Binary fission may be asymmetrical, transverse and longitudinal. Asymmetric organisms like amoeba may divide into two from any side, such fission is called asymmetric fission. When a line of cleave transversely divide an organism into two is called transverse fission. e.g., binary fission in paramecium. When an organism is divided longitudinally into two daughter organisms, this is called longitudinal fission. e.g., binary fission in planaria and euglena.

Multiple fission is a number of successive divisions of nucleus followed by cytoplasmic division in which one cell divides into many cells. e.g., amoeba, tapeworm. Amoeba form cyst under unfavorable conditions and form new amoebae by multiple fission.

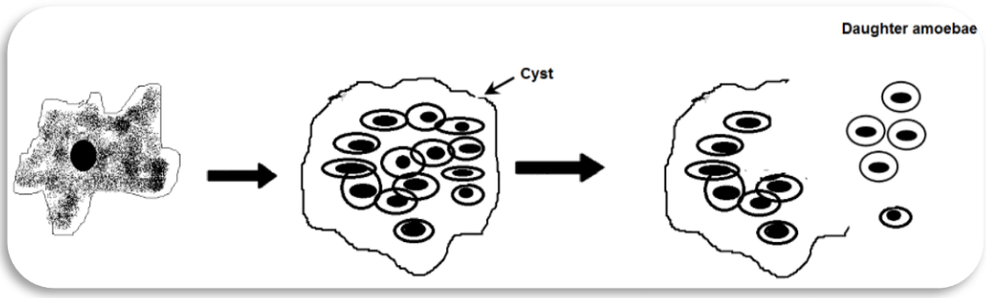


Fig. Multiple fission in amoeba.

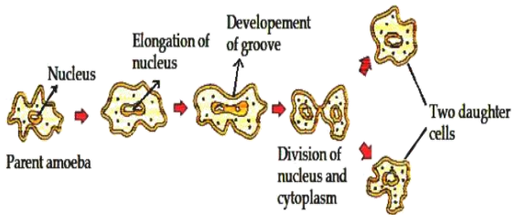


Fig. asymmetrical fission in amoeba



Fig. Longitudinal fission in planarian

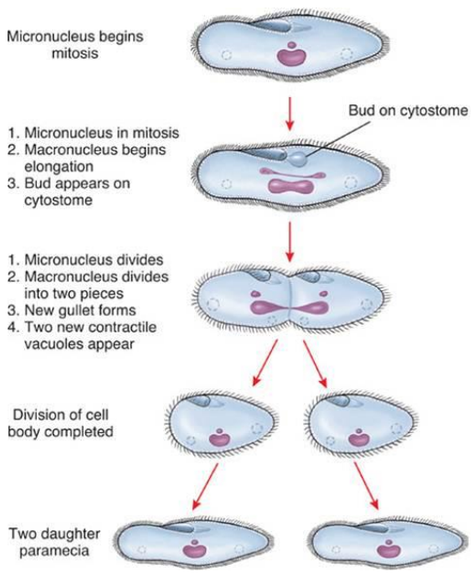


Fig. Transverse fission in paramecium

125 Methods of Asexual Reproduction: 2/4: budding

b. Budding

Asexual reproduction in which an outgrowth appears on the body of animal which grows into complete organism. e.g., hydra, obelia and sponges. In budding, a group of cells rapidly divides and form a small outgrowth called bud. This bud increases rapidly and form a complete organism. The bud may get detach from the parent or sometimes it may remain attached with the parental organism and form a closely associated group of individuals called a **colony.** Members of a colony belong to same species.In sponges internal buds are produces which are called as **gemmules.** Gemmule is a collection of cells surrounded by body wall. When parent dies, each gemmule gives rise to a new individual.

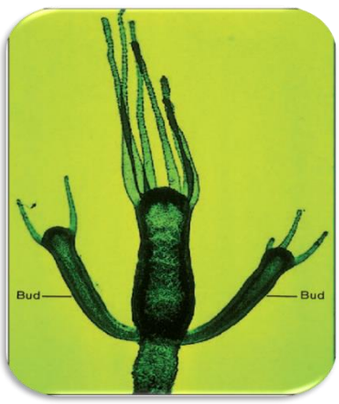


Fig. Budding in hydra

126 Methods of Asexual Reproduction: 3/4: Fragmentation

b. Fragmentation

Fragmentation is the process of asexual reproduction in which an animal breaks into two or more pieces and each piece is capable of growing into a new organism. Fragmentation is common in different groups of animals including cnidarians, Platyhelminthes and echinoderms

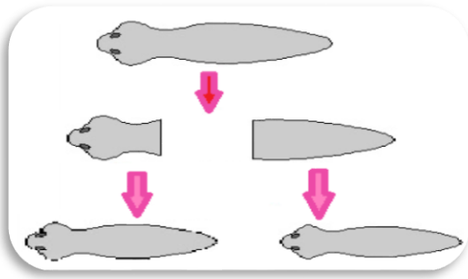


Fig. Fragmentation in planaria

127-128 Methods of Asexual Reproduction: 4/4: Parthenogenesis-I

b. Parthenogenesis



Fig. New Mexico Whiptail lizard population has only females

The process of production of an organism from an unfertilized egg. There are many patterns of parthenogenesis. Parthenogenesis is a spontaneous activation of a mature egg, followed by normal egg divisions and subsequent embryonic development. In some species, the only way of reproduction is parthenogenesis. In others, those eggs which could not fertilize undergo parthenogenesis to produce new organism. In parthenogenesis genetic material is only shared by the female. Parthenogenesis is very common in wasps, bees and ants. It is essential for social organization of these insets. Parthenogenesis have been observed in some lizards and snakes. New Mexico Whiptail lizard are exclusively females. Their population contain no male members. Eggs of this lizard are spontaneously activated and produce a female young one. Parthenogenesis has two common forms: Meiotic parthenogenesis and ameiotic parthenogenesis.

In **ameiotic Parthenogenesis** eggs are diploid. They are produced without meiosis. Young ones are the clones of their parents. This method is common in flatworm, rotifers, crustaceans and some insects. In **meiotic Parthenogenesis** eggs are produced by the meiosis process. Haploid egg starts development and diploid number restored by chromosomal duplication. However, some insects remain haploid and no chromosomal duplication occurs. In some cases, egg may be activated by sperm to start parthenogenesis. Here sperm will not fertilize egg, just activates. In others eggs are activated spontaneously.

Parthenogenesis is advantageous:

1) When population consists of only females.

2) When environment is very stable.

3) For social organization of colonies of bees, wasps, ants (for males)

The disadvantage of parthenogenesis are:

1) Less genetic variability.

2) Little chances for organisms to shift gene combination in changing environment.

129 Advantages of Asexual Reproduction

Advantages of Asexual Reproduction

1) Species survival: Asexual reproduction is specially the beneficial to those species have one sex only.
2) Minimal energy requirements: Only one sex is involved so less amount of energy is required.
3) Positive genetic combinations passed on to next generation: Because there is no genetic recombination, the good traits acquired of an organism are transmitted to next generation

4) Rapid Increase in Population: Under stable conditions animals reproduce asexually causing a rapid increase in population

130 Disadvantages of Asexual Reproduction

Disadvantages of Asexual Reproduction

1) Low genetic variability: No genetic recombination. Very little variations and population is genetically same.
2) Susceptibility against diseases: As population is genetically same, so equally susceptible to disease or changing environmental conditions. This may result in the death of whole population at once.
3) All mutations pass on to next generation: Most of the mutations are detrimental. In Asexual reproduction all mutations are passed on to next generation

131 Sexual reproduction

B. Sexual Reproduction

It the types of reproduction in which new offspring are produced by the fusion of gametes. In sexual reproduction normally two sexes are involved. Offspring get one set of chromosomes from male and other from females. This results in new genetic combinations. Gametes produced by male are called sperms while gametes produced by female are called eggs. Sperms are produced in testes while eggs are produced in ovaries. Both gametes are haploid and produced by the process of meiosis. The process of sperms production is called **spermatogenesis** while the process of eggs production is called **oogenesis.**

132 Internal and External Fertilization

Internal and external fertilization

The process of fusion of gametes is called fertilization. The process in which fusion of gametes occurs outside the body of animals is called external fertilization. Normally, it occurs in water. Female releases many eggs in the water, the process is called broadcast spawning. Male spray sperms on the released eggs. So, fertilization occur in water and the development is also outside the body of female i.e., external. In case of **Internal Fertilization,** fusion of gametes occurs inside the body of animals. In mating process male introduces sperms into female body, sperms are motile they approach to female gametes within the body of female. Many sperms are packed into a ball like structure called **spermatophore**. Which a accepted by female. Sperms are protected in spermatophore and there is minimal loss of sperms.

Species with external fertilization produced large number of gametes while few gametes are produced in case of internal fertilization.

133-134 Hermaphroditism

Hermaphroditism

It is the condition in which both male and female sex organs are present in the same animal. E.g., earthworm, mangrove killifish and hamlet fish. Hermaphroditism is more beneficial to sessile animals. Hermaphrodites are also called monoecious (Greek. *Monos,* single and *oikos,* house). Monoecious means one room for two sexes. Some undergo self-fertilization, but mostly cross fertilization is the rule. Two animals come close to each other, both donate and receive sperms. Sequential hermaphroditism is a special type of hermaphrodites.

Sequential hermaphrodites have one sex in first phase of life and opposite sex in other phase of life. Means if animal is male in first phase of life, it will be female in second half of life. Those sequential hermaphrodites which are female in first phase of life and male in second phase of life called **protogynous** (femle first) e.g., wrasses. Those animals which are male in first phase of life and female in second phase of life are called **protandrous** e.g., clone fish.

135 Bisexual Reproduction

Bisexual reproduction

It is the type of reproduction in which two separate individuals are involved. They are also called **dioecious** (Gr. *Di,* two and *oikos*, room). Most of the animals have bisexual reproduction method. One benefit of this is that the offspring have new gene combination and new genotypes appear. Sexual dimorphism is the condition in which both male and female are phenotypically different. The difference is mainly in secondary sex characters, size, colour and weight. e.g., many of the males in birds are brightly coloured and larger in size.

136 advantages and disadvantages of sexual reproduction

Sexual reproduction is consider as more advantageous than asexual reproduction. Important benefits of sexual reproduction are:

1) Genetic variability: Genetic recombination occurs in sexual reproduction which give rise to new combination of traits. This may help in changing environment or in case of disease.
2) Foundation for evolution: In sexual reproduction, variation appear in offspring which provide the foundation of evolution
3) Elimination of Deleterious Traits: Genetic recombination helps in elimination of deleterious traits.

Sexual reproduction also has certain disadvantages, some of which are as follows:

1) Asexual animals: sexual reproduction is not feasible for animal which rely on asexual means only, especially population with one sex only.
2) Adaptive genes combination dismantle: In sexual reproduction new genetic combination arise. This will upset the already established adaptive gene combination.
3) Wastage of energy: in sexual reproduction a large number of gametes are formed. Many of which are not fertilized. This will cause a significant loss of energy.

137 reproductive strategies in amphibians

Reproductive Strategies in Amphibians

Class Amphibia has three orders: order caudate, order gymnophiona and order anura. Oder caudate and gymnophiona has internal fertilization while order anura has external fertilization. **Eggs** have no hard covering. Eggs may be given as single, clumps or strings. Majority of member of gymophiona retain eggs within body while members of order anura lay eggs in water, moist soil and tree trunks. **Breeding** in temperate region is linked to temperature while in tropical region, it is linked to rainy season. **Courtship behviour** helps to locate breeding sites, identify potential mates, prepare them for reproduction and ensures egg fertilization. Animals use olfactory, visual, vocalization and tactile cues for courtship behavior. **Development** include aquatic larva or internal in case of gymnophiona.

138 reproductive strategies in reptiles, birds

Reproductive Strategies in Reptiles and Birds

Reptiles and birds are dioecious. They have internal fertilization. Fertilization in the upper portion of oviduct. Sperms may be stored in seminal vesicles e.g., up to 4years in some turtle and 6years in some snakes. **Eggs** range from soft to hard shells. Eggs are desiccation resistant and contain enough amount of yolk. Reptiles and birds eggs are called amniotic eggs. Amniotic eggs have three extraembryonic membranes: amnion, chorion and allantois. **Breeding behavior** in reptile include copulation in winter during hibernacula and fertilization in spring while birds have breeding activity in spring. **Courtship Behviour** in reptiles is very complex and function for sex recognition. Male actively seek out female. Birds show territorial behavior. Males attract females by sounds (wood pecker), dancing (cranes, peacocks). More than 90% monogamous (pair with single female). Polygamy occurs in species where young ones require less parental care. **Development** in reptiles and birds is mostly external (oviparity), fertilized eggs remain unattended while in some cases eggs are cared for humidity. Crocodile is a special case where young ones get post hatching care. In birds there is external development. Eggs are incubated and protected from predators. Young ones get extensive parental care.

139 Reproductive strategies in mammals

Reproductive Strategies in Mammals

Mammals are also dioecious. Eggs contain little amount of yolk except egg laying mammals. They have internal fertilization. Fertilization in the upper portion of oviduct. Some bats have delayed fertilization.

Mammals also have complex **courtship behavior**. Male actively seek out female. Males show territorial behavior. Vocalization is very common among mammals. Mammals show **reproductive cycle.** They have definite time for egg maturation. Females undergo estrus cycle during which it is behaviorally and reproductively receptive to males. Some females undergo one estrus cycle during one year they are monoestrus, other have many estrus cycles and are known as polyestrus. **Development** in mammals is internal (viviparity) except montremes (oviparous). In marsupial mammals fetus get nutrition from uterine milk, placenta is primitive and gestation period is short. In eutherian mammals embryo is implanted deeply, fetus get nutrition through placenta, gestation period 20 days to 19 months.

140 Examples of reproduction in Fishes and Amphibians

Examples of Reproduction in Fishes and Amphibians

Thousands of eggs are fertilized in bony fishes. Many of them get fungal and bacterial infection. Many others are contaminated with silt. Still others eaten by predators. Bony fishes have little amount of yolk which can support development only for short time. That's why fishes have rapid development and hatchlings actively feed and they achieve maturity within short time. In some fishes embryos are cared properly. Sea horse and pipefish males carry embryo in their ventral pouches.

Amphibians mostly have external fertilization in water in few internal. Eggs are cared for aeration and infected eggs are removed from clutches. *Pipa* female carry eggs on back. *Rheobatrachus* females carry egg and tadpole in stomach. Amphibians contain little amount of yolk and longer time for development. That's why they have larval stage to complete development. During development they undergo metamorphosis.

141 Examples of reproduction in Reptiles and Birds

Examples of Reproduction in Reptiles and Birds

Reptiles are true terrestrial vertebrates. They have shelled eggs with extraembryonic membranes. Chorion and amnion give protection and provide aqueous environment. Allantois plays its role in gaseous exchange and to store excretory products. Reptiles contain large amount of yolk. Complete development takes place within shelled egg and there is no larval stage. Some snakes are ovoviviparous because they keep eggs within body and young ones are hatched out.

All birds except water fowl lack penis. They deposit semen against cloaca and fertilization is internal. Fertilization before shell formation. Hard shelled eggs are incubated by parents. There is more parental care and feeding than amphibians, reptiles and mammals.

142 Examples of reproduction in Mammals

Examples of Reproduction in Mammals

Among mammals monotremes (duckbilled platypus, spiny anteater) lay eggs. Duckbilled platypus lay eggs in nest/burrows while spiny anteater incubate eggs in ventral pouch. Marsupial mammals like koala and kangaroo keep the young one in ventral pouch (marsupium). They have short gestation period (8- 40 days) while feeding from mammary glands (suckling) is for 60 to 120 days. Most of the mammals are placental mammals (deer, lion, monkeys etc.). They have internal development and long gestation period. After birth young one is nourished through mammary glands. The longest gestation period is seen in elephants which lasts for 22 months.

143 Reproductive Patterns in Animals

Reproductive Patterns in Animals

All invertebrates and vertebrates have one of the three reproductive patterns.

a. Oviparous

These are the animals which lay eggs and development takes place outside the body of animals. Fertilization may be internal or external. Majority of invertebrates and many vertebrates are oviparous.

b. Ovoviviparous

These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. Both fertilization and development is internal. Many members of annelids, brachiopods, gastropods, fishes and reptiles.

c. Viviparous

These are the animals in which eggs develop into the body and the young one gets its nourishment from mother. Both fertilization and development is internal. Scorpions, most mammals, elasmobranch fishes, few amphibians and reptiles are viviparous.

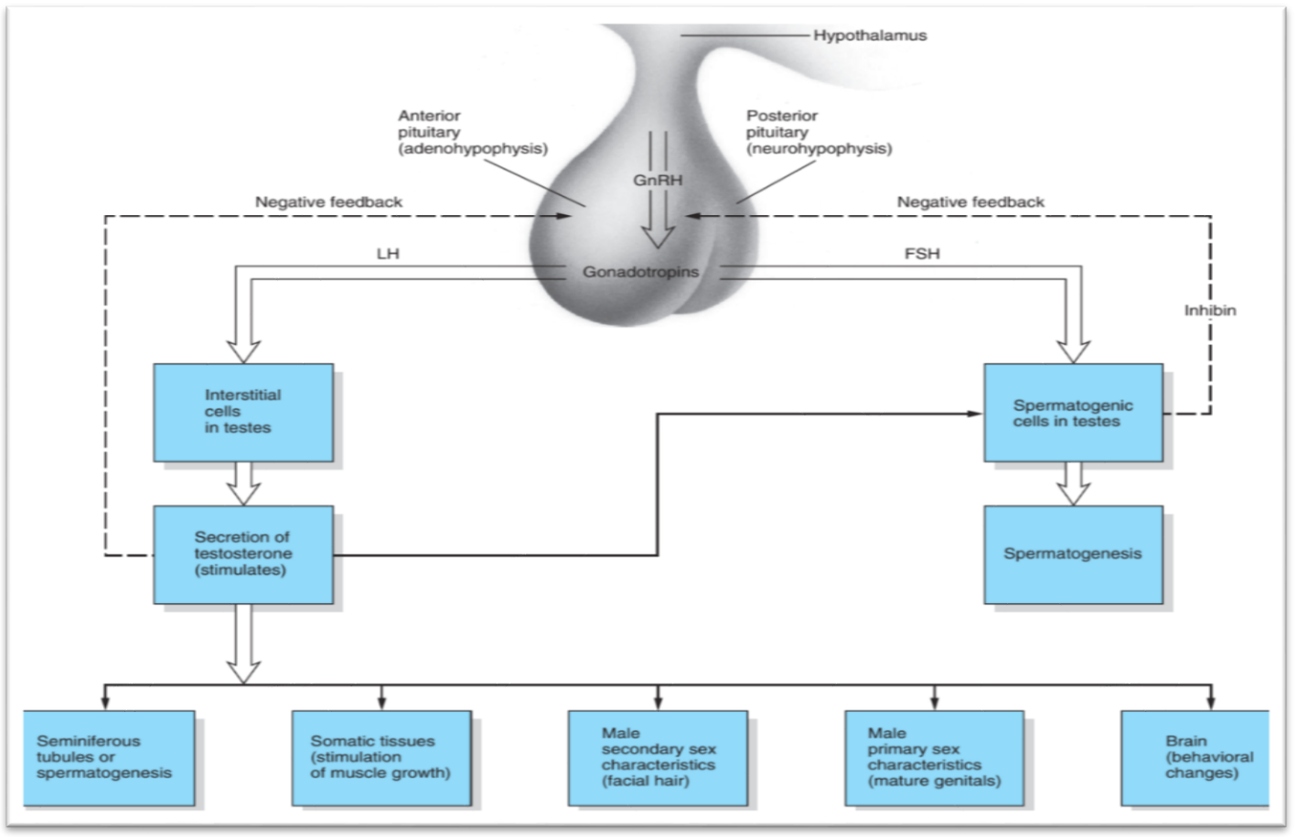


Fig. Hormonal control of male reproductive function

144 Invertebrate Reproductive system

INVERTEBRATE REPRODUCTIVE SYSTEM

Invertebrates may have a simple reproductive structure which release gametes into water for external fertilization. Other may have complex structures for internal fertilization. Some invertebrates like polychaetes have no permanent reproductive organs. Cells lining body cavity produce gametes which are released through coelomic or nephridial duct. Sometimes body wall may rupture. However most of the vertebrates have male and female reproductive organs.

Male reproductive system is meant for production and safe transfer of sperms. Male reproductive system has testes for sperm production. Sperm pass through sperm duct/vas deferens to seminal vesicle for storage. Then sperms are passed out through ejaculatory duct. Accessory glands add seminal fluid.

Female reproductive system has ovaries where egg or ova is produced. Mature eggs pass through oviduct to a common genital chamber, from where they enter into female bursa (vagina). Then eggs are stored in seminal receptacles. In some species, sperms from one mating are stored and used all the life.

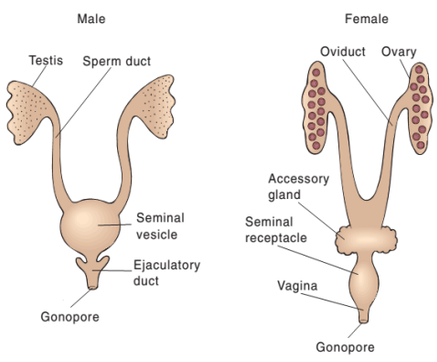


Fig. generalized structure of invertebrate reproductive system

145 Human Male Reproductive System

Human Male Reproductive System

Male reproductive system have two basic function: Production of sperms and Transfer of sperm to female reproductive tract. Important components of male reproductive system include scrotum, testes, rete testis, epididymis, duct deferens and ampulla. Seminal Vesicle. Male reproductive system consists of a pair of testes enclosed in a sac like structure called scrotum. Each testes has a large number of seminiferous tubules. These are the sites for sperm production. Seminiferous tubule join a network of tubules called rete testis. After that there is a dilated tubule called epididymis. Epididymis has three functions: 1) storage and maturation of sperms. 2) Smooth muscles that help to propel sperm during ejaculation and 3) Transfer sperms to duct deferens. Duct deferens transfer comes out of scrotum and passes through the lower part of abdominal wall via inguinal canal. Sometime this point is weakened and intestine part may protrude out. This condition is called inguinal hernia. Around urinary bladder duct deferens enlarges to form ampulla. Next to ampulla is called ejaculatory duct. After combining with urinary tube it moves in penis as urethra.

There are different glands present in reproductive system. Important glands are seminal vesicles, prostate gland and bulbourethral gland.

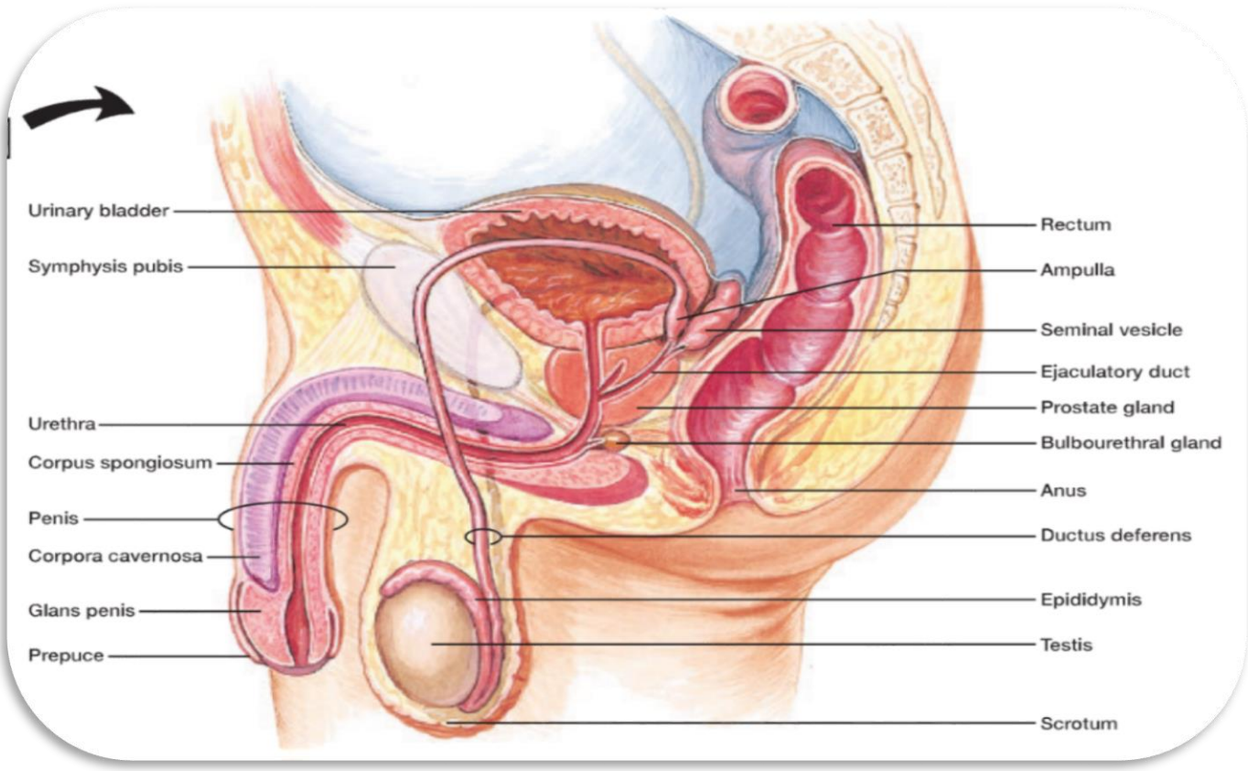


Fig. The human male reproductive system

146 Production of Sperms in Human

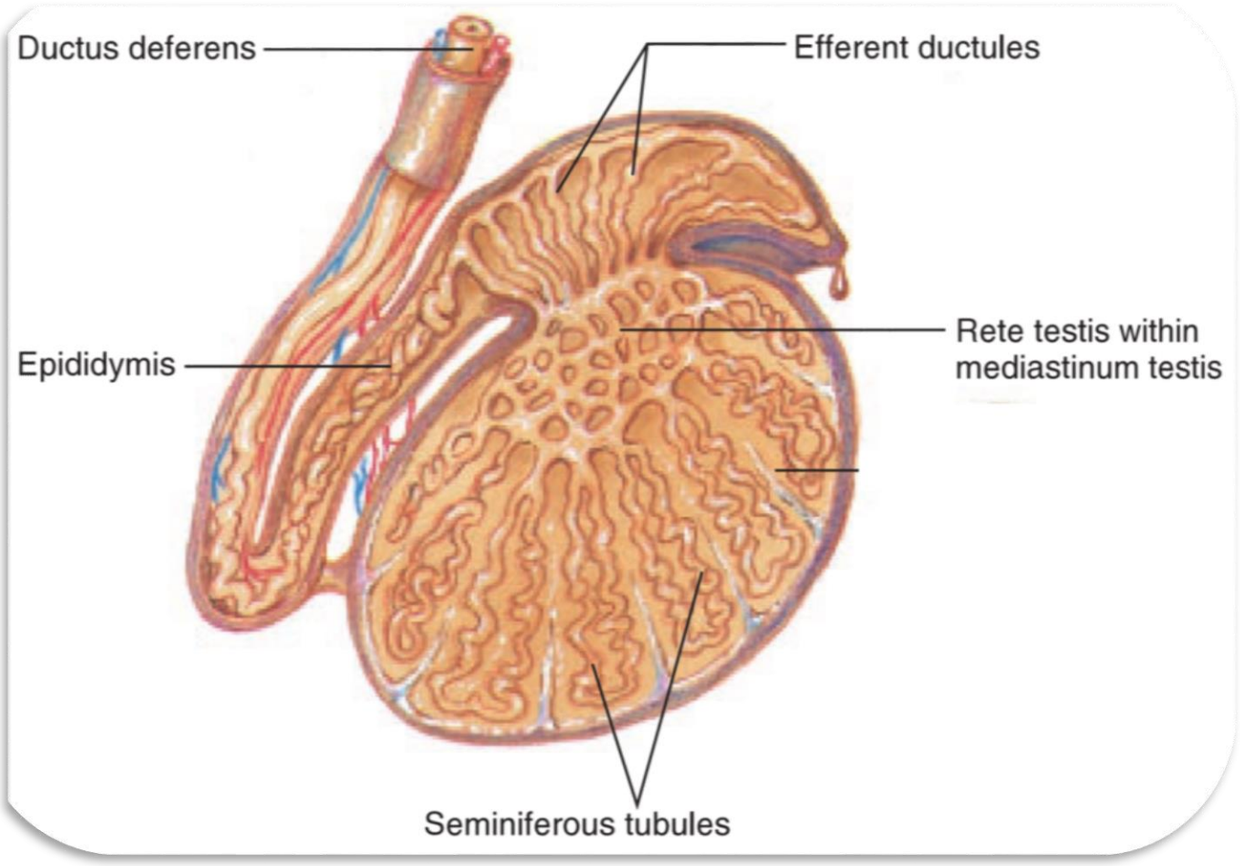
Contents

1. Testes
2. Seminiferous Tubules
3. Spermatogenesis
4. Structure of Sperm

1. Testes

- A pair of testes
- Scrotum
- 34ºC for sperm production and survival

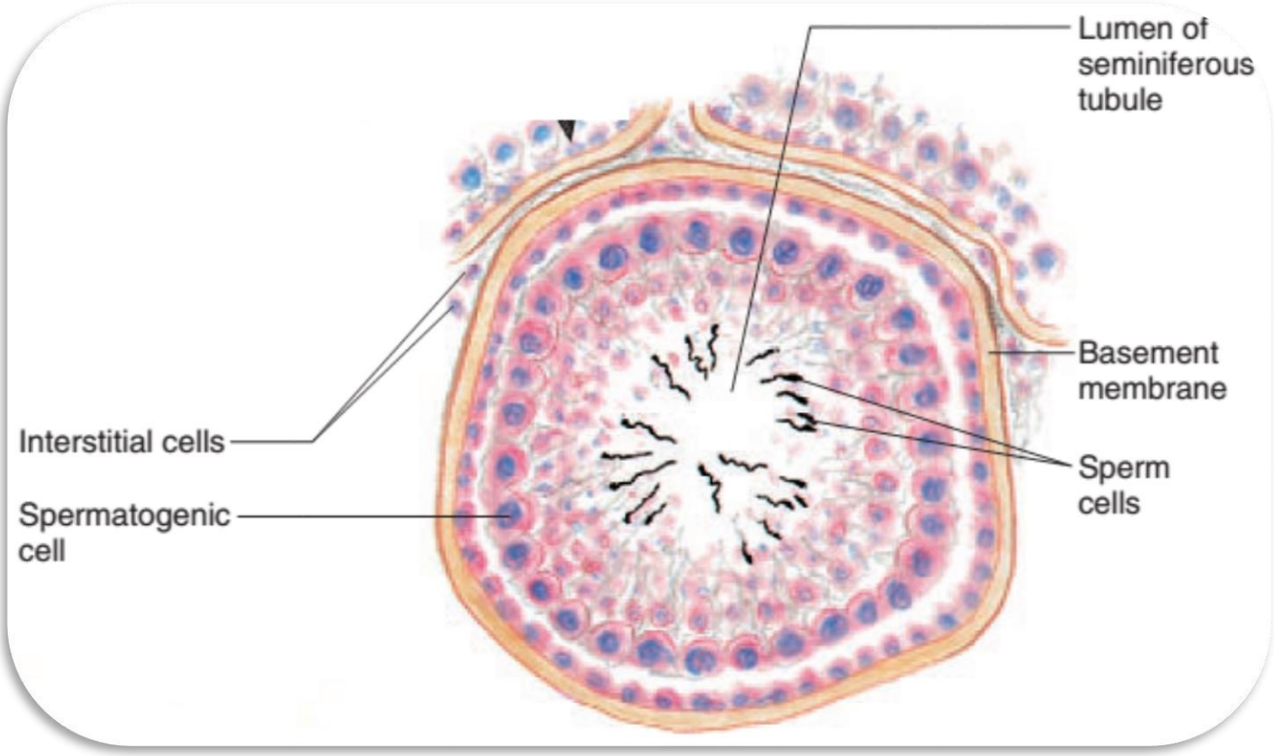
1. Testes



2. Seminiferous Tubules

- About 800 in each testes
- Seminiferous tubules walls contain Two types of cells:
  - Spermatogenic cells
  - Sustentacular cells
- Clusters of interstitial cells between seminiferous tubules.

2. Seminiferous Tubules



3. Spermatogenesis

- Thousands of sperms are produced in each second in seminiferous tubules

- The process of sperm production is called spermatogenesis
- Spermatogenic cells are present in the wall of seminiferous tubules

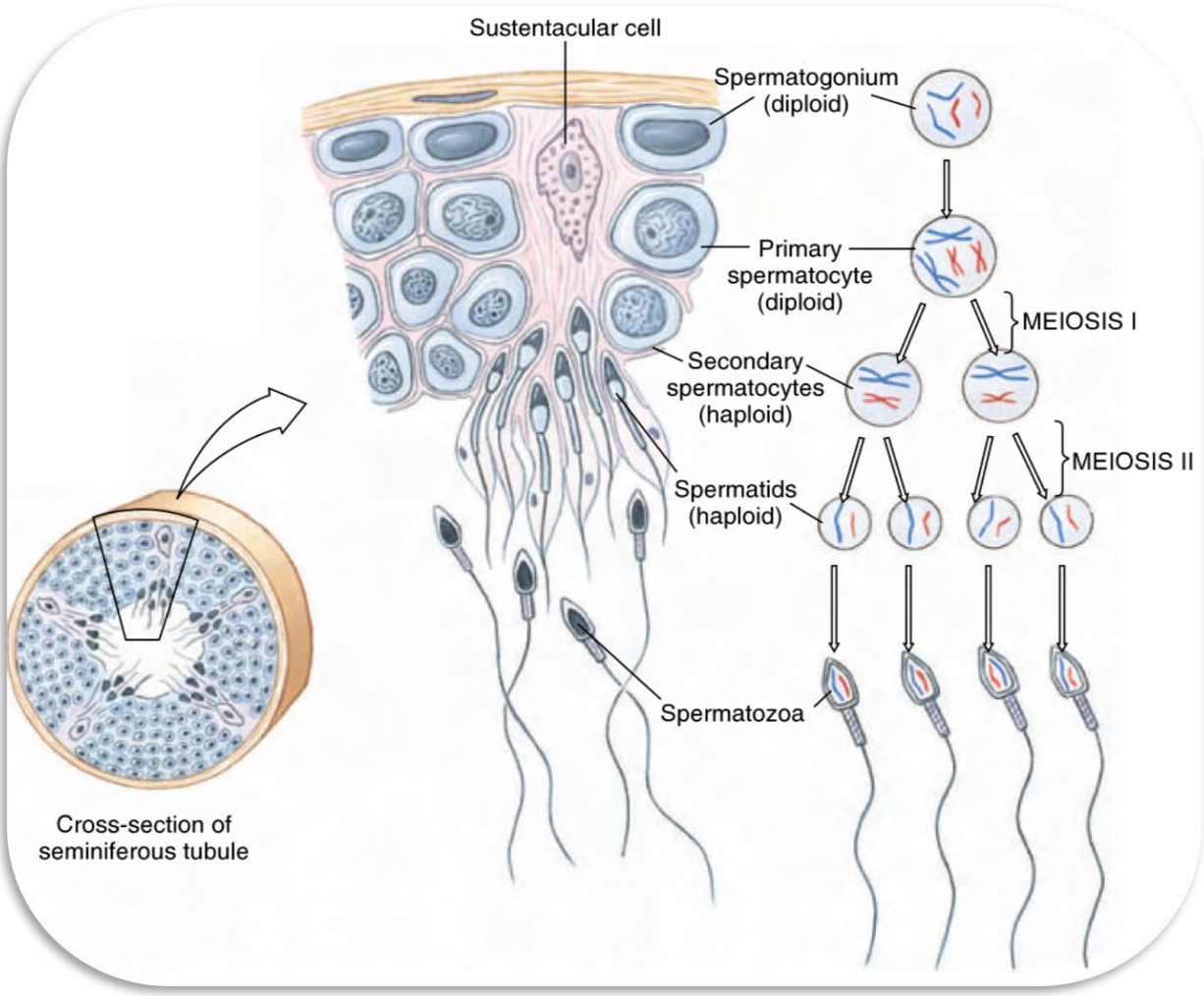
3. Spermatogenesis

- Spermatogenic cells are also called spermatogonia
- Spermatogomium increase in size and becomes primary spermatocyte
- Meiosis-I results in two secondary spermatocytes

3. Spermatogenesis

- After meiosis-II four haploid spermatids are formed
- Spermatid undergo morphological changes and are transformed into sperms which are motile

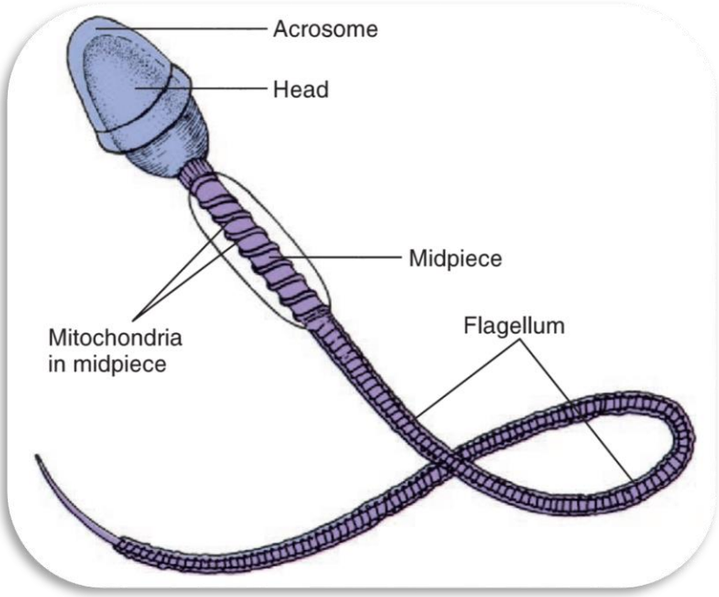
3. Spermatogenesis



4. Structure of Sperm

- A mature sperm consists of:
- **Head:** contain haploid nucleus and acrosome cap having acrosin.
- **Midpiece:** have mitochondria
- **Tail:** microtubules that bend for whiplike movements

4. Structure of Sperm



147 Transport of sperms in Human

Transport of sperms in Human

Spermatogenic cells produce sperms while ustentacular cells secrete fluid for nourishment of sperms. Sperms from seminiferous tubules comes into a network of called rete testis and then into epididymis where they are store for some time. Muscular contraction propel sperm forward. Sperms through duct deferens are transported to ampulla where stored till ejaculation. Seminal vesicles **S**ecrete water, fructose, prostaglandins and Vitamin C. This is energy source for sperm motility and helps to neutralize pH of female reproductive tract. Prostate gland secretes water, enzymes, cholesterol, buffering salts and phospholipids. Bulbourethral glands **s**ecrete clear alkaline fluid that lubricates urethra to facilitate ejaculation. Fluid that results from the combination of sperms and glandular secretions is called semen. Average human ejaculation contain 3 to 4 ml of semen having 300-400 million sperms.

148 Hormonal control of male reproductive Function

Hormonal Control of Male Reproductive Function

Male sex hormones are collectively called androgens (e.g., Testosterones). The hormones which travel from brain and pituitary to testis and ovaries are called gonadotropins (e.g., FSH and LH). Testosterone is secreted by interstitial cells it has role in development of testis, prostate and secondary sex characteristics.

Low level of testosterone is detected by hypothalamus. Hypothalamus release GnRH which effects the pituitary gland to release FSH and LH (ICSH). FSH stimulate spermatogenesis and LH stimulate the production of testosterone. Testosterone inhibits the secretion of LH while inhibin produced by sustentacular cells inhibits the secretion of FSH.

149 Human Female Reproductive System

HUMAN FEMALE REPRODUCTIVE SYSTEM

Reproductive role of human female is more complex. The main functions of human female reproductive system are egg production, to nourish, carry and protect developing embryo. Female also nurse the young one after birth.

Human female reproductive system consists of ovaries, uterine tubes fimbriae and uterus, vagina, external genitalia and mammary glands. Ovaries produce eggs. They are also responsible for estrogen and progesterone production. Uterine tubes carry eggs from ovaries to uterus. Fimbriae are finger like projections which receive the egg into uterine tube. Uterus is the place where embryo is implanted for development. Vagina receives sperms during sexual activity. External genitalia are important in sexual arousal and have protective role while mammary glands are to feed the milk to newborn.

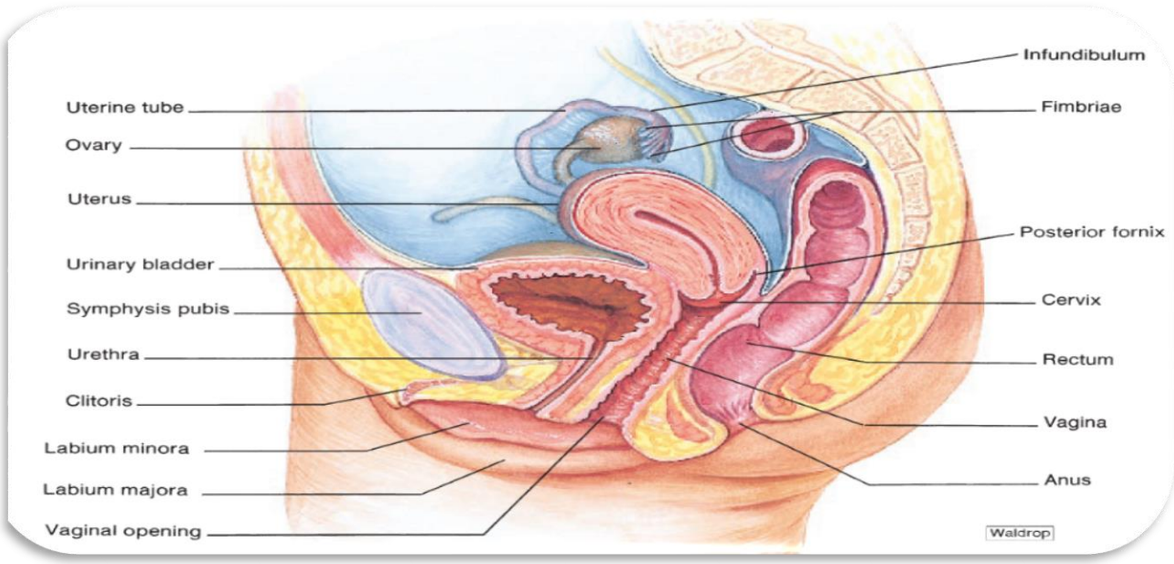


Fig. human female reproductive system

150 Menstrual Cycle in Human

MENSTRUAL CYCLE IN HUMAN

It is the monthly cycle of changes in ovaries and preparation of uterus for implantation of embryo. The cycle begins at puberty and stops at the age of 45 to 55 years.

1. **The Follicular Phase:** Hypothalamus releases GnRH which stimulate the release of FSH and LH from anterior pituitary gland. FSH starts the development of oocyte in the ovarian follicles.
   i. **The Follicular Phase:** Developing follicle produces estrogen. Estrogen builds up and prepares uterus lining to become thicker (endometrium) for implantation of embryo.
   ii. **Ovulation:** Elevated level of estrogen increases the release of LH from pituitary gland and inhibits FSH. LH cause the mature follicle to enlarge rapidly and release secondary oocyte. This process is ovulation.
   iii. **The Luteal Phase:** Mature follicle after release of oocyte becomes corpus luteum. Corpus luteum secretes estrogen and progesterone. They complete the development of endometrium and maintain it for 10-14 days.
   iv. **Menstruation:** If the oocyte is not fertilized, corpus luteum disintegrates. Estrogen and progesterone production stops. As a result endometrium breaks and menstruation starts.

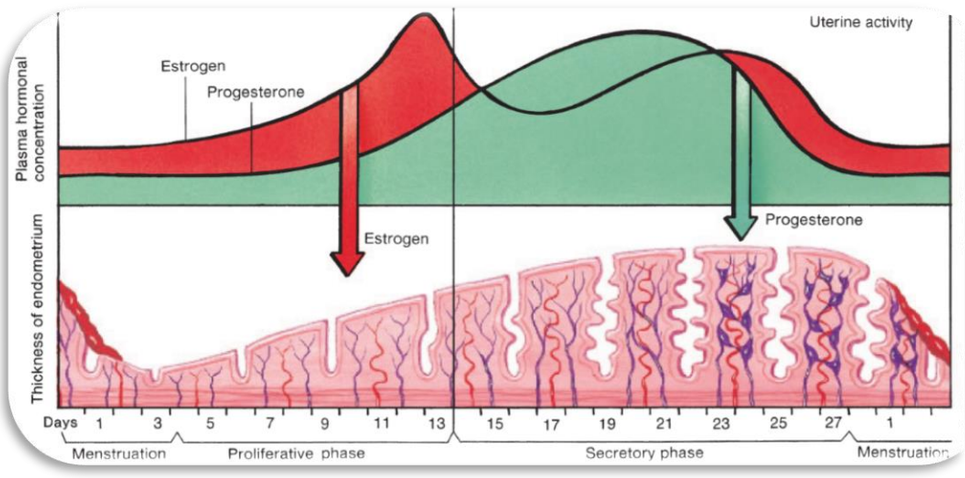


Fig. the human menstrual cycle

151 Follicle development in human

Follicle development in human

An ovary consists of an inner medulla and outer cortex. Medulla has blood vessels and nerves while cortex has developing follicles. During first five months of development many small follicles are formed in the ovary of female fetus. These are called primordial follicles. At the time of puberty during each menstrual cycle some primordial follicles began to develop further and become primary follicles which then converted to secondary follicles and graafian follicle. Many follicle start development in each cycle but only one completes all the stages. The follicle before release of egg is called graafian follicle. LH

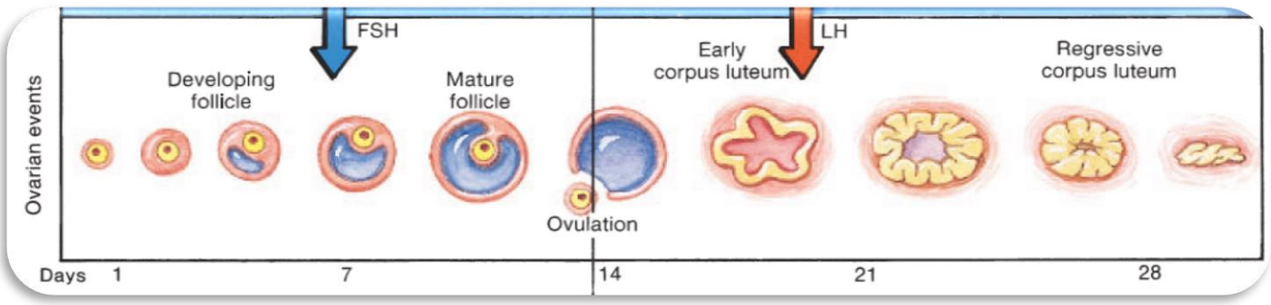


Fig. Different stages of follicle development

hormone causes the rupture of follicle to release egg. Graafian follicle after release of oocyte becomes corpus luteum which remain active for 10-14 days. If fertilization does not occur corpus luteum is converted to corpus albicans which ultimately disintegrates.

During each menstrual cycle several follicles start development but only one matures. Others disintegrates by the process atresia. In this process cells are disintegrated by apoptosis.

152 Production and Transport of Eggs in Human

Production and Transport of Egg in Human

Egg production starts in female while still at the developmental stages. A definite number of eggs development takes place in specific structures called follicles. Egg development is halted at early stage of meiosis-I. At puberty during menstrual cycle meiosis-I is completed and meiosis-II starts which is completed after fertilization.

Egg is released from ovary by the process of ovulation. LH cause the release of oocyte from ovary. Ovary is encircled by finger like ciliated fimbriae. The ciliary action and muscular contraction cause the entry of oocyte into fallopian tube also called uterine tube. Fertilization takes place in upper one third portion of uterine tube. Oocyte completes meiosis-II after fertilization. Fertilization should take place within 24 hour of ovulation.

Egg moves due to ciliated cells and muscular contraction of uterine wall. Egg takes about 5 days to reach to uterus. If fertilized, embryo implanted into uterus otherwise egg is released out.

153 Hormonal Control of Female Reproductive Function

Hormonal Control of Female Reproductive Function

Low level of estrogen stimulate hypothalamus to release gonadotropin-releasing hormone (GnRH). GnRH acts on pituitary gland to release follicle stimulating hormone (FSH) and luteinizing hormone (LH). Both of these hormones act on ovary. FSH start the development of follicle. 15-20 follicles start development but only one matures. Ovary also starts to produce estrogen. Estrogen causes the thickness of endometrium of uterus

Higher level of estrogen triggers LH and inhibit FSH form pituitary gland. LH from pituitary causes the ripening of ovarian follicle and release of oocyte from follicle. LH also convert collapsed follicle into corpus luteum after ovulation. Corpus luteum secretes estrogen and progesterone. Estrogen and progesterone have two roles: they maintain endometrium for completing development and also inhibit LH and FSH. If no fertilization occur, FSH and estrogen level drops and new cycle is started.

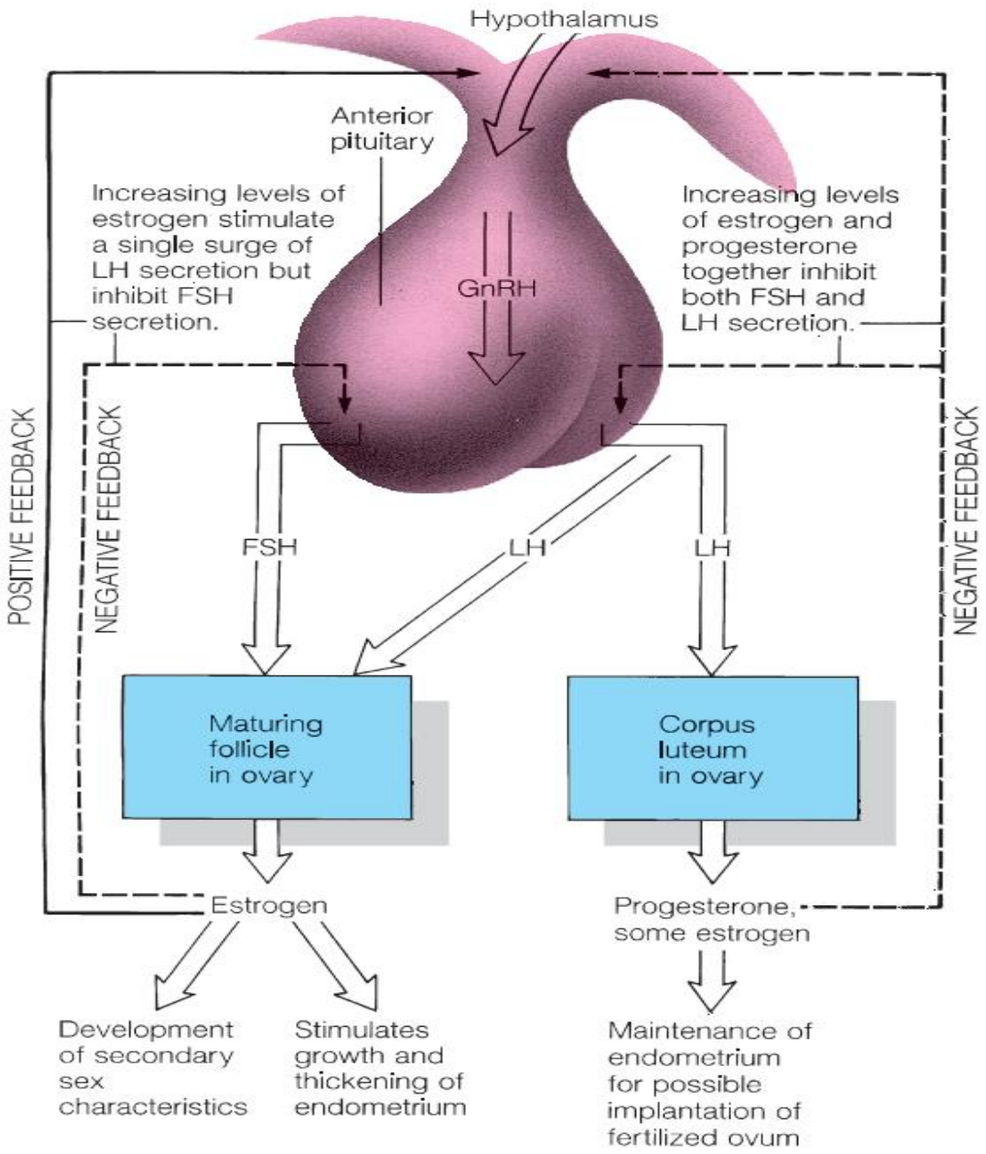


Fig. hormonal control of human female reproductive system.

154 Historic Perspective of Embrology

Historical Perspective of Embryology

Preformation theory: This theory says that gametes contain miniature version of all the element present in an adult. This means that every organ in a very small size is already present. Most biologist of 17th and 18th century believe in it.

Epigenesis theory gained popularity in mid-18th century. This theory says that embryo contain material from which embryo is gradually build. Much like a carpenter builds a building from nails, lumber and bricks.

Birth of modern embryology: Modern embryology emerged with the proposed germ layer theory. Embryo form 2-3 layers from which different body parts are formed.

Development Biology: It the study by which animals and plants grow. Developmental biologists use the tools of molecular biology to investigate the genetic control of development.

155 Gametogenesis: spermatogenesis

Gametogenesis

Gametogenesis is the process of formation of gametes. It is of two main types: spermatogenesis and oogenesis.

Spermatogenesis: The process of sperms production in male gonads through meiosis is called spermatogenesis. Spermatogenesis takes place in testes. In human the entire process of

spermatogenesis and its transportation takes about 3 months. However, it is different in different animals.

Testes contain diploid stem cells. Sperm mother cell is called spermatogomium. It enlarges in its size and become primary spermatocyte. Primary spermatocyte undergo meiotic cell division. After meiosis-I two secondary spermatocytes are formed. Then meiosis-II results in four haploid spermatids. Spermatids do not undergo any further cell division. They undergo morphological changes to produce sperms. Spermatids are non-motile while sperms are motile cells.

156 Gametogenesis: oogenesis

Oogenesis: The process of egg production in female gonads through meiosis is called oogenesis. Oogenesis takes place in ovaries. Egg formation and maturation takes place in follicles of ovary. Ovarian follicles contain oogonia. Oogonium increase its size and becomes primary oocyte which is a diploid cell with duplicated chromosome. Primary oocyte undergo meiotic cell division. After meiosis-I a secondary oocyte and a polar body is formed. Meiosis-II results in a haploid ootid and another polar body. Ootid undergo differentiation process to form ovum.

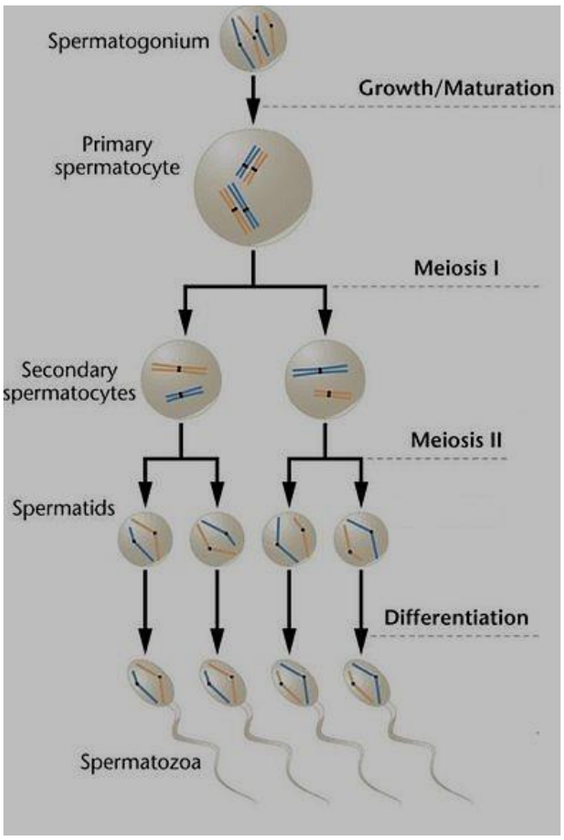


Fig. Spermatogenesis in animals

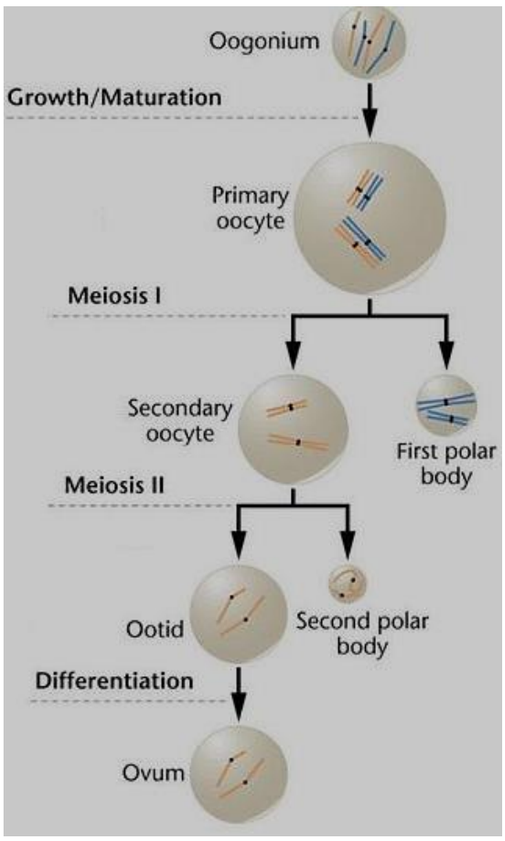


Fig. Oogenesis in animals

157 Types of Eggs

TYPES OF EGGS

Eggs may be classified on the basis of amount of yolk or distribution of yolk.

Classification on the basis of amount of yolk: On the basis of amount of yolk eggs may be alecithal, microlecithal, mesolecithal and macrolecithal. Eggs containing negligible amount of yolk are called alecithal egg. e.g., placental mammals. Eggs containing very little amount of yolk are called microlecithal eggs. e.g., sea urchin, amphioxus and marsupial mammals. Eggs with moderate amount of yolk are called mesolecithal eggs. e.g., lung fish and amphibians. While eggs with large amount of yolk are called macrolecithal eggs. e.g., reptiles and birds.

Classification on the basis of distribution of yolk: Eggs may be classified on the basis of distribution of yolk. Eggs with equally distributed yolk are called homolecithal eggs. e.g., sea urchin and mammals. Eggs with yolk in the center and cytoplasm in periphery are called centrolecithal eggs. e.g., insects. Eggs in which yolk is present at one end while cytoplasm at other end are called telolecithal eggs. e.g., frogs, reptiles and birds. The region of egg containing cytoplasm, its organelles and less amount of yolk

is called animal pole while the region of egg containing more amount of yolk and less cytoplasm is called vegetal pole.

158 Fertilization: The Process

FERTILIZATION

Fertilization is the process in which egg and sperms fuse to form a diploid zygote which is the first representative of next generation. Fertilization is of two types: external fertilization and internal Fertilization. External fertilization takes place outside the body mainly in water, while internal fertilization takes place within the body of animal. Fertilization may be via copulation, cloacal connection or via spermatophore.

The structure surrounding plasma membrane of egg is called vitelline membrane. It is made up of fibrous proteins and receptors needed for sperm binding. The sperm contain an acrosomal head. The acrosome have specific enzymes called lysins which dissolve egg membrane during fertilization.

Many sperms approach to egg but only one successful. In some species only nucleus of sperm fuses with egg. In others, mitochondria, centriole and flagella of sperm may get entry into egg.

Egg Activation: It is a series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development. Normally many sperms approach to egg but only one sperm should fertilize the egg. For this, second fertilization is stopped by changes in plasma membrane and outer region of cytoplasm.

159 Fertilization: Membrane and Cortical Events

Membrane and corticular events: As sperm contacts, **microvilli** from egg plasma membrane wrap around the sperm. Contraction of microfilaments causes the sperm to draw into the egg. Within milliseconds of sperm penetration, the **charges on plasma membrane** are changed. This makes plasma membrane unresponsive to other sperms. **Fertilization membrane** is formed by the discharge of granules of egg cortex. These granules are discharged between plasma membrane and vitelline layer. These granules loosen the contacts between plasma membrane and vitelline layer. Water molecules enter into the space and proteins of corticular granules thicken and strengthen the vitelline layer. Fertilization membrane formed within 1 to 2 minutes. After this, outer cytoplasm begins rotational and sliding movements. In amphibians, this rotation of cytoplasm results in the formation of **gray crescent** which is important in development.

160 Fertilization: Metabolic and Nuclear Events

Metabolic and nuclear events: After fertilization ionic changes occur in cytoplasm causing an increase in **intracellular pH**. This triggers the developmental process in zygote. The zygote starts the cleavage (cell division) process. For this, **DNA replication** starts as two copies of DNA are required for two daughter cells. Rate of **protein synthesis** is also boosted. This is because more and more proteins and enzymes are required for cell division and daughter cells. mRNA already present in the cell is activated to start protein synthesis. The mRNA in zygote used to protein synthesis is maternal in origin. Thus bulk of initial proteins is synthesized by maternal mRNA. This influence is called **maternal dominance**.

161-162 Cleavage and its Patterns-I

CLEAVAGE AND ITS PATTERNS

Cleavage refers to cell divisions that occur during embryonic development. Initially all the cells of embryo divide at the same time this is called **synchrony**. As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**. Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres. Egg with unevenly distributed yolk result in unequal size blastomeres. There are following important patterns of cleavage:

a. Holoblastic Cleavage

The pattern of cleavage in which egg is completely divided into two cells. Egg contain little or moderate amount of yolk. Cleavage in fertilized eggs of Amphibians, mammals, echinoderms, flatworms and nematodes.

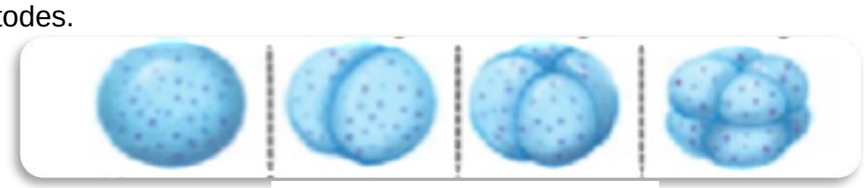


Fig. Holoblastic cleavage

b. Meroblastic Cleavage

The pattern of cleavage in which egg is not completely divided into two cells. This is due to large amount of yolk. In these embryo development takes place around or on the top of yolk. e.g., cleavage in fertilized eggs of birds and reptiles.

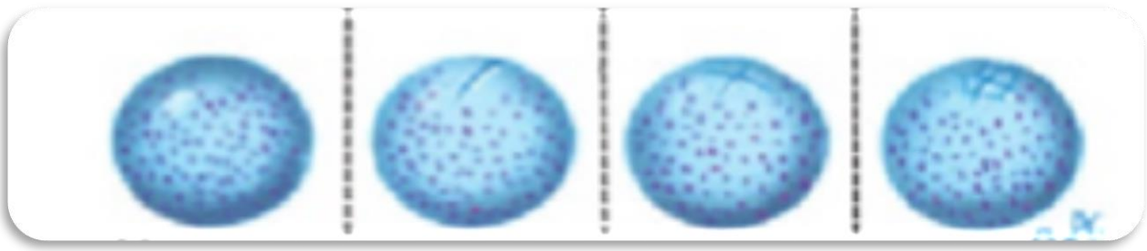


Fig. Meroblastic cleavage pattern

c. Spiral Cleavage

It is pattern in which cells of animal pole are not exactly laying above the vegetal pole cells. Cleavage takes place at oblique angle forming a spiral pattern. Example: protostomes cleavage

d. Radial Cleavage

It is pattern in which cells of animal pole are exactly laying above the vegetal pole cells. Upper tier of cells is exactly on the lower tier of cells. Example: deurostomes cleavage

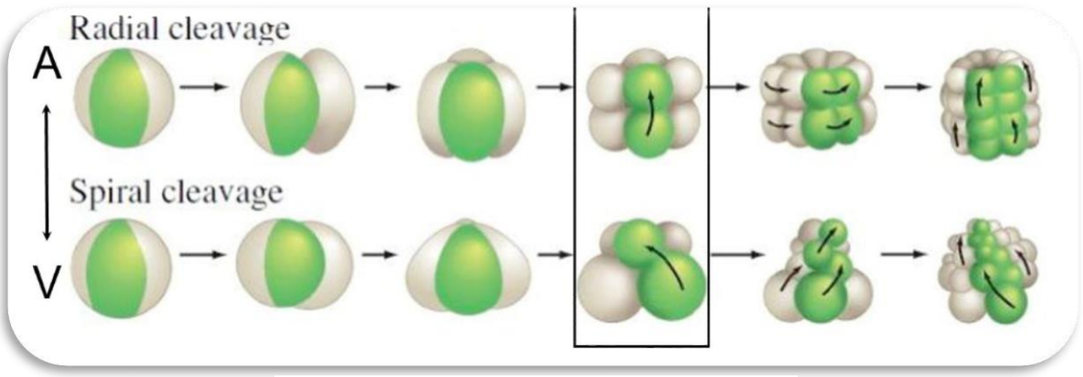


Fig. Radial and spiral cleavage pattern

163 Echinoderms Embryology: General Concepts

ECHINODERM EMBRYOLOGY

General Concepts

Echinoderms include starfish, sea cucumber, brittle star and sea urchin. Echinoderms are extensively studied group. This is because they are easily maintained in laboratory. Echinoderms lay eggs in water. In echinoderms fertilization is external and it takes place in water. Egg of echinoderm contain very little amount of yolk (microlecithal eggs).

164 Echinoderms Embryology: Cleavage

Cleavage: Cleavage is holoblastic and completely divides into two. Cleavage lines pass in such a way that cells lie exactly one above the other. This type of cleavage is called radial cleavage. Radial cleave is the characteristic of deuterostomes.

First two cleavage are from animal to vegetal pole (meridional) and perpendicular to each other. Third cleavage is equatorial and right angle to first two. It separates animal pole and vegetal pole. Fourth cleavage is medional. Initially division is equal, however, at sixteen cell stage division becomes unequal.

165 Echinoderms Embryology: Morulation and Blastulation

Morulation and Blastulation: Cleavage results in a ball of cells called morula. Morula is a solid ball of cells without any cavity. At about 120 cell stage, a fluid filled cavity **blastocoel** is formed in the embryo. There is a single layer of cells around the blastocoel. Embryo at this stage is called blastula. At blastula stage embryo comes out of the fertilization membrane. At blastula stage cilia develop and embryo start swimming. At late blastula stage, some cells breakoff and position themselves within blastocoel. These cells are called primary mesenchyme and will form skeletal elements (spicules) of embryo. Late blastula is about 1000 cells of different size.

166-167 Echinoderms Embryology: Gastrulation-I

Gastrulation: Gastrulation is the process in which blastula is changed into gastrula. During this process primary germ layers are formed. Gastrulation begins at a point in vegetal half where cells start invagination. The point where invagination starts is blastopore. The invagination forms a new cavity in the embryo called **archenteron**. Archenteron gradually replaces blatocoel. Blastopore will form anus and archenteron is future gut of echinoderms. Some cells at the tip of developing archenteron get apart and form secondary mesenchyme cells. Secondary mesenchyme cells play a central role in pulling archenteron up to blastocoel wall. Initially embryo is round in shape. During gastrulation it begins to elongate and attains pyramidal shape (Prism) and ultimately form pluteus larva. The larvae have a preferred direction of movement. Pluteus larva swims freely and feeds on smaller plants and animal. It has bilateral symmetry, although adults have radial symmetry.

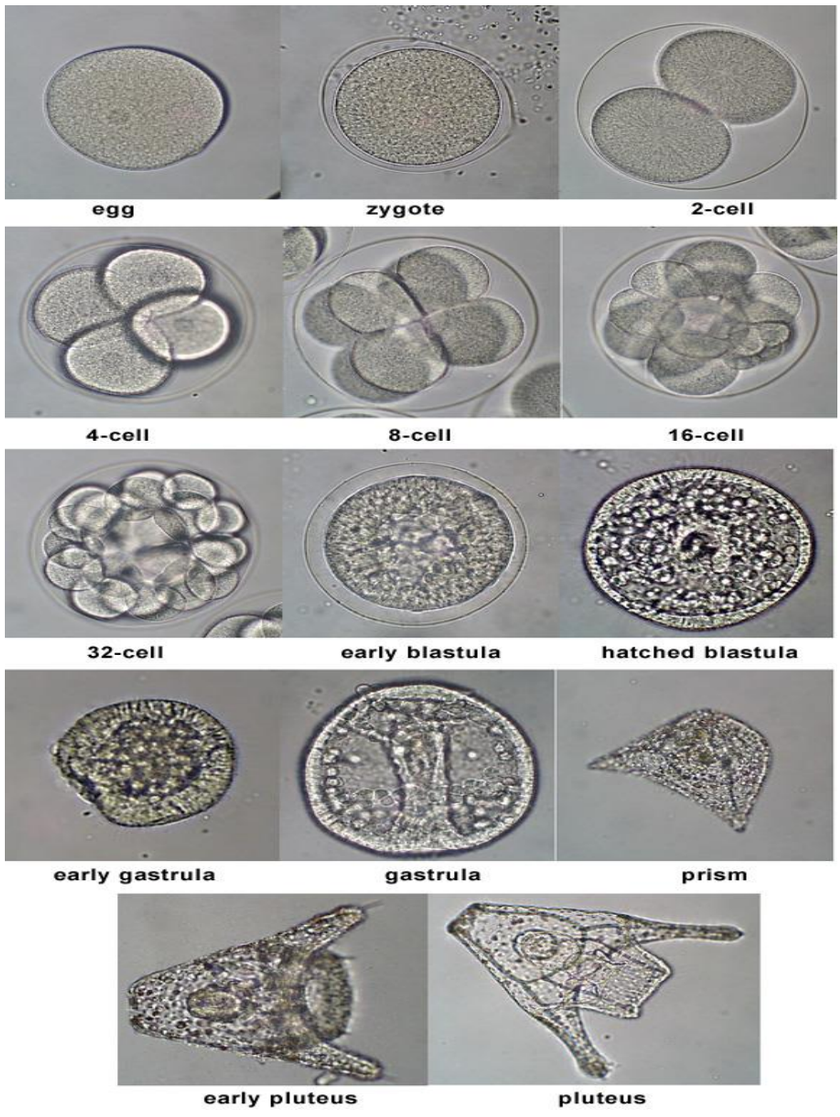


Fig. Echinoderm embryology

168 Amphibian embryology: General Concepts

AMPHIBIAN EMBRYOLOGY

Most of the amphibians lay laid in water. Eggs are mesolecithal as they have moderate amount of yolk. The upper pigmented portion is called animal pole while the lower pole with large amount of yolk is called vegetal pole. Darkly pigmented animal pole on upper side camouflage from predators as it blends with bottom of water body. Light colour of vegetal pole blends with sky when viewed form below. Dark colour of animal pole also absorbs heat which promotes development. Fertilization takes place in water. Usually very little care by parents.

169 Amphibian embryology: Cleavage

Cleavage: Amphibians have mesolecithal eggs. Cleavage completely divide cell in two cells. Due to yolk at vegetal pole cleavage is slow. Size of cells at animal pole is smaller while larger at vegetal pole. First cleavage start from animal to vegetal pole. Cleavage at vegetal pole is slow due to large amount of yolk. It also divides gray crescent from middle. Second cleavage is perpendicular to the first and divides embryo into four. Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole. Fourth cleavage is meridional. Cleavage lines pass in such a way that cells lie exactly one above the other (radial cleavage). This radial cleavage is the characteristic of deuterostomes.

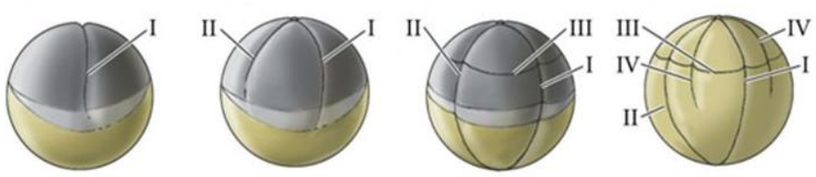


Fig. Cleavage in amphibians

170 Amphibian embryology: Morulation and Blastulation

Morulation: Cleavage results a ball of cells called morula. Morula is a solid ball of cells without any cavity. Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk.

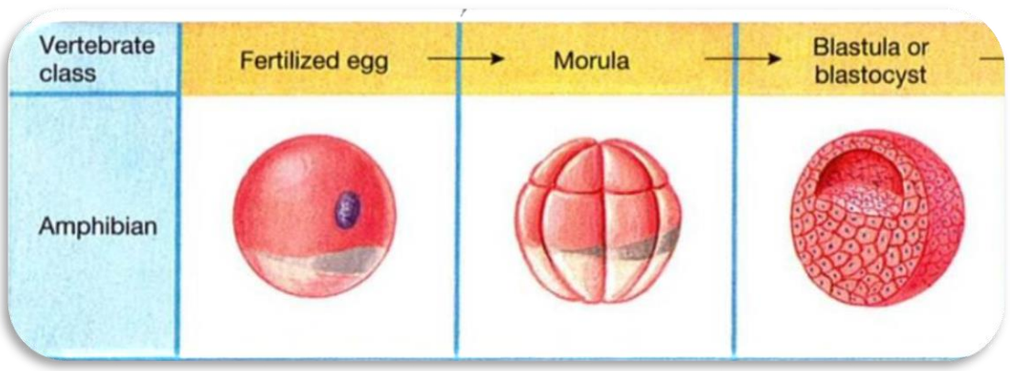


Fig. Morulation and blastulation

Blastulation: After morula formation, a fluid filled cavity **blastocoel** is formed in the embryo. Cell form multiple layers around the blastocoel. In amphibians blastocoel is formed towards animal pole due to yolk at vegetal pole

171-172 Amphibian embryology: Gastrulation-I

Gastrulation: Gastrulation is the process during which cells arrange themselves into different layers called primary germ layers. Embryologist use dyes and carbon particles to mark different cells. Cells forming different structures are grouped on the surface of blastula. Later these migrate during gastrulation and form different structures. So these cells are named according to their future structure such as presumptive notochord, presumptive mesoderm and so on. Gastrulation begins with a slit like opening **blastopore** between gray crescent and ventral pole. Animal pole margin of blastopore is called its dorsal lip. Cells at the bottom of groove move anteriorly and groove spreads transversely. Cell at the surface begin to roll over the dorsal lip. This process is **involution**. Migrating cells spread across the slit like blastopore transversely and downward to the vegetal pole. Due to this migration one end of slit meets the other end. The ring like blastopore surrounds the yolk cells and they protrude out of the

blastopore. This is called yolk plug. Later blastopore cells enclose yolk plug and blastopore is considered as closed.

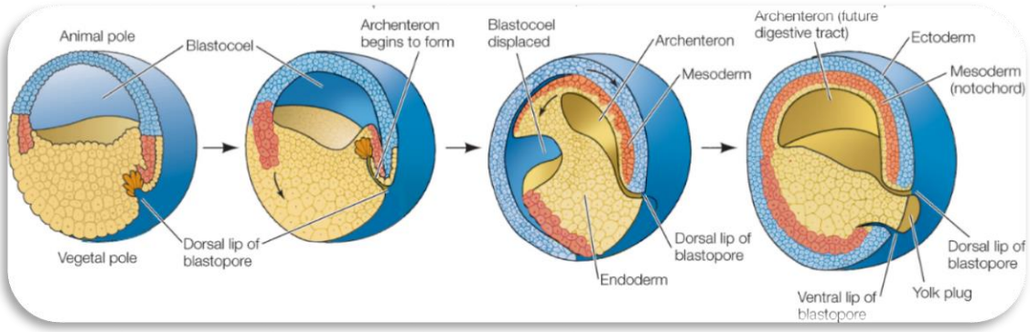


Fig. Gastrulation in amphibians.

Rolling over cells from blastopore form archenteron. Blastocoel becomes smaller and archenteron become larger. Archenteron is future gut and yolk cell make floor of gut. Some of the animal pole cell are rolling into the blastopore, they will form endoderm and mesoderm. The remaining animal pole cells are called ectoderm. These ectodermal cells migrate towards blastopore and spread over entire embryo. This is called **epiboly.** Some of the last cells rolling over the blastopore are presumptive notochord and presumptive mesoderm. Initially they form dorsal lining of archenteron near blastopore. Later they detach from endoderm and move between endoderm and ectoderm in the region of blastopore. This is called **chordamesoderm** which differentiates into notochord. Lateral to notochord mesodermal cell spread and thicken along sides of embryo. These thickenings are called **somites.** Somites appear as row of pumps from side view. Eventually somites spilt to form coelom

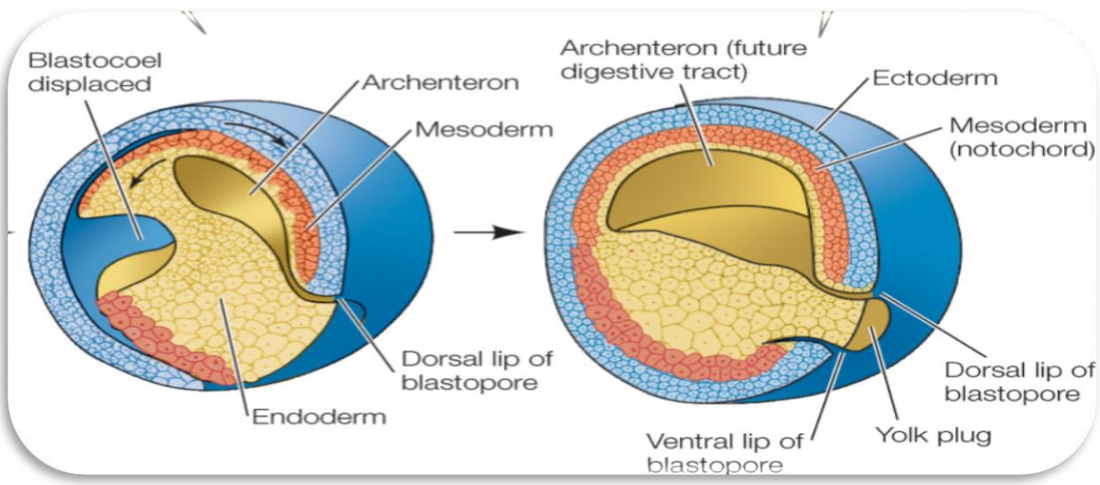


Fig. Mesoderm formation in amphibians

173 Amphibian embryology: Neurulation

Neurulation: The process of forming neural tube is called neurulation. An oval shape area on the dorsal

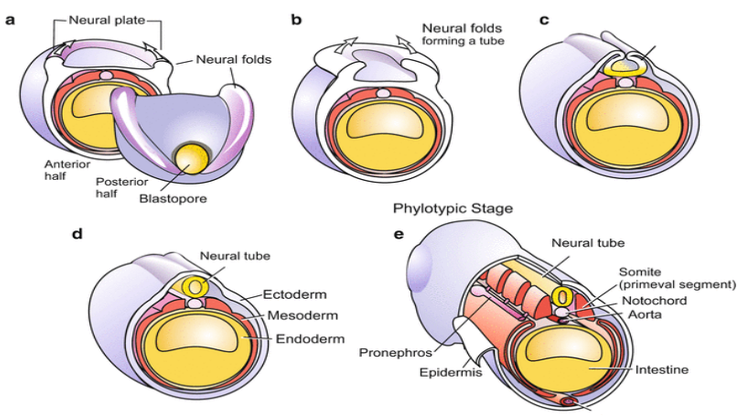


Fig. Neurulation in amphibians

surface of embryo is formed. Microfilament flatten and thicken this area. This region is neural plate. The

edges of the neural plate roll up and over the midline of neural plate. These raised edges in longitudinal direction are called neural folds. The raised neural folds meet with each other longitudinally over the midline of neural plate. Now neural plate is converted into a tubular structure called neural tube. The portion of neural tube that is close at the end will form brain

174 Avian embryology: General Concepts

AVIAN EMBRYOLOGY

Chicken development is used as a model for the development of birds and reptiles. Birds egg is macrolecithal and contains large amount yolk. Due to large amount of yolk egg nucleus is present at one side, so they are called as telolecithal eggs. Yolk is also protected by albumin $CaCO_3$ shell. In birds fertilization is internal and it takes place in upper portion of oviduct. Development starts right after fertilization while egg is still in oviduct. All developmental stage are completed within shelled egg. In chicken development is completed in approximately three weeks.

175 Structure of Avian Egg

Structure of Avian Egg

Membranes and fluids are collected around the egg while moving through oviduct. Water and albumin is deposited around the egg. This protect from desiccation and mechanical damage. Albumin is used for protection and a source of nutrition. Two denser strands of albumin called chalaza are attached to the egg shell. They suspend egg in the center of albumen. Shell gland secrete $CaCO_3$ which deposit around the egg. It is also impregnated with proteins. Thousands of tiny pores (40-50µm) are for exchange of gases. Inside the shell two membranes are present. At the round end of egg an air pocket is present between two membranes. Air pocket enlarges with time. Near the time of hatching the chick breaks air pocket, the air inflates its lungs and the chick start breathing.

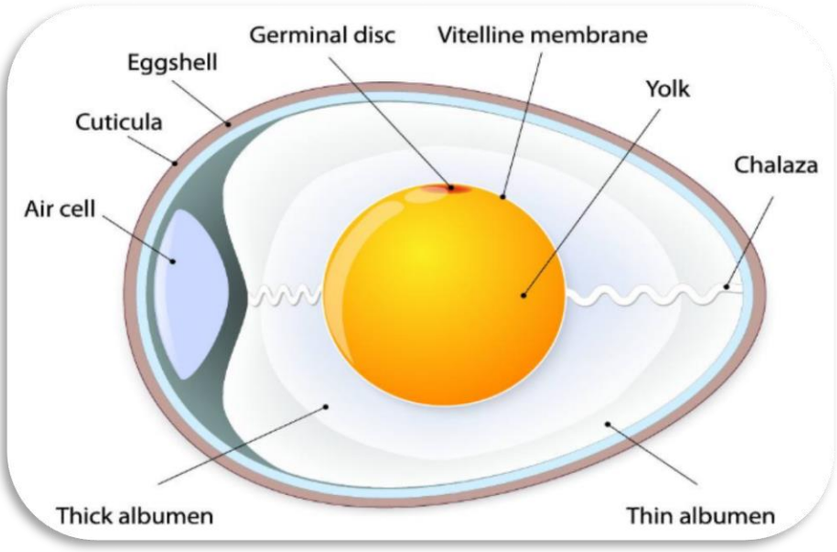


Fig. Structure of chicken egg

176 Avian embryology: Cleavage

Cleavage: Due to large amount of yolk cleavage do not divide cell in two. There is no division in the large amount of yolk. All the divisions are at one end of egg forming a disc shape structure. Such a cleavage is called discoidal cleavage. Cleavage is radial as cleavage lines pass in such a way that cells lie exactly one above the other. Radial cleave is the characteristic of deuterostomes. First two cleavage are vertical and perpendicular to each other. Third cleavage is equatorial and right angle to first two. Fourth cleavage is vertical.

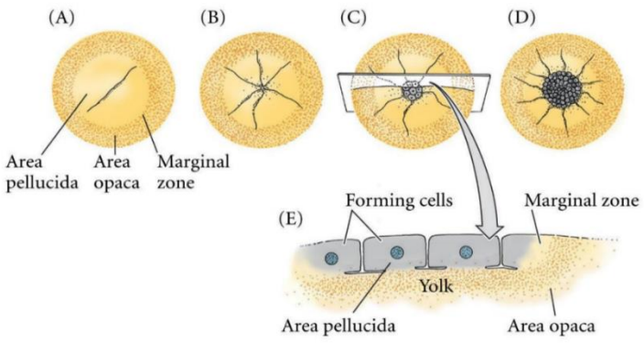


Fig. discoidal cleavage in chick embryo

177 Avian embryology: Blastulation

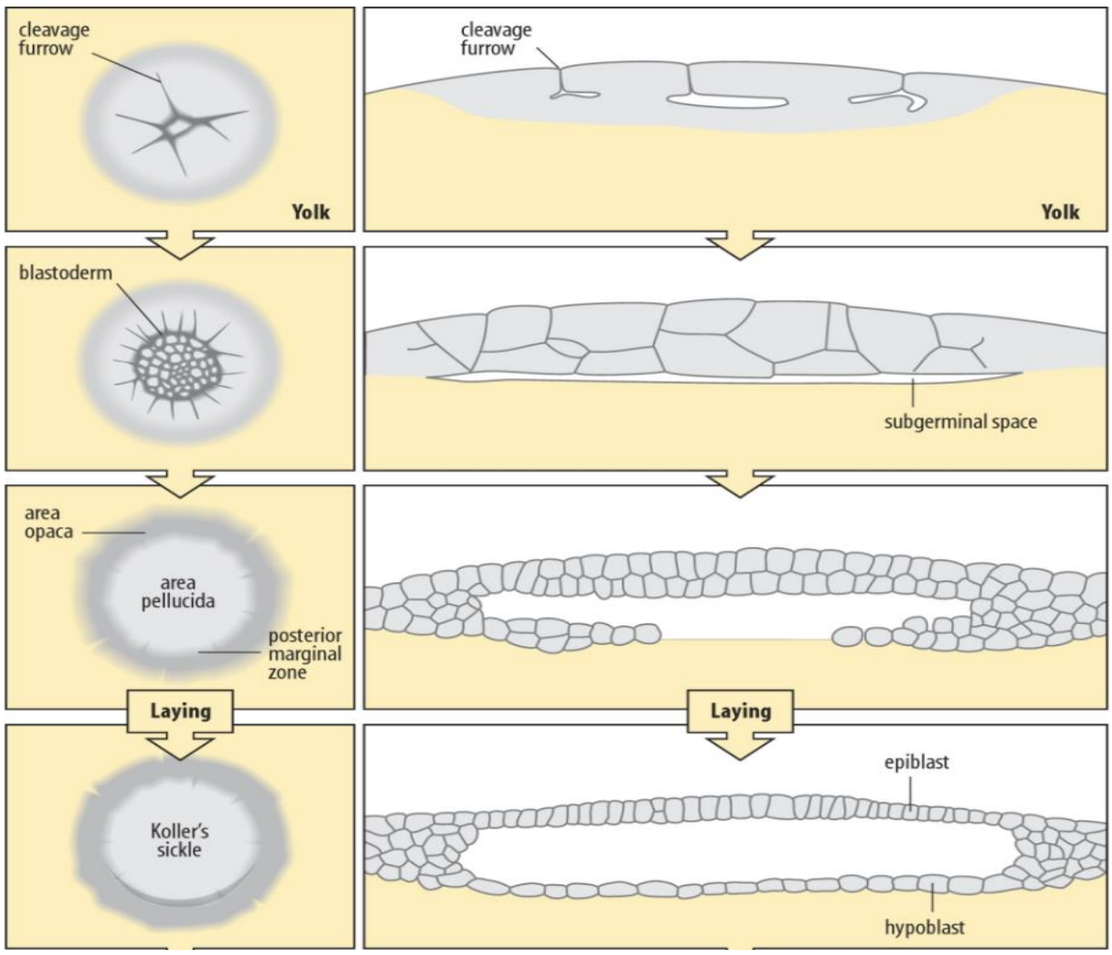


Fig. Blastulation in chick embryo

Blastulation: The cells in the animal pole dived. When the cell are approximately 60,000, they form a blastoderm. The yolk below blastoderm is used which forms a fluid filled cavity called blastocoel. Blastoderm splits into two layers: Outer layer is called epiblast while the inner layer is hypoblast.

178-179 Avian embryology: Gastrulation

Gastrulation: The dorsal cells of blastoderms form a medial linear invagination called primitive streak. Primitive streak gradually extends anteriorly. It defines the axis of embryo. It also separate right and left side. The anterior margin of primitive streak there is a regional thickening called Henson's node.

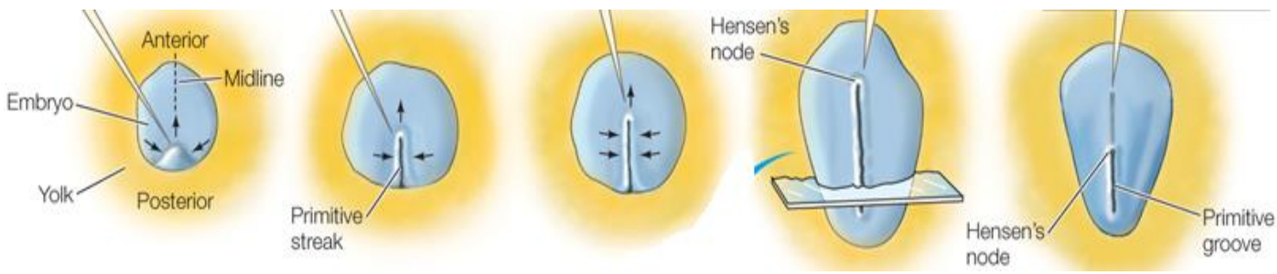


Fig. primitive streak and Hensen's node in chick embryo

Henson's node contains funnel shaped depression through which cell enter between epiblast and hypoblast. Henson's node is functionally equivalent to blastopore of amphibians. First cells migrating through Henson's node move anteriorly and ventrally to form pharyngeal endoderm of foregut. Hypoblast cell are moved anteriorly. Next migrating cells move anteriorly, but not far away. They remain between endoderm and epiblast to form prechordal plate mesoderm. They form head process. After that the migrating cells form chordamesoderm which forms notochord. Cells that ingress through middle

part of streak give rise to middle part of somites and to heart and kidney. Cell which ingress through posterior end give rise to lateral plate and extra embryonic mesoderm.

After migration, the cell remaining outside form ectoderm. Ectodermal cells close to streak form neural tube and the cells farther from streak will form epidermis. Henson's node attains a more posterior position due to regression of primitive streak. Now the epiblast cells are all ectodermal cells. The ectodermal cell migrate and enclose the yolk. This process is called epiboly. Embryo lift it off the yolk. Only a connection is retained between embryo and yolk. Blood vessels develop in yolk stalk to carry nutrients from yolk.

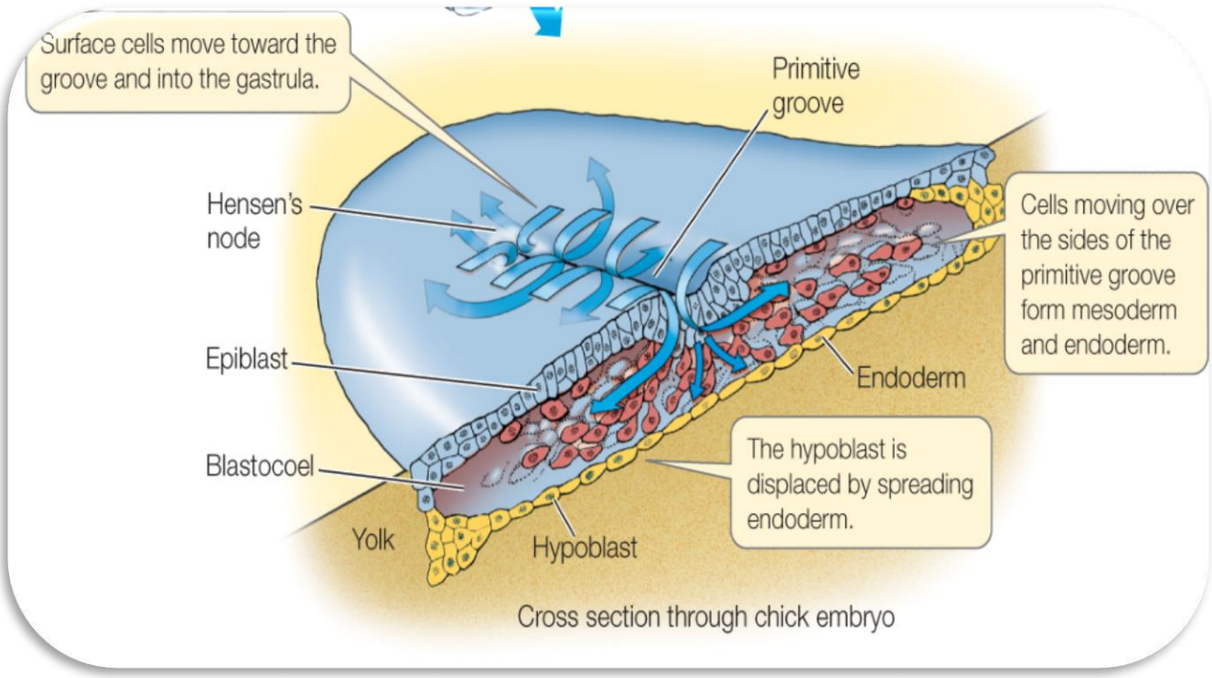


Fig. Cross section through chick embryo

180 Mammalian Embryology: General Concept

MAMMALIAN EMBRYOLOGY

Class mammalia is divided into three subclasses. Prototheria are egg laying mammals. Their eggs have large amount of yolk. Marsupial mammals are pouched mammals. Their eggs have verty little amount of yolk (microlecithal eggs). Placental mammals have amost no amount of yolk (alecithal eggs). Placental mammals egg is smallest in animals kingdom. Mammals have internal fertilization. Placental and marsupial mammals have internal development and both give birth to young ones.

Marsupial mammals have premature birth. They keep the newborn in marsupium and nourish with their nipples. Placental mammals have long gestation period. Fetus is fed through placenta.

181 Mammalian Embryology: Cleavage and blastulation

Cleavage and Blastulation: Mammalian cleavage is slowest in animal kingdom. Blastomeres have unique orientation. There is rotational cleavage. First cleavage is meridional while in second cleavage one is meridional and other is horizontal. Cell division is asynchronous in the very beginning. Cell are not 2-4-8, but may be odd number. Cell switche from maternal to zygote control at 2 cell stage for proteins synthesis. Through 8 cell stage, loose arrangement with plenty of space. After

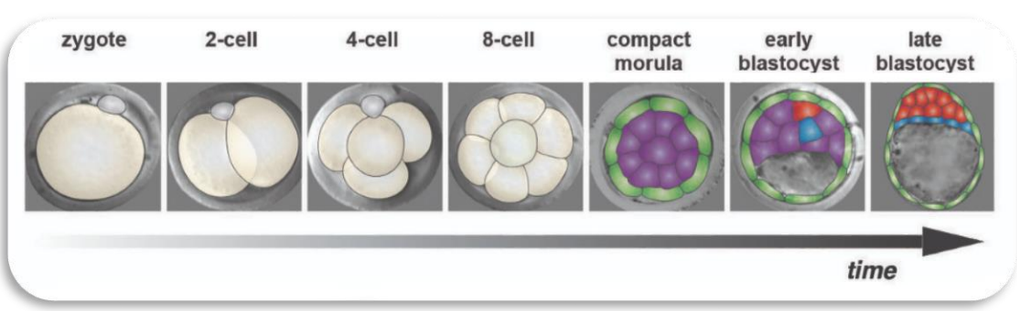


Fig. Cleavage and blastulation in mammals

third cleavage the cells tightly huddle together to form a compact ball. Eight cells divide to produce 16 cell **morula**. A small group of internal cells is called **inner cell mass** (ICM) while the outer cells form **trophoblast.** ICM form embryo while trophoblast forms chorion for placenta formation. 8 cells stage blastomere can form both ICM and trophoblast, they are called **totipotent.** ICM cells can form any type of embryo cells, they are called **pluripotent**. Trophoblast cells of morula secrete fluid that create blastocoel. As blastocoel expands ICM and trophoblast take their positions. At this stage blastula is called **blastocyst.**

182 Mammalian Embryology: Gastrulation

GASTRULATION: Both mammals and birds originate from reptile. Despite difference in amount of yolk both have striking similarity in development. First segregation of cells within inner cell mass to form hypoblast and hypoblast. Both epiblast and hypoblast form a structure **bilaminar germ disc.** Hypoblast cells form extraembryonic endoderm which form yolk sac. Epiblast form embryonic endoderm and amnion cavity. Epiblast cells move through primitive streak for form endoderm and mesoderm. The cell anterior and lateral to primitive streak form ectoderm.

183 Hormonal Regulation in Pregnant Female

HORMONAL REGULATION IN PREGNANT FEMALE

The cells developing embryo and placenta release **human chorionic gonadotropin** (hCG). hCG maintains the corpus luteum so it may not disintegrate. High level of hCG is used as immunological test for pregnancy by 2 weeks. **Progesterone** is released by corpus luteum. Progesterone maintains the endometrium lining to retain pregnancy. **Prolactin** is secreted by pituitary gland. It is produced in pregnant female to start the process of milk production and eject milk after childbirth. **Oxytocin** is also produced by pituitary gland. It caused contraction of smooth muscles to start labour and expulsion of young one. It also stimulate nipples to release milk. **Prostaglandins** secretion increase during labour. They also cause smooth muscle contraction.

184 Prenatal Development in Human: 1st Trimester

PRENATAL DEVELOPMENT IN HUMAN

In human the development can be divided into two phases: prenatal (before birth) and postnatal (after birth) phases. After two weeks of fertilization till the end of 8th week it is called embryo and from 9th week till birth is called fetus. After the birth it is called newborn or baby. The total gestation period in human is 280 days. This duration is divided into 3 months periods called trimesters.

First Trimester starts from fertilization. Zygote divides to become morula. By 4th day it is converted into 50- 120 cell blastocyst. As embryo reaches into uterus, it implants itself onto uterine wall. Trophoblast invade the endometrium. Gastrulation is similar to birds and reptiles. Once gastrulation completed different organ formation starts called organogenesis. By middle of first trimester all body organ begin to develop and embryo also grows in size.

185 Prenatal Development in Human: 2nd Trimester

Second Trimester: It ranges from 14 to 27 week. Almost all major and vital organs have been completed during this trimester. Growth is magnificent. Heart beat can be heard by the end of fourth month (18 Week). Eyes become open and bones ossification started.

186 Prenatal Development in Human: 3rd Trimester

Third Trimester: It start from 7th month till birth. Fetus has potential to survive if born premature in this trimester. Hair and nail grow in this trimester and the fetus weighs almost double.

187 The Placenta

THE PLACENTA

Placenta is s temporary organ which connects developing fetus to mother via umbilical cord. Human have placenta formed form chorion and allantois. The tiny chorionic villi sent out from blastocyst form placenta . Placenta formation starts from day 6 and is completed by 1st trimester. Embryonic blood vessels of placenta do not merge with maternal blood vessels. Placenta is about 22cm in length. Placenta is used for nutrient uptake, thermoregulation, waste elimination, gaseous exchange, fight against internal infection and placental hormone production. Placenta is connected to fetus through

umbilical cord which contains two fetal umbilical arteries and one fetal umbilical vein. They spiral about each other in umbilical cord. Average size of umbilical cord is 55cm with 1-2cm diameter.

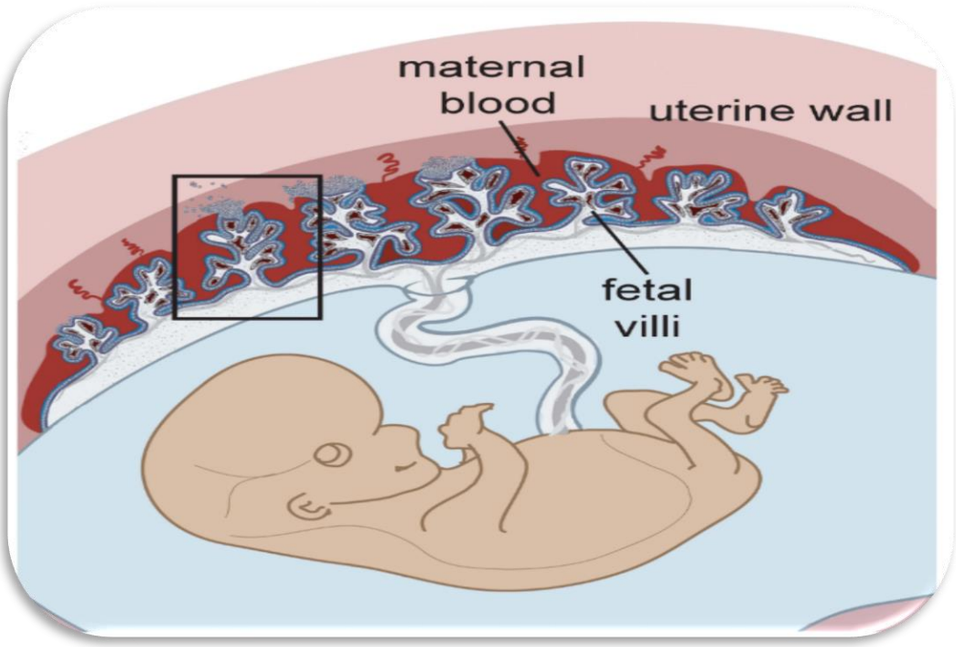


Fig. placenta and umbilical in human

188 Birth in Human

BIRTH IN HUMAN

Gestation is the total time period which embryo or fetus spends within body of mother. In human it is 280 days or 40 weeks. The process of birth is called parturition. During parturition, mother's uterine muscles begin to contract and cervix begins to dilate. Mother ovaries and placenta produce relaxin hormone. Relaxin causes mother pelvic bone to slightly separate, so the baby may pass from birth canal. During birth baby pituitary gland secretes Adrenocorticotropic hormone (ACTH). ACTH stimulates adrenal glands to secrete prostaglandins which work along with oxytocin. Stretch in uterine wall stimulate pituitary to secrete oxytocin hormone in mother. Oxytocin and prostaglandins cause powerful muscular contractions. Usually within an hour of muscular contraction baby is expelled out from uterus. After baby comes out the process of contraction continue to expel placenta, the process is called afterbirth.

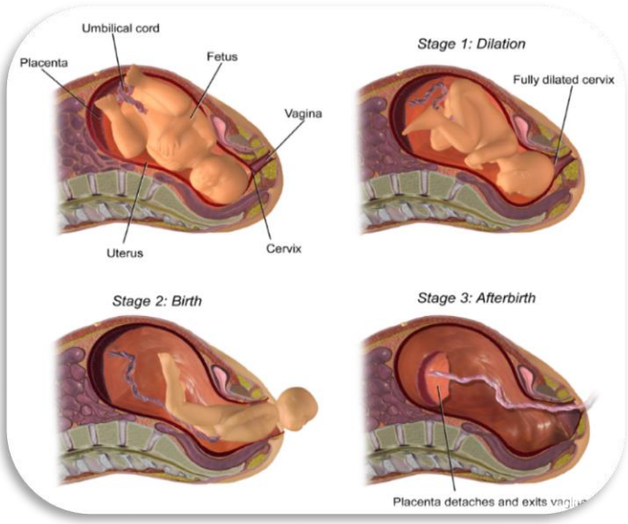


Fig. Stages of birth in human

189 Multiple Births

MULTIPLE BIRTHS

The organisms which give birth to only one offspring at a time are called **uniparous**. *e.g.,* human, cows, bufallos. Those animals which give birth to more than one young ones are called **multiparous.** *e.g.,* mice, rat, dogs, cats. Usually the young ones are both male and female. However armadillo give birth to four young ones at same time, all males or females.

Monozygotic Twins are from single zygote and are identical twins. They are strikingly alike and of same sex. About 33% have separate placenta. Majority have separate amniotic sac. Small proportion have one placenta and one amniotic sac which indicate separation occur after 9 day of pregnancy. There is conjoining risk, the condition is Siamese twining.

Dizygotic Twins are produced from two zygotes. They are called as fraternal twins. They do not have striking similarities and may be of same or different sex. They have separate placenta and separate amniotic sac.

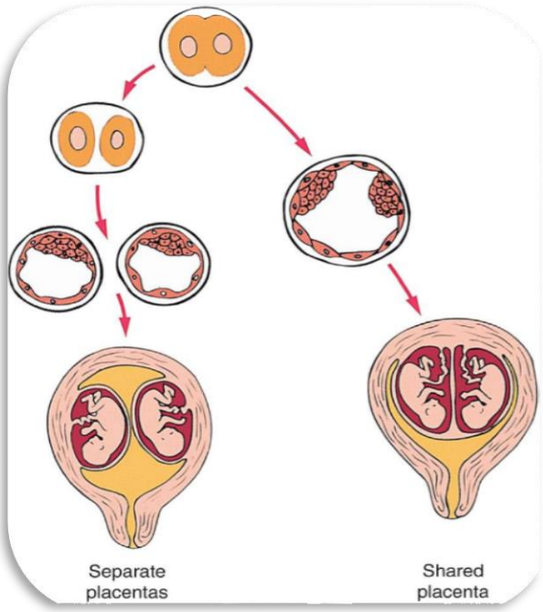


Fig. Monozygotic twins in human

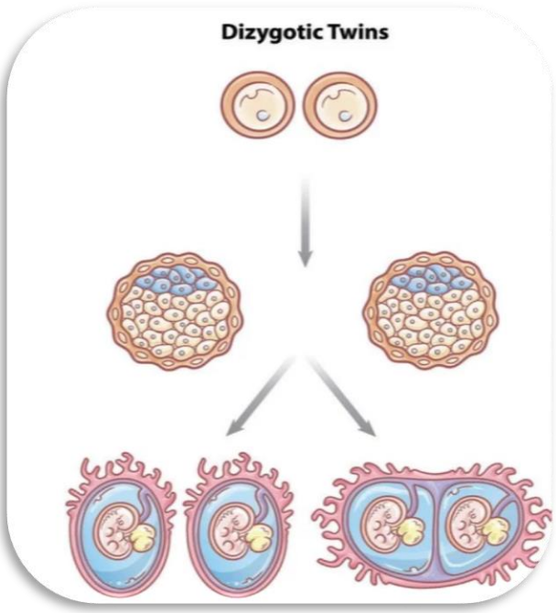


Fig. Dizygotic twins in human

190 Milk Production and Lactation

MILK PRODUCTION AND LACTATION

Mammary glands are unique feature of mammals. They are evolved from sweat glands in the skin. The process of milk production by the mammary glands is called lactation. During pregnancy breasts enlarges due to hormone prolactin. Estrogen and progesterone inhibit the secretion of milk before birth. After birth both decrease and milk production starts. Newborn stimulate pituitary to release oxytocin and prolactin. Prolactin produce milk while oxytocin triggers its release from mammary glands. The higher protein fluid in breast at the time of birth is called **colostrum.** It contains maternal antibodies to boost immune system. Colostrum works as laxative to remove meconium (stool of infant). Mother's breast don't release milk for 1-3 days.

191 Extraembryonic membranes

EXTRAEMBRYONIC MEMBRANES

Extraembryonic membranes are yolk sac, amnion, chorion and allantois. Reptiles, birds and mammals embryos have amnion, chorion and allantois are called amniotes whereas eggs of fishes and amphibians having amnion, chorion and allantois are called anamniotes. Amnion encloses embryo in a fluid filled sac. Protects against shock and desiccation. Chorion is near to shell, mechanical support, highly vascular, gaseous exchange. Allantois is for storage and excretion of toxic wastes, also for gaseous exchange. Yolk sac is involved in distribution of nutrients to developing embryo.

Ectoderm and mesoderm on both side of embryo lift off the yolk and grow dorsally, they fuse to form chorion (outer side) and amnion (inner side). A ventral outgrowth of gut tract give rise to allantois. It gradually increase in its size along with development. Yolk sac develops from proliferation of endoderm and mesoderm around yolk.

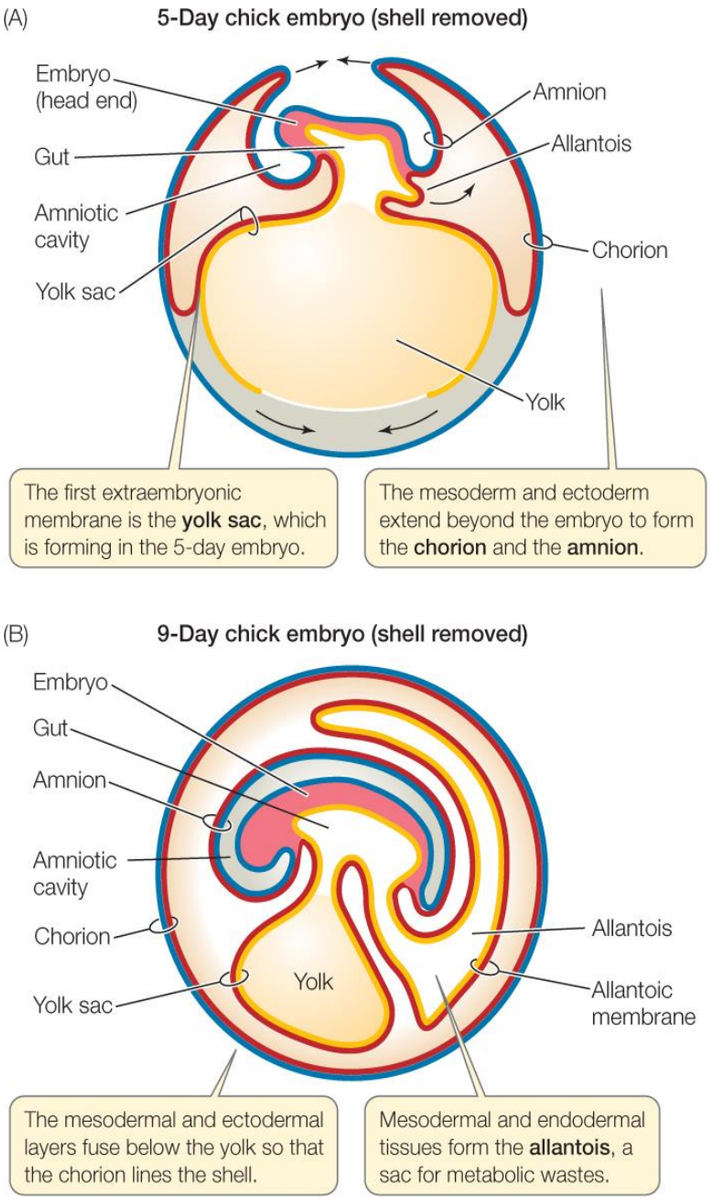


Fig. Development of extra embryonic membranes. (Ref. macmillanhighered.com)

192 Primary Germ Layers: Ectoderm derivatives

PRIMARY GERM LAYERS

During gastrulation three germ layers are formed. These are ectoderm, mesoderm and endoderm. These germ layers give rise to different body parts.

Ectoderm: It is the outer most layer of cells formed during gastrulation. Some of the ectodermal cells aligned themselves into a columnar pattern to form **neural plate** which is converted into neural tube. Neural tube will form central nervous system while cells of neural crest will form peripheral nervous system. After neurulation all ectodermal cells form epidermis. Epidermis also give rise to hair, sebaceous glands, sweat glands, nails, claws and hoofs.

193 Primary Germ Layers: Endoderm derivatives

Endoderm is the inner most primary germ layer formed during gastrulation. The entire gastrointestinal tract except mouth, pharynx and anus develop from endoderm. Endoderm also give rise to the lining of all glands which are part of digestive system e.g., liver, gall bladder and pancreas. Endoderm also give rise to respiratory tract, thyroid, parathyroid, thymus, pancreas, auditory tube, tympanic cavity and urinary system.

194 Primary Germ Layers: Mesoderm derivatives

Mesoderm is one of the primary germ layer formed during gastrulation. It is present between ectoderm and endoderm. Mesoderm give rise to skeletal system, urinogenital structures, circulatory system, muscular tissues, skin, tendons, ligaments and adipose tissues. Coelom is also formed by out pouching or splitting of mesoderm.

195 Mechanism of Development: Nuclear Equivalence

MECHANISM OF DEVELOPMENT

Nuclear equivalence

In nineteenth century, scientist were working that how developing embryo develop multiple types of cells? Roux-Weismann gave the hypothesis that genome is broken down into smaller and smaller fragment. This is the reason that different cells give rise to different types of cells. In 1892, Hans Driesch Discovered that if two cell stage sea urchin embryo is mechanically shaken to get cells apart, both develop into normal larvae. This shows that both cells contain same genetic material. Hans Spemann **p**laced a minute ligature of hair around salamander zygote. Nucleus containing portion remain dividing while the other didn't. Later at 16 cell stage, one nucleus move to anucleated portion and it also started cell division. When separated both developed into normal embryos. Hans Spemann further concluded that both halves developed normally only if both receive gray crescent. This means cytoplasmic contents paly their role in formation of different types of cells.

196 Mechanism of Development: Cytoplamic Specification

Contents

1. Morphogenetic Determinants
2. Tunicates Example
3. Vertebrate Embryo
4. Mosaic Development
5. Regulative Development

1. Morphogenetic Determinants

- Cytoplasmic components that are unequally distributed within egg
- These determinants control the development of particular cell type
- Fate of blastomeres is specific by type of cytopalsm

2. Tunicate Example

- Some tunicates contain different colours in cytoplasm
- **Yellow:** muscle cells
- **Gray equatorial:** notochord and neural tube
- **Clear:** larval epidermis
- **Gray vegetal:** gut

3. Vertebrate Embryo

- Cytoplasmic specification is relatively less important
- Gray crescent form dorsal lip of blastopore, is essential for gastrulation

4. Mosaic Development

- When a particular blastomere is isolated form embryo, it forms characteristic structure
- In the absence of that cell the structure is missing
- Mosaic development is characteristic of most protostomes

5. Regulative Development

- In many animals, fate of cell depends upon interaction with neighbors rather than cytoplasm itself
- Each cell is able to produce entire embryo
- If a blastomere is removed, the other change their fate to compensate this.

197 Mechanism of Development: Embryonic Induction

Embryonic Induction

The capacity of some cells to evoke developmental process in others is called embryonic development. It is wide spread phenomenon in development. Spemann and mangold first reported embryonic induction. When the dorsal lip of blastopore was transplanted to lateral or ventral side of other embryo, it developed notochord, somite, ectoderm of host and formed the whole system formed. Dorsal lip of blastopore can induce the development of whole embryo, it is called as primary organizer. The phenomenon of induction by blastopore is called primary induction. Formation of other type of cell by inductive process is termed as secondary induction.

198-199 Development: Protostome vs Deuterostome-I

PROTOSTOME VS DEUTEROSTOME

Protostomes belong to group protostomia of animal kingdom. They are part of grade bilateria along with deuterostomes. Platyhelminthes, rotifera, nematoda, annelida, arthropoda and Mollusca are the member phyla of protostomes.

Deuterstomes belong to group deuterostomia of animal kingdom. They are part of grade bilateria along with protostomes. Echinodermata, hemichordate and chordata are the member phyla of deuterostomes.

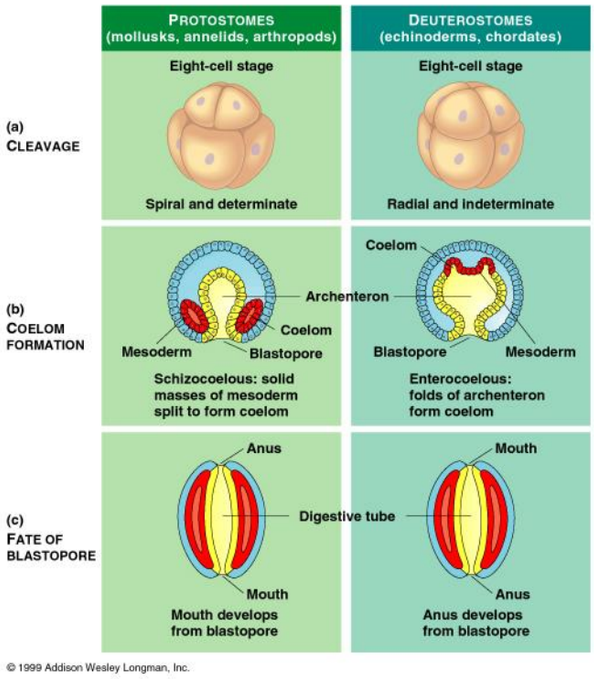


Fig. some major differences between protostome and deuterostome

a. Both differ in cleavage pattern. In protostomes cleavage is spiral while in deuterostomes cleavage is radial.
b. In protostomes cleavage is determinate as the fate of blastomeres is established at very early stage *i.e,* 2 or 4 cell stage. In deuterostomes cleavage is indeterminate as the fate of blastomeres is established at later stages of development.
c. In protostomes blastopore develops into mouth while in deuterostome blastopore develops into anus.
d. In protostomes coelom is formed by splitting of mesoderm. This pattern is called schizocoelous while in deuterostome coelom is formed by out pouching of primitive gut (archenteron). This pattern is called enterocoelous.
e. Development of many protostome include a top shape larva called trochophore larva while some deuterostome have a Kidney-bean shaped larva called dipleurula.

200 Metamorphosis

METAMORPHOSIS

Metamorphosis is an abrupt change in structure and function from immature stage to adult. It takes place in insects, mollusks, crustaceans, cnidarians, echinoderms, tunicates, fishes and amphibians. During metamorphosis animals undergo a series of molts to grow, and each stage between molts is called an instar e.g., mosquito larvae have four instars and butterfly larvae have five instars.

Metamorphosis is beneficial for animals because animal uses different types of food at different stages. This reduces the competition between juvenile and adult stages e.g., tadpole of frog is herbivore while adult is insectivore.

The disadvantages of metamorphosis include short life of adult while metamorphosis stages are pretty long. As larva and adult do not share same food, this may be disadvantageous in food poor environment.

Hormonal Control of Metamorphosis: Metamorphosis is controlled by thyroid hormones in vertebrates. In insects prothoracicotropic hormone from insect brain stimulates ecdysone hormone

released by prothoracic gland. At larval stage juvenile hormone is also released this results in next larval instar. At final stage of larva, juvenile hormone is not released and ecdysone converts larva into pupa.

Metamorphosis is of two types: complete metamorphosis (holometabolous) and incomplete (hemimetabolous) metamorphosis.

201 Holometamorphosis

Holometamorphosis (Complete metamorphosis) occurs in about 88% insects. It is the type of metamorphosis which separates the physiological process of growth from differentiation and sexual stages. There are four stages i.e., egg, larva, pupa and adult stage. Each process is completed without intervention of other stage.

Early embryonic development completed at egg stage. Larval stage is motile and worm like form. Larva actively feed and grow in size. It never possess wings and have simple eyes. Larva undergo metamorphosis to become pupa. It is quiescent, non-feeding and mostly very little motile. During pupa stage, there are drastic changes in morphology and physiology. Adults have wings and reproductive organs. Most insects adult grow very little. Some do not feed and entirely focus on mating and reproduction.

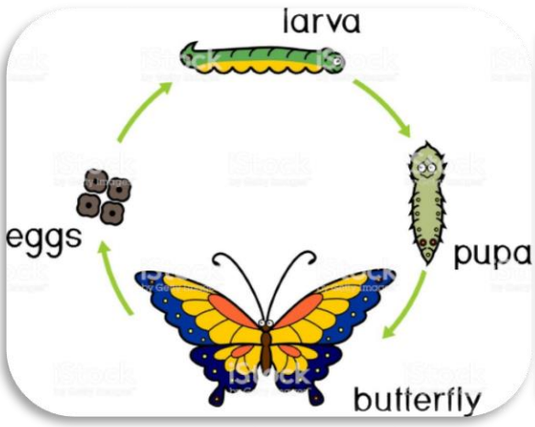


Fig. Complete metamorphosis

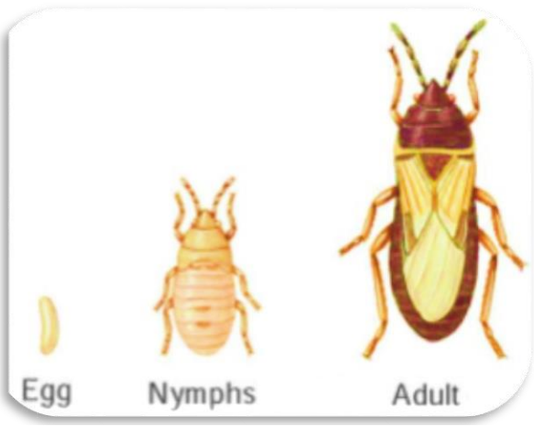


Fig. Incomplete metamorphosis

202 Hemimetamorphosis

Hemimetabolous metamorphosis (Incomplete metamorphosis) is the mode of development of certain insects like aphid, grasshopper and cockroach. They have three distinct stages i.e., egg, nymph and adult.

Initial stages of development are completed at egg stage. Nymph is an immature form hatched out from the eggs. It lacks wings and have underdeveloped reproductive organs. They adult have wings and reproductively mature.

203 Hierarchy of Ecology

Contents

1. Ecology
2. Ecological Hierarchy
3. Organism
4. Population
5. Community
6. Ecosystem
7. Biosphere

1. Ecology

- It is the branch of biology that deals with the relationship of organisms to one another and with their physical environment

2. Ecological Hierarchy

- It is the way in which ecological components interact with each other
- There are five levels
    - Organism
    - Population
    - Community
    - Ecosystem
    - Biosphere

3. Organism

- Organism is the first level in ecological hierarchy
- All the living things come at this level
- E.g., human, lion, tiger, cow, tree, shrub
- The way how organism interact with its environment is studied at this level

4. Population

- Group of organism of same species living at same geographical location at same time
- At this level we study how the members of group live together and interact with each other

5. Community

- Groups of different species living together in a given area
- The number of species represent the diversity of community
- Community ecology include
    - Predation
    - Parasitism
    - Competition

6. Ecosystem

- It is the study of all population in a community and their physical environment
- Biotic components and abiotic components
- How living organism make the use of abiotic components
- How they get energy
- Effect of physical factors on organisms

7. Biosphere

- Biggest of all levels
- All the area of earth and atmosphere occupied by living organisms
- Main focus is on:
    - Geology
    - Climate
    - Water bodies
    - Latitude
    - Longitude

204 Biotic and Abiotic Factors of an Ecosystem

Contents

1. Ecosystem

2. Components of Ecosystem
3. Biotic Factors
4. Abiotic Factors

1. Ecosystem

- It is the unit of ecology where living organisms interact with their physical environment
- Transfer of energy and material between organisms takes place within an ecosystem
- The incorporation of energy and materials in an ecosystem is called its productivity

1. Ecosystem

- Ecosystems are very dynamic
- They are controlled by different external and internal factors

2. Components of Ecosystem

- Ecosystem has two main components/factors
- Biotic factors/components
- Abiotic factors/components

3. Biotic Factors

- All living organisms in ecosystem make biotic factors.
    - Producers
    - Consumers
    - Decomposers

4. Abiotic Factors

- All nonliving things in ecosystem make abiotic factors.
    - Temperature
    - Moisture
    - Light
    - Soil
    - Air
    - water

205 Abiotic Factors: Temperature

Contents

1. Temperature
2. Loss of Heat to Environment
3. Gain of Heat
4. Effect on Habitat Requirement
5. Body Size and Metabolism

1. Temperature

- It is the measure of heat present in a body
- Animals extend a part of their energy on regulating temperature
- Temperature influence metabolic rate

2. Heat Loss to Environment

- Conduction
- Convection

- Radiations
- Evaporation

3. Heat Gain

- Radiations
- Conduction
- Metabolism

4. Effect on Habitat Requirement

- Thermoregulatory needs influence habitat requirement:
- Availability of food
- Availability of water
- Availability of shelter

5. Body Size and Metabolism

- Larger animals have low metabolic rates
- Smaller animals have higher metabolic rate
- Mouse have 20 times higher metabolic rate than elephant
- This is probably due to large surface to volume ratio for heat exchange

206 Abiotic Factors: Moisture

Contents

1. Moisture
2. Gain of Water in Animals
3. Loss of Water in Animals
4. Measures to Reduce Water Loss
5. Water Regulation in Aquatic Animals

1. Moisture

- It is the amount of water diffused in environment as vapours, within solid or condensed on surface
- In air the amount of water is also called humidity
- High moisture….low water loss
- Less moisture….more water loss

2. Gain of Water in Animals

- Animals acquire water from:
- Food
- Drinking
- Metabolism

3. Loss of Water in Animals

- Water is lost along with
- Urine
- Feces
- Gaseous exchanges
- Evaporative cooling

4. Measures to Reduce Water Loss

- Animals opt following strategies to reduce water loss

- Dry feces
- Concentrated urine or semisolid excretory wastes
- Nocturnal behavior
- Ability to store water

5. Water Regulation in Aquatic Animals

- Freshwater animals live in hypotonic environment: conserve ionic loss
- Marine animals live in hypertonic environment: avoid salt accumulation and dehydration

207 Abiotic Factors: Light

208 Abiotic Factors: Soil

Contents

1. Soil
2. Soil Characteristics
3. Soil as Habitat
4. Effect on Plant and Animal Life

1. Soil

- Soil reflects the geological and biological history
- It effects the animals life directly and indirectly

2. Soil Characteristics

- Characters include
    - Texture
    - Amount of organic matter
    - Fertility
    - Water holding ability
- They directly affect the diversity of life on or in the soil

3. Soil as Habitat

- Soil is the habitat of large variety of animals including
    - Protozoa
    - Nematodes
    - Earthworms
    - Arthropods
    - Salamanders
    - Snakes
    - Moles

4. Effect on Plant and Animal Life

- Soil have obvious effect on plant growing in that area
- Different kinds of plants influence the animal life whether they are living in or on the soil

209 Population Growth

Contents

1. Populations change over time

2. Survivorship Curves
3. Types I Curve
4. Type II Curve
5. Type III Curve

1. **Population Change Over Time**

- Population are dynamic in nature
- With the passage of time they are subjected to change due to:
    - Birth
    - Deaths
    - Migration

2. Survivorship Curves

- Individuals within population are estimated by survivorship curve
- It is the logarithmic plot between survivors and their age
- Curves types are:
    - Type I
    - Type II
    - Type III

3. Type I Curve

- Also called convex population survivorship curve
- Individuals of type I, population survive to an old age, then rapidly die
- Environmental factors rarely cause mortality
- Individuals live their potential life span
- E.g. human population

4. Type II Curve

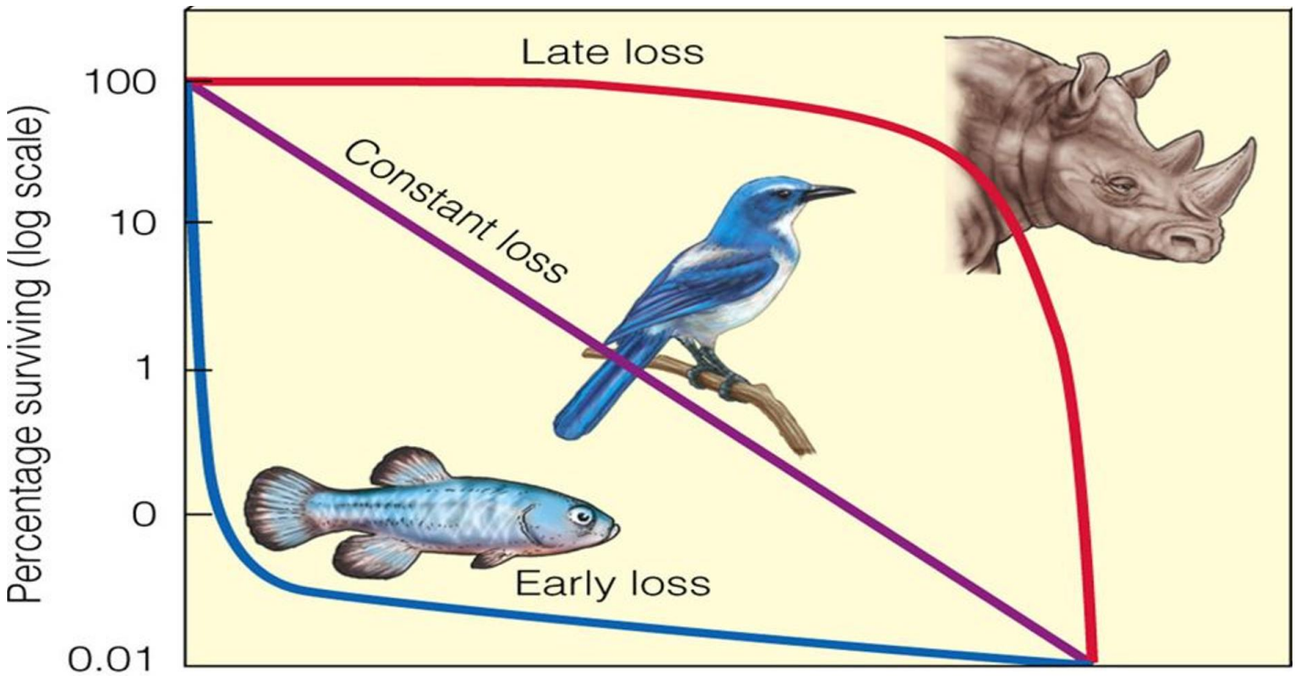
- Also called diagonal population survivorship curve
- Individuals of type II, population have constant probability of death throughout their life
- Environmental factors influence mortality
- E.g., populations of birds and rodents

4. Type II Curve

- Mortality in natural populations is often high
- For Example, each cohort of birds experience 50% mortality each year
- Potential life span of American robin is 11 years, however average life is 1.4 years

5. Type III Curve

- Also called concave population survivorship curve
- Individuals of type III, population have very high juvenile mortality
- Adults have much lower mortality rate
- E.g., fishes and many invertebrates



210 Exponential Growth

Contents

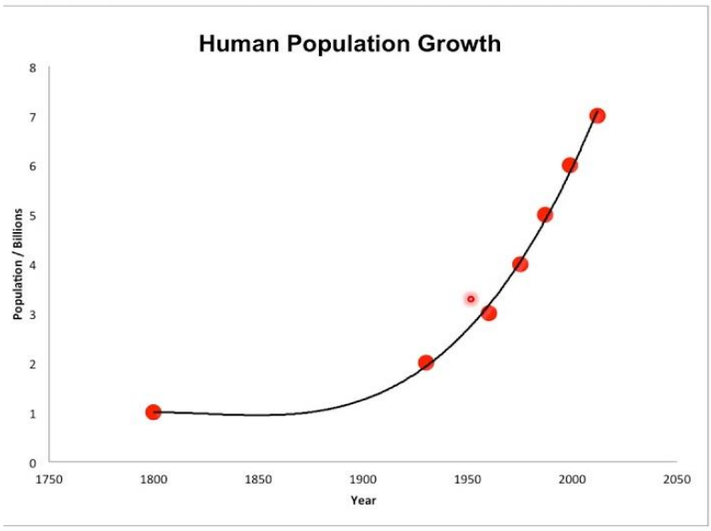
1. Exponential Growth
2. Exponential Growth in Fruit Flies
3. Factors Affecting Reproductive Potential
4. Biotic potential

1. Exponential Growth

- Population growth in which number of individuals increase by a constant multiple in each generation
- Population increases by same ratio per unit time

2. Exponential Growth in Fruit Flies

- A few fruit flies in a large jar with abundant food
- 1 female lay 50 eggs
- 2 weeks for maturity
- Each female of 2nd generation produce 50 more flies and so on



3. Factors Affecting Reproductive Potential

- Number of offspring produced
- Survival to reproductive age
- Reproductive period duration
- Length of time taken to maturity

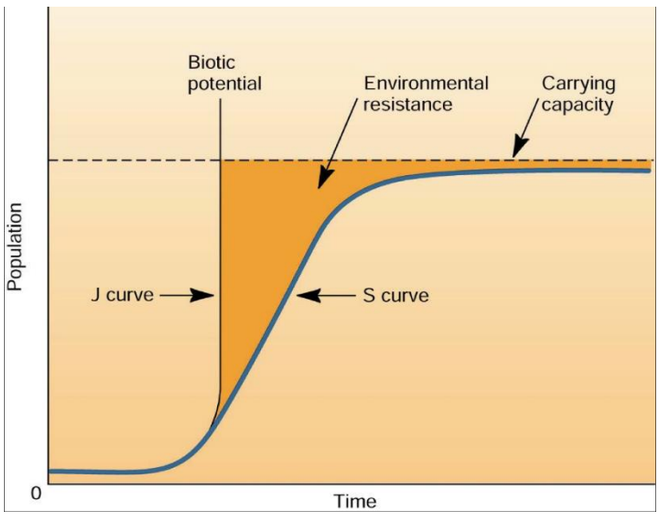
4. Biotic Potential

- The capacity of a population to increase maximally is called biotic potential
- It is also called intrinsic rate of growth

211 Logistic Population Growth

Contents

1. Environmental Resistance
2. Carrying Capacity
3. Logistic Population Growth



1. Environmental Resistance

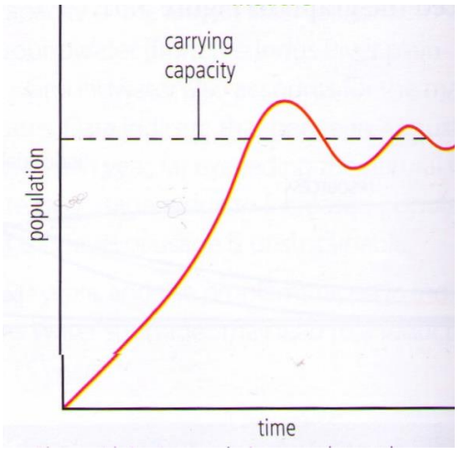
- The constrain of climate, food, space and other environmental factors is called environmental resistance
- Due to these factors exponential growth can not occur indefinitely

2. Carrying Capacity

- The size of population that a particular environment can support is called its carrying capacity
- It is symbolized by *K*

3. Logistic Population Growth

- S-shaped curve: At carrying capacity population growth levels off
- J-shaped curve: when population overshoot carrying capacity. Growth curve may fluctuate around K or assume J-shape



212 Population Density

Contents

1. Population Density
2. Density-independent Factors
3. Density-dependent Factors

1. Population Density

- It is the number of individuals living per unit area.
- Population density impact populations in predictable ways

2. Density-independent Factors

- These factors affect population regardless of their number
- Weather conditions
- Earth quake
- Pollutants
- Deforestation
- Hurricane

- Fire

3. Density-dependent Factors

- These are the factors which are more severe when population density is high
- Competition for resources
- Disease
- Predation
- Parasitism

213 r and K selection

214 Animal Inactivity

Contents

1. Animal Inactivity
2. Torpor
3. Hibernation
4. Winter Sleep
5. Aestivation

1. Animal Inactivity

- When food is scarce or environmental conditions are severe, the animals stop feeding and metabolic activities are decreased.
- Torpor
- Hibernation
- Winter Sleep
- Aestivation

2. Torpor

- Period of decreased metabolism and lower body temperature in daily activity cycle
- Hummingbird and bat should eat constantly
- At the end of torpor body temperature and metabolism increase quickly

3. Hibernation

- Time of low body temperature and decreased metabolism from weeks to months
- The set point of thermoregulatory center drop to 20°C
- Thermoregulation not suspended in homeotherms
- Rodents, shrews and bats
- Ectotherms may not regulate

4. Winter Sleep

- Period of winter inactivity
- Body temperature do not drop substantially
- Animals can wake and become active
- e.g., Bears

5. Aestivation

- inactivity period of some animals to withstand in drought
- Animals live in burrows

- Do not eat or drink
- Emerge after moisture returns
- E.g., many invertebrates and frogs

215 Interspecific Interactions: Competitive Exclusion

Contents

1. Competitive Exclusion
2. Universality of Competitive Exclusion
3. Example of Competitive Exclusion

1. Competitive Exclusion

- It states that two species competing for same requirement of food, habitat, nesting sites and other conditions of life can not coexist

2. Universality of Competitive Exclusion

- It influenced the scientists for more than 50 years
- Numerous studies in nature failed to document its universality
- This is because no two species have identical requirements

3. Example of Competitive Exclusion

- In California Golden-naveled scalesucker is a wasp parasite of red scale insect.
- Scientists introduced lingnan scalesucker from China
- Within 2 years lingnan replaced golden-naveled scalesucker

216 Interspecific Interactions: Coexistence

Contents

1. Coexistence
2. How Coexistence Occurs?
3. Coexistence May Turn into Competitions



1. Coexistence

- It is the phenomenon in which competing species live together

2. How Coexistence Occurs

- Coexistence can occur when species utilize resources in slightly different way
- MacArthur studied 5 species of warblers that use same caterpillar from different region of trees

2. How Coexistence Occurs

3. Coexistence May Turn into Competition

- By partitioning resource can be divided
- However, when resources are not enough then similarities become limiting factors and competitive exclusion may occur

217 Interspecific Interactions: Coevolution

Contents

1. Coevolution

2. Role of Natural Selection
3. Flowering plants and Pollinators
4. Hummingbird and Passion Flowers

1. Coevolution

- It is a phenomenon in which ecologically related species sometimes coordinated such that each species exert a strong selective influence of the other

2. Role of Natural Selection

- Coevolution…..when competing for same resources or …... predator-prey relationships
- Natural selection favours the development of selective characters
- If balanced, no winner

3. Flowering Plants and Pollinators

- Plants display adaptation to attract specific pollinators
- Elaborate olfactory and visual adaptations such as bright colour, petal arrangement
- Flowers red and tubular while hummingbird has long beak and see red colour

4. Hummingbird and Passion Flower

- *Ensifera ensifera* is a hummingbird with extremely long beak
- Passion flowers store nectar deep within tubular flower
- Competition for nectar feeding is limited

218 Interspecific Interactions: symbiosis

Contents

1. Symbiosis
2. Facultative and Obligate Symbionts
3. Ectosymbiosis
4. Endosymbiosis

1. Symbiosis

- Symbiosis is a close and long term association between two different organisms
- Organism may get benefit or harm
- Mutualism
- Commensalism
- Parasitism

2. Facultative and Obligate Symbionts

- Organisms are entirely dependent on symbiosis are obligate. *E.g.,* Fungus of lichen
- Which can survive as independent and in association. *E.g.,* algae of lichen

3. Ectosymbiosis

- It is a symbiosis in which symbionts live on the body surface of the host or in the digestive tact
- E.g., Lice, barnacles and cleaner fish

4. Endosymbiosis

- It is the relationship in which one symbiont live within the tissue of other.
- E.g., nitrogen fixing bacteria in root nodules, algae inside the corals

219 Interspecific Interactions: Parasitism

Contents

1. Parasitism
2. Ectoparasites and Endoparasites
3. Intermediate Host and Definite Host
4. Parasitoids

1. Parasitism

- It is a relationship in which on organism (parasite) live in or on other organism (host).
- Parasite gets benefit while host is harmed.
- Host usually survives in this interaction

2. Ectoparasites and Endoparsites

- Parasites living on the body surface of host are ectoparasites. E.g., lice, leeches
- Parasites living within the body of host are endoparasites. E.g., tape worm, liver fluke, plasmodium

3. Intermediate Host and Definite Host

- Some parasites have complex life histories with multiple host
- Intermediate host: Immature stages and asexual reproduction of parasite
- Definite host: host where parasite spends its mature life and reproduce sexually

4. Parasitoids

- These are the animals that deposit eggs or spend developmental stages on the other animals.
- As development proceed they kill the host.
- They have properties of predator and parasite
- E.g., parasitoid wasps

220 Interspecific Interactions: Commensalism

i) Commensalism

It is the symbiotic relationship between two organisms in which one member benefited while other is neither helped nor harmed. It is a small fish with disk on head. By these disks they get attached with large sharks and whales. When shark or whale feed they eat the scrapes of food. Similarly, tree frogs live in green leaves of tree. They feed on insects and are protected from predators in green leaves of plants. A commensal do not harm its host. However, sometimes environmental factor change and a commensal becomes parasite. e.g., *Entamoeba gingivalis* lives in mouth of human may cause teeth and gum decay occasionally.

221 Interspecific Interactions: Mutualism

i) Mutualism

It is the type of symbiotic association in which both animals get benefit from each other. For example, clownfish and sea anemone live together in a close association. Sea anemone protect the clownfish while clownfish provide food to sea anemone. Similarly, in cleaning mutualism small fishes or invertebrates may swim into the mouth of larger fishes to remove ectoparasites while cleaning fishes find safety from predators.

222 Interspecific Interactions: Predation

a. Predation

It is type of interaction in which on animal kills and eat other animals. The animal which eats is called predator and the animal being eaten is called prey. Interaction between predator and prey are complex. A large population of prey supports more number of predators. A decreased population of prey results in a decreased population of predators. Predators also effect the distribution of prey. For example, the lake trout of North America has never moved to freshwater of Siberian freshwater system. This is probably due to predation by Pacific sea lamprey.

223 Interspecific Adaptations: Camouflage

b. Camouflage

It is the phenomenon in which the colour pattern of animal helps to hide the animal from others. There are two major types of camouflage: cryptic coloration and countershading. In **cryptic coloration** camouflage animals takes pattern according to surroundings. Some prey animals are camouflage from predators. e.g., arctic hare white fur coat in winter takes pattern of snow and brown in spring takes pattern of soil. Similarly predators use camouflage to avoid detection by prey. For example, Asian tiger's rufous-red coat with black strips help it to undetected during stalking. **Countershading** camouflage is common in eggs of frog and toad. Eggs are darkly pigmented on top while light pigmented on the bottom. From top dark top mixes with dark bottom. From bottom light colour blends with light air water interface.

224 Interspecific Adaptations: Aposematic Coloration and Mimicry

a. Aposematic Coloration and Mimicry

Phenomenon in which animals protect themselves by advertising themselves dangerous or distasteful is called **aposematic coloration**. They actually warn the predators to not attack on them. Dandrobatid frogs of central and South America have poisonous secretion and are brightly colored. Their bright colour is a warning to the predators.

Mimicry is the phenomenon in which one species gains resemblance with other species and gets protection from predators. This resemblance may be with one or more than one species. In **batesian mimicry,** one species resembles with other species and protected by aposematic coloration. Flies belonging to family Syrphidae resemble with bees or wasps, thus protected. Here flies are mimic and bee or wasps are model. **Mulerian mimicry** both mimic and model both are distasteful. If predator eats any one of these, it will not prey again any one of them.

225 Community structure

COMMUNITY STRUCTURE

Communities are not the random mixture of populations but they are organized entities. Some members of community have greater importance in organizing community. Species that is responsible for determining community characteristics is called **dominant species**. e.g., a stream having a large population of rainbow trout that will determine the population of certain invertebrates. Those species on which several other species are dependent are called **keystone species**. Removal of keystone species will cause the death of dependents. Cyclic activities also characterize the community. Community structure based on cyclic activities of daily or seasonal basis is called temporal structure of community. Ecosystems where layering is present are have spatial structure. Different populations are present at different layers of forests. Variety of animals present in a community is called **community diversity**. It is also called community richness. Stable climate, more resources, high productivity, moderate predation promote diversity.

226 Community stability

Community Stability

Communities are born and they die. Some changes in community are not permanent. They are replaced by next generation. Whereas some changes are permanent.

It is the sequence of community types during maturation of an ecosystem is called ecological succession. It is the change in community structure in a predictable way. It is the first community established in an area devoid of life is called pioneer community. e.g., a lake formed by glacier and the first community establish in this lake will be **pioneer community**. During ecological succession and ecosystem is occupied by different communities in a sequence and each successional stage is called a seral stage. The final community during an ecological succession is called **climax community**. They are complex and have more species diversity.

The process of succession in an area that was not supporting any life previously is called **primary succession**. E.g., Ecological succession in a lake formed by glacier or land exposed by land sliding. The process of succession where previous life destroyed by some natural catastrophs (e.g., fire) is called **secondary succession**.

227 Ecological Niche

ECOLOGICAL NICHE

It is the role of an organism in an ecosystem which include all the attributes of life style of an animal i.e., where it lives? what it eats? where it search for food? what physical parameters it prefers?

Ecological niches of two species may overlap. This will result a competition between them. Animals will compete for same resources. Competition between species may be reduced if resources are divided. One part of resources is used by one species and second part by other species. This mechanism is called **niche diversification**.

228 Food Chains

FOOD CHAINS

It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism. It tells us the sequence to eat and being eaten. Food chain is important: i). To understand the feeding relationships between organisms. ii). To explore the energy flow mechanisms and matte circulation. Iii). To understand the problems of biomagnification.

Food chains are mainly of two types: grazing food chain and detritus food chain.

a. Grazing food chain

It is the food chain that starts from the living green plants. Green plants are eaten by herbivores which are eaten by carnivores. Such food chains are directly dependent on sunlight.

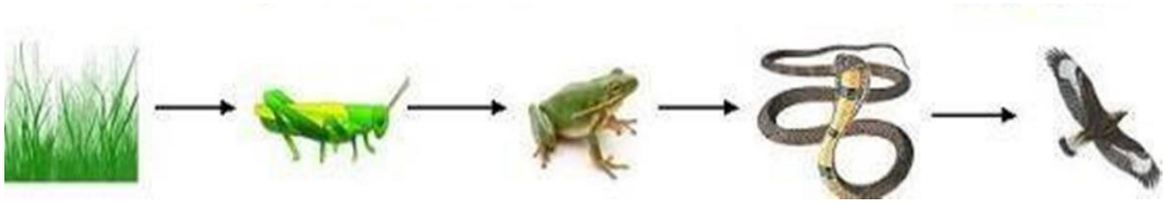


Fig. Grazing food chain of an ecosystem

Detritus Food Chain

It is the food chain that starts from dead organic matter. Organism of detritus food chain are

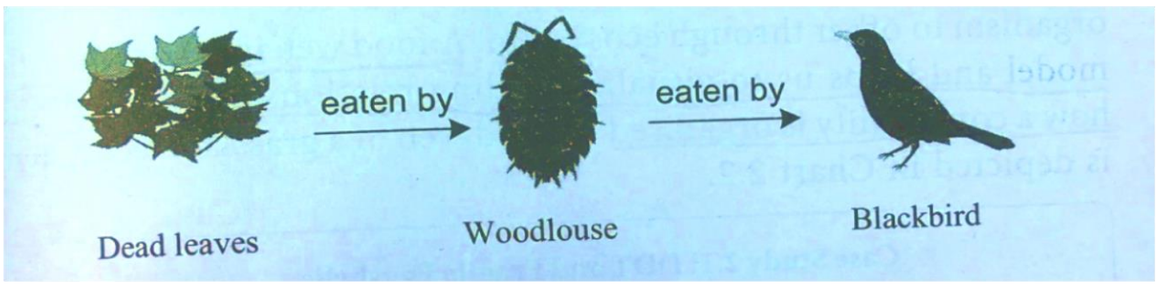


Fig. Detritus food chain of an ecosystem

229 Food Webs

Food Webs

It is a complex network of organisms through which energy and matter is transferred in an ecosystem. Food web consists of many interconnected food chains**.** Grazing food webs are the food webs where plants make the starting point. These food webs based on algae and green plants. The detrital food webs are the food webs where decomposers make the starting point. Bacteria and fungi move the food material to the carnivores.

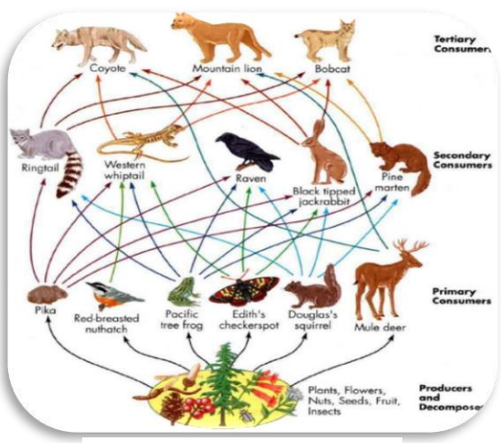


Fig. the grazing food web

230 Trophic Structure: producers

TROPHIC STRUCTURE

As food webs are very complex. The animals feeding in a food web are distributed into different trophic levels. Autotrophs make 1st trophic level.

a. Producers

These are the autotrophs which obtain nutrition from inorganic compounds. They form the first trophic level of an ecosystem. Producers are mainly of two types: Photoautotrophs and chemolithotrophs.

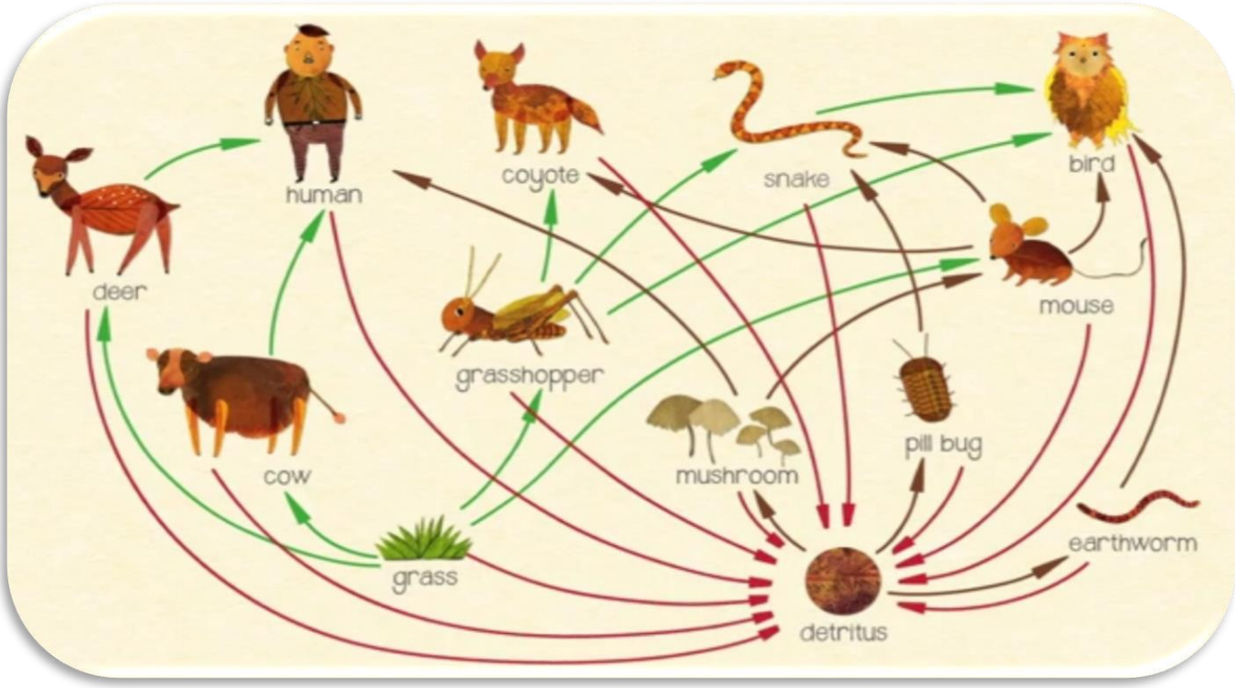


Fig. detritus food web

Photoautotrophs use sunlight as energy source for the synthesis of food from inorganic compounds. They include green plants, algae and photosynthetic bacteria. Photosynthesis transform sunlight into chemical energy. Out of total sunlight reaching to earth 30% reflects back, 20% is absorbed by atmosphere, 50% absorbed by earth and less than 1% of sunlight is used by photosynthesis. **Chemolithotrophs** obtain energy by oxidizing inorganic compounds. They usually live in extreme environments. e.g., sulfur bacteria.

231 Trophic Structure: consumers

a. Consumers

Organisms which cannot synthesize their food from inorganic compounds. They eat other organisms to get nutrition. All animals are consumers. Those organisms which feed on plants are called **primary consumers**. They are also called herbivores. E.g., insects, grain eating birds. Different reptiles, birds and mammals which feed on leaves and other vegetation are also herbivores. Those animals which feed on herbivores are called **secondary consumers.** They are also called carnivores. e.g., lion, tiger and frog. Those animals which feed on secondary consumers are called **tertiary consumers.** e.g., a snake when eating a frog, a jaguar when hunting for a crocodile.

232 Trophic Structure: Decomposers Ecological efficiencies

a. Decomposers

These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds. They are also called saprotrophs. Chief decomposers of an ecosystem are bacteria and fungi. They release extracellular enzymes which breakdown substrate molecules. Organisms at all trophic level when die are used by decomposers. If there are no decomposers mounts of dead bodies will be everywhere. All useful minerals will remain lock in dead bodies. There will be little space to live. Decomposition also recycle the nutrients in ecosystem.

233 Ecological efficiencies

ECOLOGICAL EFFICIENCIES

It is the efficiency by which animal at a trophic level convert food into new biomass. A large amount of food is not digested by the animals. Herbivores have lot of plant biomass, but they are less efficient. Carnivores are more efficient. At an average there is 10% biomass conversion efficiency at each trophic level. Biomass conversion also depends upon the maintenance requirement of animals. Efficiencies range for less than 1% for most herbivores to more than 35% for ectotherm carnivores.

234-235 Ecological pyramids

ECOLOGICAL PYRAMIDS

It is the graphical representation of relationship between organisms at different trophic levels of an ecosystem. There are three most common types of pyramids: Pyramid of number, pyramid of biomass and pyramid of energy.

a. Pyramid of Numbers

It is the graphical representation of total number of different organisms present at each trophic level. This also shows the population of organisms in a food chain.

For example in a grass land the survey was conducted to check the trophic structure. The number of producers was 5.842 x 10^6, primary consumers 7.08x10^5, secondary consumers: 5.4x10^4 and tertiary consumers was one.

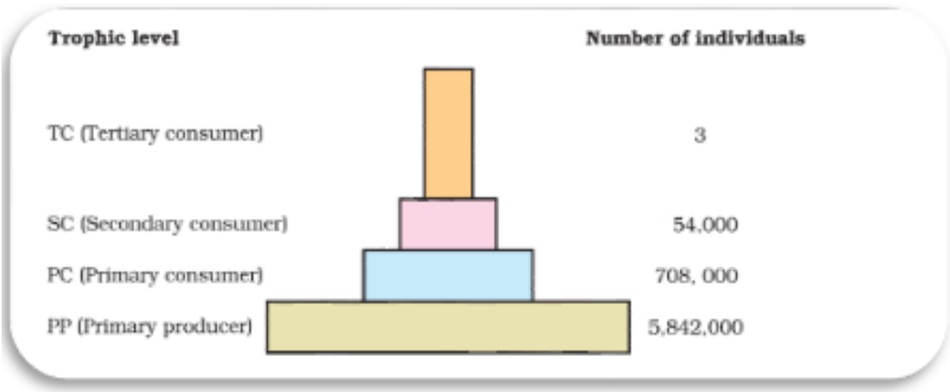


Fig. Pyramid of number

b. Pyramid of

It is the representation of total mass of living matter present at each trophic level. Biomass decrease at each ascending trophic level. Biomass is measured in grams of dry mass per square meter. For example in a pond the survey was conducted to check the trophic structure. The biomass of producers was 809 g/m^2, primary consumers was37 g/m^2, secondary consumers was 11 g/m^2 and tertiary consumers was 1.5 g/m^2.

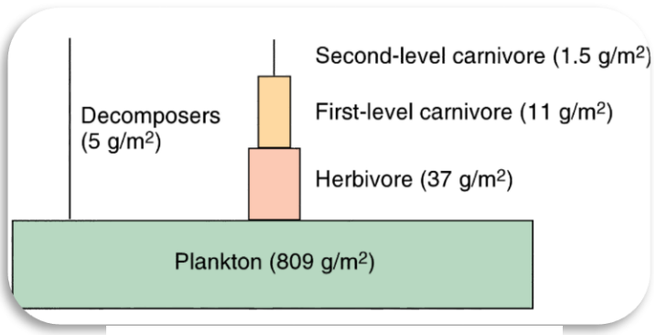
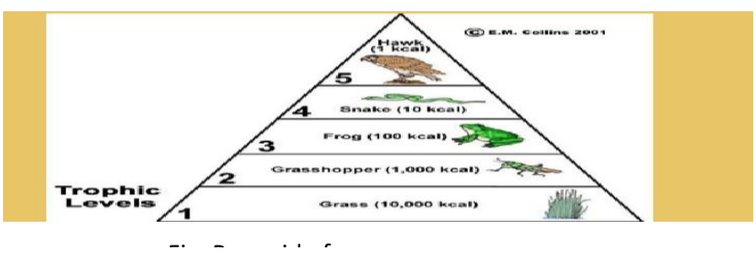


Fig. Pyramid of biomass

c. Pyramid of Energy

It is the representation of total amount of energy present at each trophic level in an ecosystem. It is always upright. Energy pyramids help to compare the productivity of ecosystems.



236 Energy and Matter Flow in An Ecosystem

ENERGY AND MATTER FLOW IN AN ECOSYSTEM

1. Energy Budget

It is the balance sheet of energy gained and energy expenditure of an organism. An organism is an open thermodynamic system which exchange energy with its surrounding in three ways *i.e.,* heat, work and potential energy of biomolecules. Energy budget allocation varies for ectotherms and endotherms. Ectotherms rely on environment as heat source while endotherms rely on metabolic heat to maintain body temperature.

When an animal eats, food passes through its gut and nutrients are absorbed. Total amount of energy assimilated by an organism when an animal eats food is called **gross productivity**. Most of the energy assimilated is used in metabolic demands which include cellular metabolism and regulation of body temperature. The amount of energy stored in the form of biomass is called **net productivity**.

4. Energy Flow

Organisms require energy in one form or the other. Energy flow in an ecosystem is major factor for survival for great number of organisms. Energy flows via food chain and food web. Source of energy is sunlight. Normally 10% of energy flows to next trophic level. Energy flow is unidirectional.

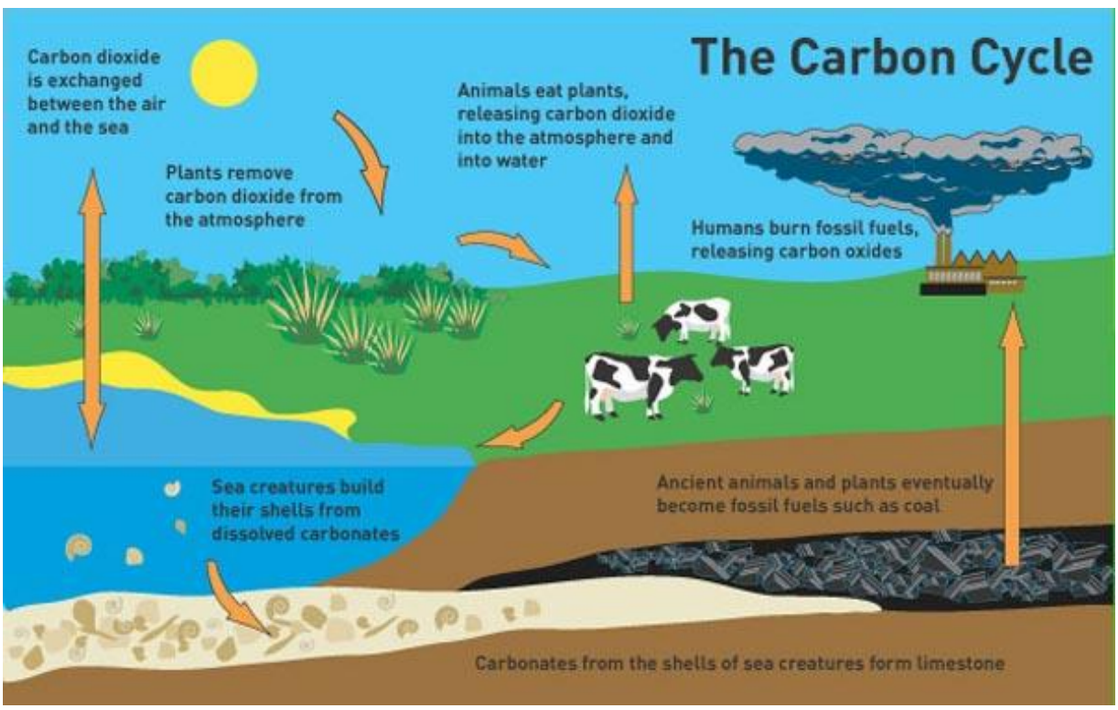
All elements essential for life are derived from environment. They make all form of molecules for life. When animals die and their dead bodies are decomposed, these materials are recycled in the ecosystem. Nutrient (material) flow in ecosystem in a cyclic way.

237-238 Carbon cycle

CARBON CYCLE

Carbon is the basic element of all organic molecules. All biomolecules have a carbon backbone. Carbon has ability to bond with four other atoms at the same time. Carbon is plentiful on earth and is rarely limiting factor. Reservoir for carbon is CO_2 in atmosphere and water. Photosynthetic plants use this carbon dioxide of air and convert organic compounds. These plants are being eaten by animals. Animals utilize these organic compounds for the synthesis of different biomolecules and also use as a source of energy. When organic matter is completely oxidized, it releases CO_2 into atmosphere. In aquatic environment, CO_2 combines with water to form H_2CO_3. When CO_2 level in water decreases H_2CO_3 breaks to release CO_2. Some carbon is tied up in the form of calcium carbonate ($CaCO_3$). This $CaCO_3$ is in the part exoskeleton and endoskeleton of mollucs, echinoderms, and other invertebrates. They ultimately form limestone reservoirs.

A large supply of carbon is also tied up in fossil fuels. During 65 million year of Carboniferous period (280mya), organic matter got buried and converted into coal, oil and gas. When fossil fuels are used, carbon returns to atmosphere.



239-240 Nitrogen cycle

LESSON 239-240

NITROGEN CYCLE

Nitrogen is an essential component of proteins, nucleic acids and ATP. It is also found in membrane lipids and modified carbohydrates. Atmosphere is main reservoir nitrogen. About 78% nitrogen present in air, but this is not directly available to plants. Plants can use nitrogen in the form of NH_3 and NO_3^-. Animals obtain nitrogen from plants. Atmosphere nitrogen converted to NH_3. This process is called **nitrogen fixation**. Aqueous form of NH_3 is NH_4^+. Nitrogen fixation is mainly done by bacteria in the roots of leguminous plants and cyanobacteria. Lightening is another source of nitrogen fixation. In industry such as fertilizer industry nitrogen fixation is done by artificial method.

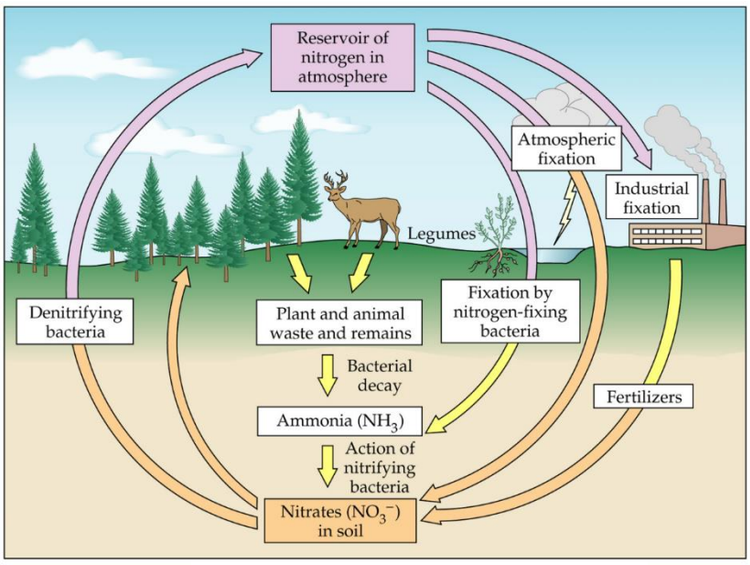


Fig. The nitrogen cycle

In the process is **nitrification** during which NH_3 is converted to nitrate (NO_3^-). It is a two step process. In first step NH_3 is converted to nitrite (NO_2^-) and in second step NO_2^- is converted to NO_3^-. *Nitrococcus* and *Nitrobacter* do this job, respectively. Plants use these nitrates to synthesize proteins and nucleic acids acid. Some plants have ability to use ammonia (ammonium ions) as nitrogen source. Animals utilize plant materials as a source of nitrogen. The process by which nitrogen is incorporated into plants and animals is called assimilation. When plants and animals die or release nitrogenous waste, this organic nitrogen is also converted into ammonia by a process called **ammonification**. Ammonia released in this process may be directly assimilated by the plants or used after nitrification. Some bacteria convert nitrates back to nitrogen is called **denitrification**. This process is completed under anaerobic conditions. *Pseudomonas* and *Aachromobacter* are the examples of denitrifying bacteria.

241 Phosphorus cycle

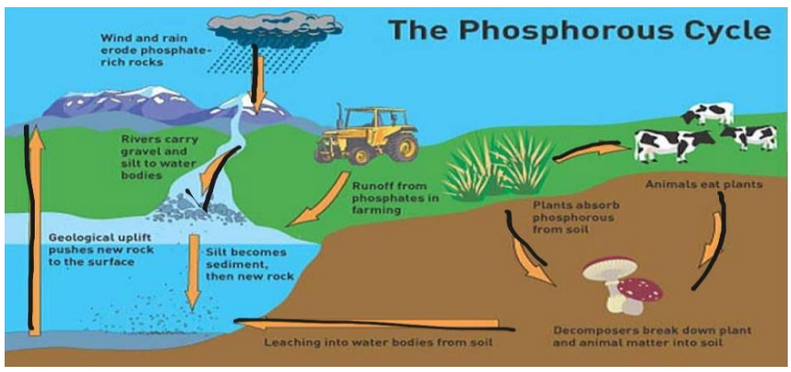
PHOSPHORUS CYCLE

1. Importance of Phosphorus

Phosphorus is an essential component of DNA, RNA, ATP, Phospholipids, Vertebrate skeleton and Teeth. Sedimentary rocks contains deposits of phosphorus. Extraction of phosphorus takes place by weathering, rain and other erosion processes. Plant, microbes and fungi absorb phosphorus from soil. Animals use phosphorus by utilizing plants. Some phosphorus also comes through water. As plants and animals die, phosphorus is return back to the environment via decomposition. Living organisms may again use this phosphorus. Some phosphorus may run along with water and become part of sediments. Human has significant impact by using phosphorus in fertilizers. Agricultural runoff takes phosphorus to water bodies which excessive growth of algae. This phenomenon is called eutrophication.

242 Sulfur Cycle

SULFUR CYCLE



Sulfur is an essential component of proteins, enzymes and vitamins. It is abundant in nature and is rarely a limiting factor. Sulfur is a bright yellow crystalline solid. Most of it is tied up in rock and buried in depth of oceans. Atmosphere also contain oxides of sulfur. The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called mineralization and the step at which sulfides and elemental sulfur are converted to sulfates is called oxidation. Sulfates are reduced by plants, fungi and bacteria and incorporated

243 Water cycle

WATER CYCLE

Water is essential for life. Life cannot exist without water. About 97% water is marine with high salt concentration, 3% freshwater, mostly glaciers and groundwater. Less than 1% of water is available to human. Water in the oceans, lakes, swamps, rivers and soil when heated is turned into vapors. Similarly water also evaporates from surfaces of plants and animals. The loss of water from plants is by transpiration. Water also evaporates directly from ice by the process of sublimation. Water vapors move up higher and condense to form clouds. Clouds are pushed around the world by moving air. Further cooling causes precipitation. Precipitation may be in the form of rain, hail or snow. About 78% of

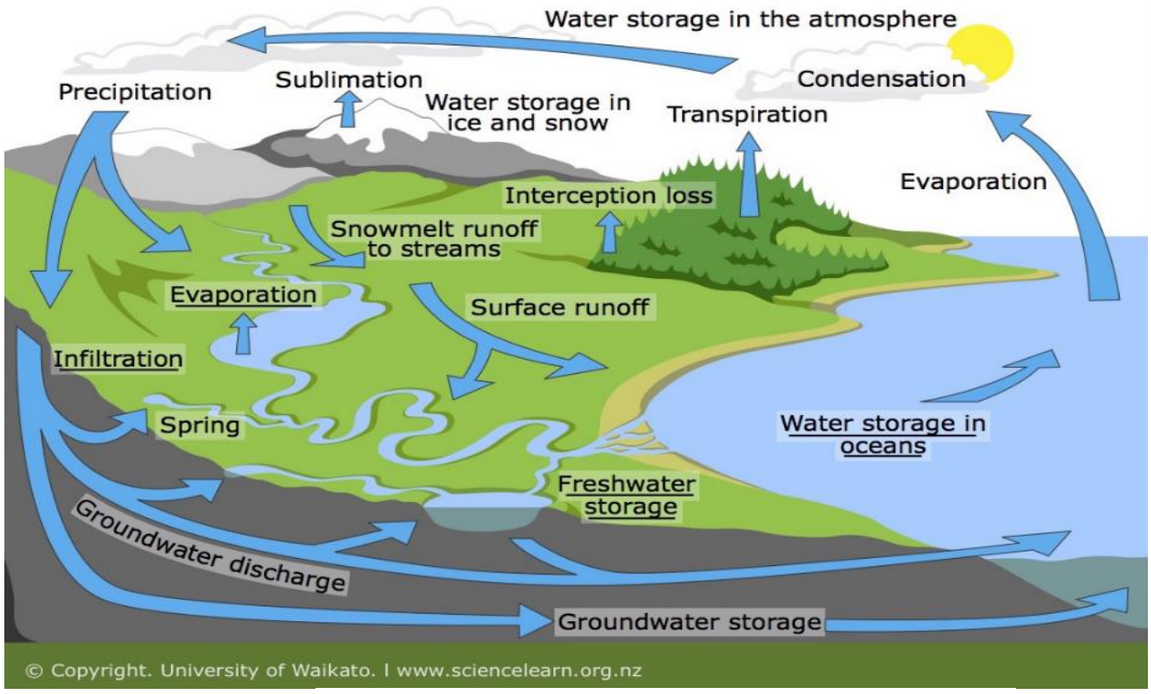


Fig. The water cycle

precipitation occurs over oceans. Rain water flows across the land is called runoff. It includes: Surface runoff and channel runoff. Flow of water from surface into ground is called infiltration. It becomes soil moisture or ground water. Underground water is present in aquifers. This water may return to surface as springs or by pumps. Under ground water moves slowly and may remain in aquifer for thousands of years. Eventually is reaches to the oceans and become part of ocean water.

244 Biosphere and its Subdivision

BIOSPHERE AND ITS SUBDIVISION

Biosphere is thin outer layer of earth which can support life. It includes all life on earth and the physical environment on which living organisms exist and interact. Life exists in soil, water and air. Biosphere is subdivided accordingly: Lithosphere, hydrosphere and atmosphere.



Fig. Biosphere: atmosphere, hydrosphere and lithosphere

Lithosphere is the rigid material on the surface of earth. Most of the terrestrial animals live on lithosphere. Lithosphere is the ultimate source of all mineral elements required by all living things. **Hydrosphere** is the water on the surface of earth. It extends into the lithosphere and atmosphere. Water is distributed on earth by global hydrological system. Water returns to oceans by precipitation. **Atmosphere** is the gaseous content of biosphere. Atmosphere extends up to 3500 km above the surface of earth. Life in atmosphere is present in lower 8-15km called troposphere. The main gases in troposphere are nitrogen, oxygen and carbon dioxide.

245 Ecosystem and Biome

ECOSYSTEM AND BIOME

Ecosystem is the interaction between living and nonliving components of the environment while an area of earth with distinctive association of plants and animals is called biome. Major biomes of the world include: tundra, coniferous forests, deciduous forests, grasslands and deserts. Botanists divided terrestrial environment long ago into different biomes. Characterization of biome on the basis of animal life is difficult. A biome is identified by dominant plant formation. Because plants support a particular type of animals, so the biomes also have characteristic type of animals. The boundaries of biome are not distinct and contain trees of both types of biomes. This indistinctive boundary having a mixture of plants is called **ecocline**.

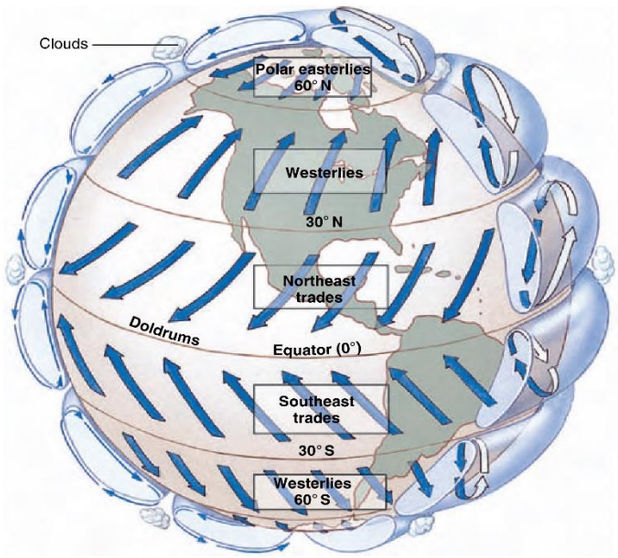


Fig. Coriolis effect and cells

Biomes are distinct by climate, pattern of rainfall, temperature and solar radiations. Warm air at equator rises and move towards poles and cold air away from pole. The earth rotation deflect this air to right and left of the poles. The effect is called **Coriolis effect**. Air circulation in each hemisphere is broken into three zones from poles: Zero to 30° latitude, 30° to 60° and when cold air move back to poles at high altitude.

246-247 Tundra Ecosystem

TUNDRA ECOSYSTEM

Treeless region found in arctic and top of mountains with severe cold climate and permanently frosted subsoil. Tundra have a long winter interrupted by a short summer. Plant can grow in this summer of 60 days. Soil remains frozen in most of the year. Subsoil is called permafrost, as it is permanently frosted. Most tundra region is covered with bogs, marshes, ponds and decayed vegetation. Average precipitation is very low approximately less than 25cm/year. Average annual temperature is -10ºC although it may drop to -50ºC in winter and rise to 10ºC in summer. There are no trees, plant life is mainly represented by lichens, grasses and dwarf woody plants. Growth of plants is very slow due to extreme conditions. Animal life is mainly represented by rodents such as arctic hare, musk oxen, arctic foxes, owls, weasels and caribou (reindeer). Tundra exists in two forms: Arctic tundra and Alpine tundra

Arctic tundra is present in northern hemisphere. It includes the areas of northern Russia and Canada. It is the home of Nganasan and Nenets people. Temperature ranges form -30ºC to 20ºC. Animal life include arctic fox, polar bear, gray wolves, caribou, snow geese and musk ox. Plant life lichen, mosses, grasses and shrubs. **Alpine tundra** is present at high elevation are too cold and windy to sport plant life. North American Cordillera, the Alps and Pyrenees of Europe, the Himalaya and Karakoram of Asia, the Andes of South America, and the Eastern Rift mountains of Africa. Flora of alpine tundra includes Lichens, grasses and shrubs. They have much more roots and rhizome than shoots which are also storage site during winter. Animal life include kea, marmot, mountain goat, bighorn sheep, chinchilla, yak and pika. Arctic tundra is changing dramatically due to global warming. Animal previously not found in arctic tundra are now competing with native animals. Shrubs are getting taller and animal like spider are more active. Permafrost is melting. It holds large amount of carbon and methane. Climate change may change tundra from carbon sink to carbon contributor.

248-249 Coniferous forests

CONIFEROUS FORESTS

Coniferous forest are dominated by cone bearing, needle shape evergreen plants called conifers. They have cool summer and cold winter. Growing season is 130 days. Soil is acidic and relatively infertile. There is moderate rainfall ranging from 40 to 100cm/year. Most precipitation in the form of heavy snow. **A**verage annual temperature is 10ºC although it may drop to -40ºC in winter and rise to 20ºC in summer. Plant life is represented by spruce, fir and pine. They are evergreen. Some primary production also on warmer summer days. Animal diversity is low but large population. Major animals are snowshoe hare, lynx, wolves, caribou, moose and summer moths.

Coniferous forests are evergreen, their primary productivity is higher than grasslands but lower than deciduous and tropical forests. Decomposition process is slow due to cool environmental conditions. Nutrients are locked up in pine needles that fall from trees. Fire is important in recycling of nutrients and promoting seed germination. Coniferous Forests are the largest terrestrial biome on earth. Stretched from North America to Europe and Asia in northern hemisphere is called taiga. Whereas, temperate Coniferous Forests are present worldwide at high altitudes in America, Africa, Europe South Asia. In Pakistan they are present at 1000 to 4000 altitude. Coniferous forests are present in Chitral, Swat, Dir, Abbot Abad, Hilly Areas of Balochistan, Murree and Azad Kashmir. Coniferous trees are used in buildings and furniture. So, these forest are seriously damaged by human. Climatic changes are affecting the coniferous forests by mild, short winter and acid rain.

250-251 Temperate deciduous forests

TEMPERATE DECIDUOUS FORESTS

These are forests with moderate climate and well define summer and winter. They drop their leaves during autumn. These forests have four seasons: Winter, spring, summer and fall. Annual precipitation ranges between 75 to 150cm/year and temperature varies from -30ºC to 30ºC. Soil is fertile and brown in colour. Decomposition of fallen leaves enrich the soil due to which primary productivity is higher than grasslands and coniferous forests. Plants are broad leaves trees. There is great diversity of animal life. Important animals include spiders, ants, white tailed deer, moose, wolves, raccoons bears and tigers. Temperate deciduous forest are located in Eastern USA, Canada, Japan, China and some part of Australia. Human get valuable timber from these forests. These forests have been mostly cut down from Europe for farming purposes. Some have been converted to commercial forestry.

252-253 The Grassland

THE GRASSLANDS

These are the areas where vegetation is mainly dominated by the grasses. They account for 20-40% of world land area. They lie between forest and deserts. Grassland is found where there is not enough rain fall. Precipitation is 25 to 100cm annually and temperature ranges from -20ºC to 30ºC. The upper layer is nutrient rich making them ideal for agricultural use. But the low water availability may eventually convert them into deserts. Grasslands of North America are called prairies while the grasslands of South America and Africa are called tropical savanna. **Praries** are treeless while **savanna** has scattered trees (less than 30%).

Plant life is dominated by different types of grasses making layers. Plant life include clover, salvia, wheat, barley, coneflowers and many species of grasses such as, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass. Grasslands support large populations of herbivores and carnivores including insects, deer, zebra, wildebeest, giraffe, bison, lions, tigers and hyenas. Important grasslands include Great Plains of North America, Pampas of South America, Veldt of South Africa, Steppes of Central Eurasia, and surrounding the desert of Australia. Continuous global warming has great impact on grasslands. Grasslands are under extensive pressure due to conversion to agricultural land and urban development.

254-255 Chaparral

THE CHAPARRAL

It is the biome fount in small patches in most of the continents. It is characterized by very hot and dry summer with temperature up to 37.5ºC. Winter is very mild and usually about 10ºC and most of the rain occur in winter. Annual precipitation is 30-75cm. Soil is thin, rocky, nutrient poor, highly susceptible to erosion and contains low amount of moisture. Plants of chaparral are low and shrubby. They have tough and waxy leaves which are broad leaves and evergreen. Plants have no thick bark so not very resistant however their recovery is very fast. There are two types of plants: sprouters and seeders. **Sprouters** are dormant meristem, storage and intact root help to regenerate quickly while **seeders** are long lived seeds, germination induced by fire. Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important animals are insects, rodents, rabbits, lizards, snakes and mule deer. Chaparral is represented by west coast of United States and South America, South Africa, western tip of Australia and coastal area of Mediterranianean.

256-257 Desert ecosystem

DESERT ECOSYSTEM

It is a region which receives annual precipitation less than 25cm. They are characterized by scarce vegetation and extreme temperature occilation. Extremely dry climatic conditions are present here. It has lowest precipitation in comparison with other biomes. There go years and years without precipitation. At day time temperature may be 38 while at night it may be 4ºC. Desert soil is dry with low organic nutrients, low fertility and low water holding capacity. Only those plants can grow which ability to survive in extreme conditions of drought. Reduced leaves, succulent stems, deep stomata and waxy cuticle are the important adaptations of desert plants. Predominant plants in desert ecosystem are different species of cacti and thorny shrubs. They have reduced foliage and drought resistant seeds. Animals of desert include different types of arthropods, reptiles (snakes, lizards), birds (road runner, cactus wren, turkey vulture and burrowing owl) and mammals (mule deer, jackrabbit, kangaroo rat and ground squirrel).

Desert is only ecosystem which is increasing with human activities. This is because the grasslands used for agriculture and ultimately converted into desert land. Important deserts of world are Antarctic Desert, Arctic Desert, Sahara Desert, Australian Desert, Arabian Desert, Gobi desert, Kalahari Desert, Thar Desert, Columbia basin and Dasht-e-kavir.

258-259 Tropical rain forests

TROPICAL RAINFORESTS

It is a hot moist biome found near the equator. They are the richest in biodiversity and most exciting areas of the earth. Tropical rainforests have warm temperature and abundant rainfall. Average temperature ranges from 20ºC to 25ºC and it remains constant. Annual rainfall is more than 200cm. There is little seasonal variation in day length. Soil is relatively nutrient poor. Most of the nutrients are tied up in organisms. High moisture and temperature make nutrient recycling rapid.

Tropical rainforest have 50% of plants and animals diversity with just 6% land. Plant life is very diverse. There are thousands of plants species. However, none of them dominating. Trees are broad leaved and non-deciduous. Plants in tropical rainforest are highly stratified. A single hectare typically contains 50 to 70 species. Some of the important plants are palm species, rubber tree, bamboo, orchid, tualang, different epiphytes and pitcher plants. Both soft bodied and hard bodied endothermic animals are abundant. Important animal groups are annelids, arthropods, amphibians, reptiles, birds and mammals. Tropical rainforest are located in South and Central America, Western Africa, Southeast Asia and Australia. Once they were 14% and now they are 6%. It is estimated that in next 40 years the remaining will be lost as well. The deforestation mean habitat loss, extinction, soil erosion and climate change.

260 Freshwater ecosystem: lotic

FRESHWATER ECOSYSTEMS

Freshwater ecosystems are classified according to water flow.

Lotic Ecosystem

It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers. Attached aquatic plants and algae provide some of the primary productivity. Most of the primary production comes from the surrounding plants by falling their materials during autumn and rainfall. Lotic systems typically have high oxygen level. Turbulence of water while flowing over rocks, logs and other objects causes oxygenation. This support the animal life in lotic system. Mainly algae and attached aquatic plants are indigenous producers. Animals include different aquatic invertebrates, larvae of insects, fishes, amphibians reptiles birds and mammals. Water Currents are the most important physical factor of lotic habitat. Animal adaptation include dorsoventral surface, strong swimmers, burrowing and Attachment with substrate.

261 Freshwater ecosystem: Lentic

Lentic Ecosystem

This is the ecosystem of standing water. Lentic ecosystem include lakes and ponds. Vertical stratification is common especially in temperate region. In summer, upper layer is warm while lower layer is cold and more dense, so no mixing. In fall, upper layer is dense so mixing of water occurs. Photosynthesis mainly in upper layer of water as light is filtered by water. Phytoplankton are main producers of lentic ecosystem. Deeper layers are depleted of oxygen in summer. In fall the mixing of entire water column supply oxygen to deeper layers. Deep, cold, nutrient poor lakes are relatively unproductive called as oligotrophic lakes. Shallow warm, nutrient rich lakes are highly productive called as eutrophic lakes. Plant life include phytoplankton, periphyton algae, emergents, floating- leaves, submersed and free floatingplants. Animal life include zooplankton, invertebrates like crabs, crayfish and shrimps, molluscs while vertebrates mainly includes variety of fishes, reptiles and birds.

262 Marine Ecosystem: Esturies

MARINE ECOSYSTEMS

Most important character used to distinguish the marine water from freshwater is salinity. Marine water typically contains 3.5% salts. Marine ecosystems are of following types.

Estuaries

Estuaries occur where freshwater meets the seawater. This dilutes the salinity of seawater. Estuaries are highly productive because the streams carry nutrients of freshwater and tides bring nutrients from ocean. Producers of estuaries include phytoplankton, attached aquatic vegetation and emergent vegetation such as cordgrass. Estuaries are nurseries for crustaceans, molluscs and marine fishes. They are the feeding grounds for wading and swimming birds. Human have drained and filled estuaries. Estuaries are polluted by oil exploration and drilling. Acids rain have also threaten the productivity of estuaries.

263 Marine Ecosystem: Intertidal Zone

The Intertidal Zone extends from splash zone ocean to low tide marks. This region is subjected to the action of both tides and waves. It is the harshest and the richest of all marine environments. Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluctuation. This zone has bounty of habitat and plenty of nutrients. Here, plants are larger. They include seagrass, red algae and

mangroves. Motile animals like fishes and crabs move with tides. Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud. Water trapped in tidal pools support a rich animal life such as echinoderms, barnacles, limpets and chitons. Attached animals close shells to avoid drought during low tides.

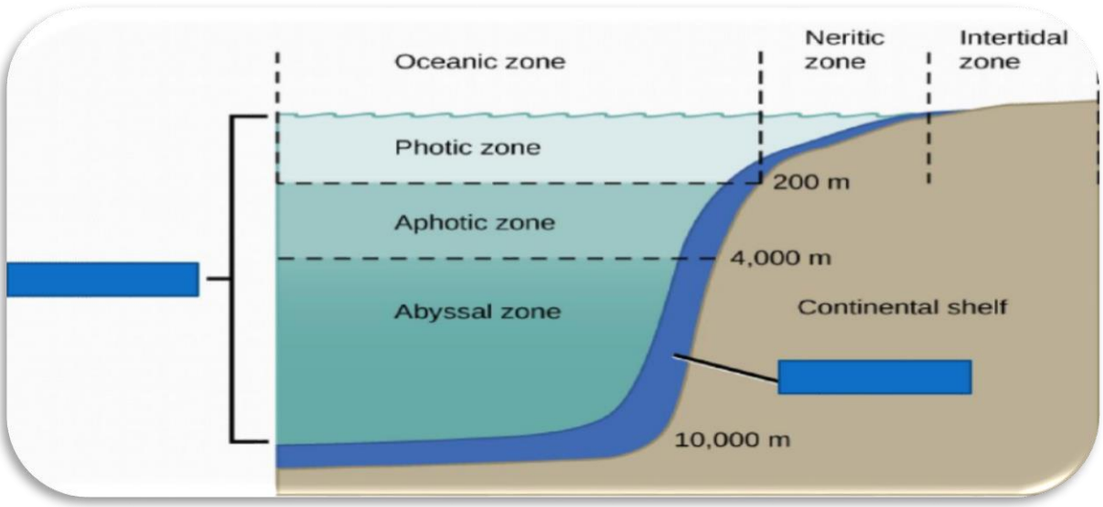


Fig. Different zones in marine ecosystem

264 Marine Ecosystem: Neritic Ecosystem

The Neritic Ecosystem extends from littoral zone to the edge of continental shelf. It is also called the shallow water. Nutrients are delivered by rivers and upwelling at the edge of continental shelf. This region is more productive than open sea. Planktonic and attached algae contribute to the primary productivity. Plants include kelps, algae, seaweeds, seagrasses and phytoplankton. Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes whereas, vertebrate fauna include different types of fishes such as code, tuna, herring, capelin and many other species.

265 Marine Ecosystem: Oceanic Ecosystem

Oceanic Ecosystem starts from continental shelf. It is the open sea. As it is nutrient poor so, it is relatively unproductive. Virtually all primary productivity is in the upper water. Phytoplankton in the upper water are the main producers of oceanic system. Invertebrate animals include worms, echinoderms and molluscs. Vertebrates include fishes from small fishes to whales and sharks. At depth no light so the animals here have luminescent organs which serve as torch light. These animals depend on nutrient coming from upper layers.

266 Marine Ecosystem: Coral Reefs

Coral Reef is the underwater ecosystem. They are formed by colonies of stony corals. Coral are members of class Anthozoa. They grow in association of zooxanthellae. Polyps of corals are held together by calcium carbonate. Reefs are formed over thousands of years in warm, clear and relatively shallow waters. The greatest depth for their growth is 50-60 m. Below this light is insufficient to support zooxanthellae. Coral Reefs are the richest source of biodiversity in oceans. Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species. Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes.

267 Ecological Problems: Human Population Growth

ECOLOGICAL PROBLEMS

Human Population Growth

Population growth is the increase in number of individuals in a population. Expanding population is the root cause of most environmental problems. More population means more need for resources i.e., food, urbanization and industrialization. The present population of earth is close to more than 7 billion.

Virtually all scientists agree that even 5 billion is more for earth to support with all facilities. Maximum carrying capacity of earth is may be 9 to 10 billion people.

Age structure shows the proportion of individuals at prereproductive, reproductive and postreproductive stage. Developing countries have more infants means high birthrates. In contrast to human, in natural population high birthrate is accompanied with high mortality.

268 Ecological Problems: Pollution

Pollution

Pollution is any detrimental change in ecosystem. Mostly is results from human activities. More human population<more demands<pollution. Toxic waste of industries, sewage waste and agricultural runoff are the important sources of water pollution. Burning of fossil fuels release CO_2, oxides of sulfur and nitrogen and Increased CO_2 are polluting air. This this increasing the threats of acid deposition, smog and global warming. Land pollution is mainly caused by oil spillage, plastics, pesticides, toxic industrial waste, garbage and buildings leftover.

Wastes and poisons enter in food chain and travel to the organisms at highest level of food chain including human. They are not biodegradable and interrupt physiological processes. They accumulate in the bodies of animals and cause neurological, developmental and different types of physiological disorders.

269 Ecological Problems: Resource Depletion

Resource Depletion

Materials or substances occurring in nature which can be exploited for human benefits are called natural resources. Important natural resources are fossil fuels, minerals, air, water, soil and biodiversity. As human population is increasing rapidly and resources are limited. For better life man is overexploitation natural resources. This will ultimately use of pollute the natural resources and human survival will be at a great risk.

270 Ecological Problems: Biodiversity Loss

Biodiversity Loss

Current documented biodiversity of earth is more than 1.6 million. Estimate is 4 to 30 million. More accurate is between 8.7±1.3million. Biodiversity of all natural areas is threatened mainly due to habitat loss, acid rain, pollution, urbanization and agriculture. Human is exploiting 40% of earth net primary production.

Important solutions are realization about biodiversity. 22nd May is World Biodiversity Day. There is the need for a system of international ethics that value natural diversity.